

SQANTI3 report

Unique Genes: 16755

Unique Isoforms: 50557

Transcript Classification

Category	Isoforms, count
FSM	50364
ISM	0
NIC	108
NNC	59
Genic Genomic	7
Antisense	6
Fusion	4
Intergenic	9
Genic Intron	0

Gene Classification

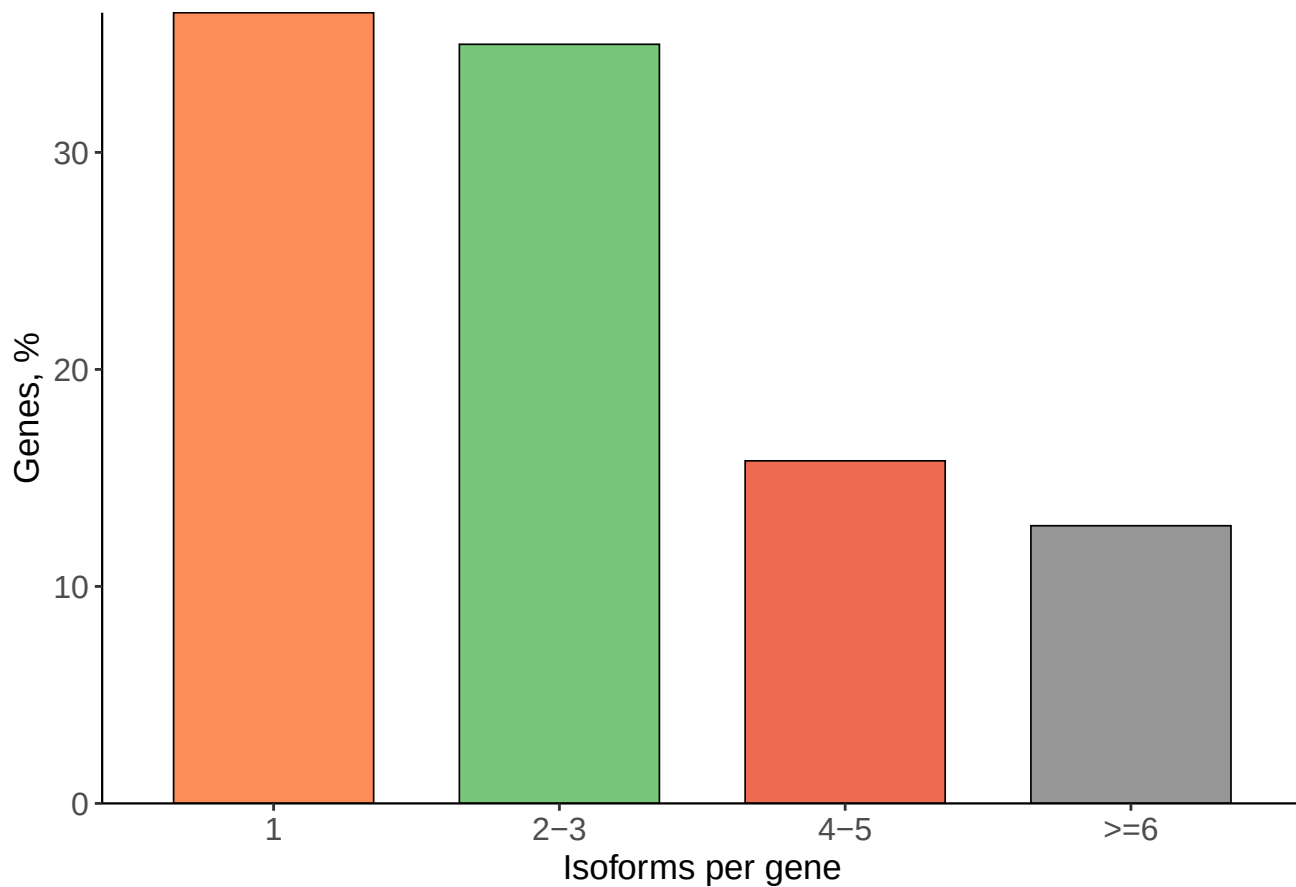
Category	Genes, count
Annotated Genes	16740
Novel Genes	15

Splice Junction Classification

Category	SJs, count	Percent
Known canonical	177263	99.84
Known Non-canonical	145	0.08
Novel canonical	132	0.07
Novel Non-canonical	3	0.00

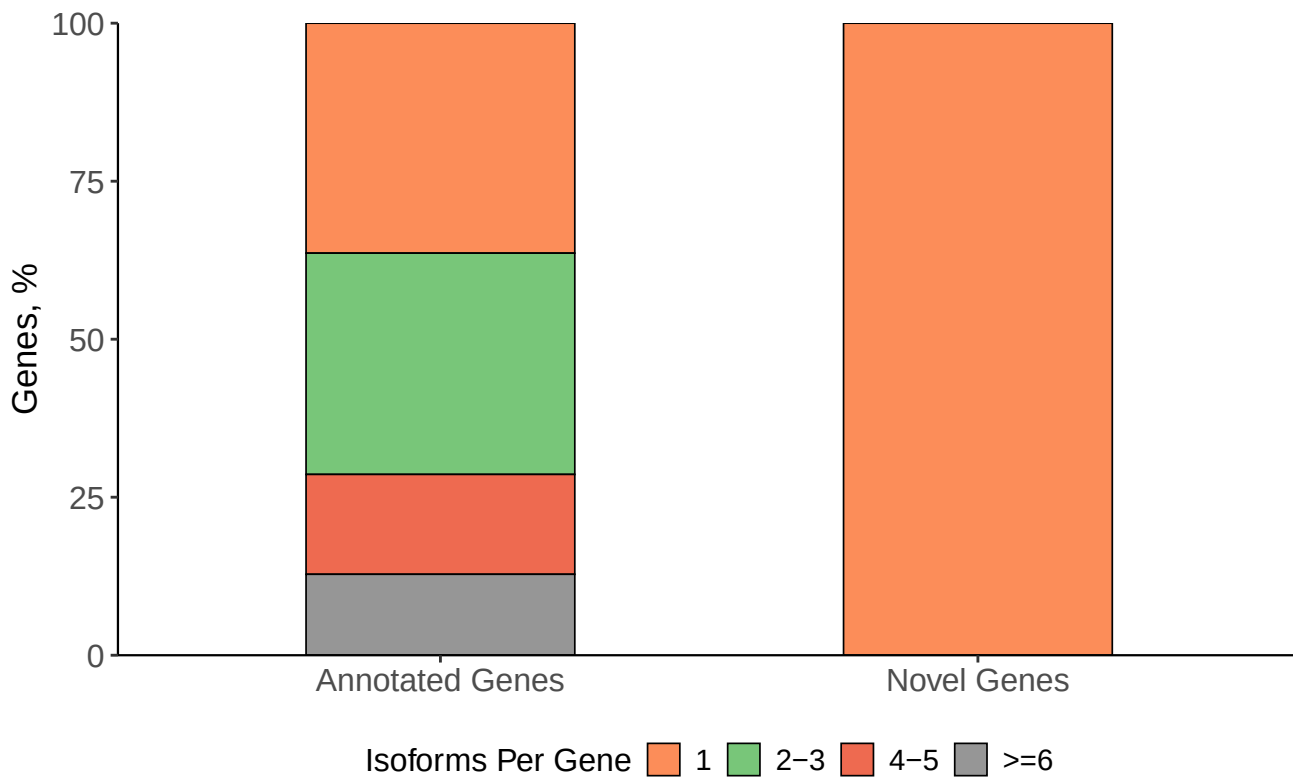
Gene Characterization

Number of Isoforms per Gene

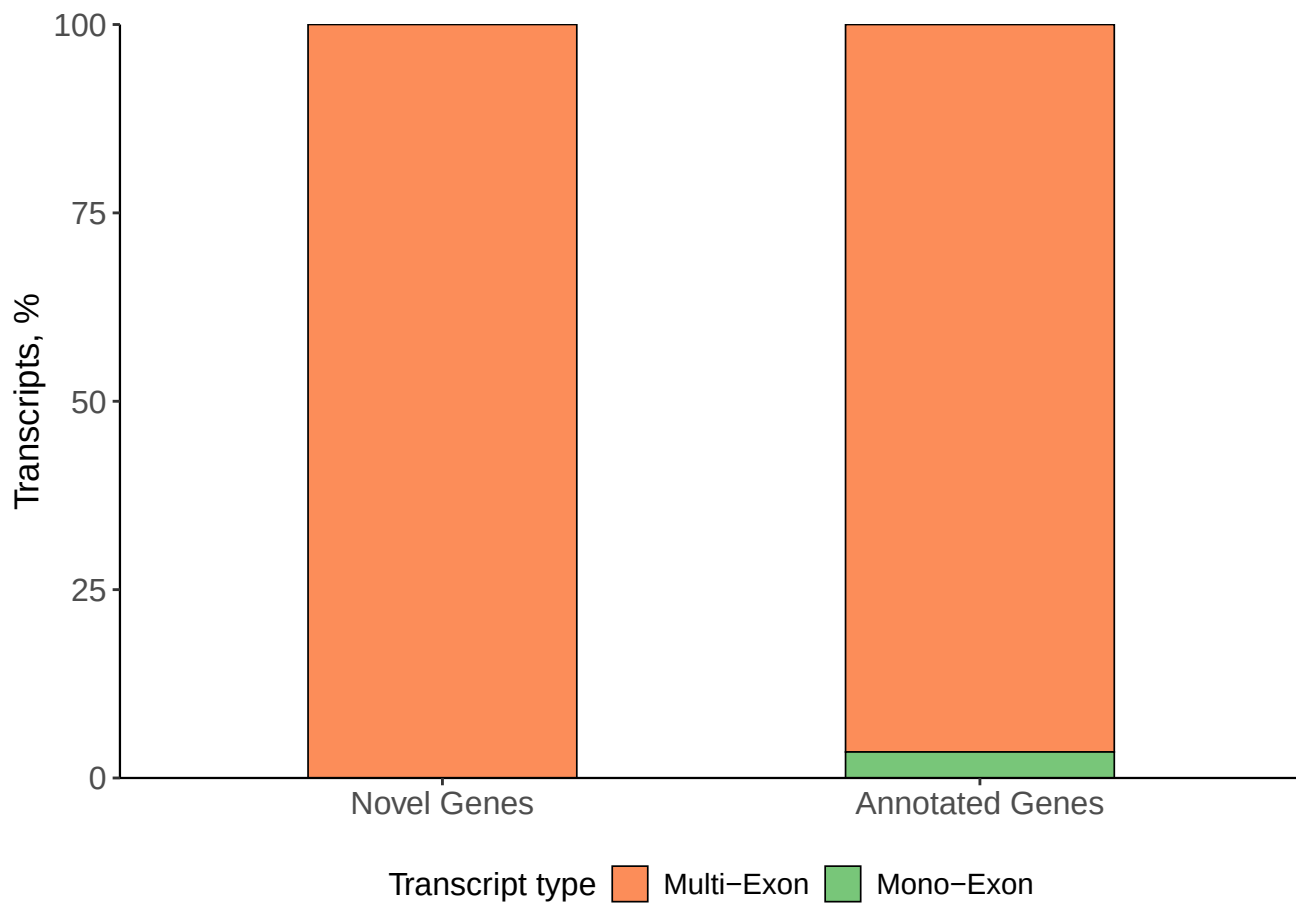


Number of Isoforms per Gene

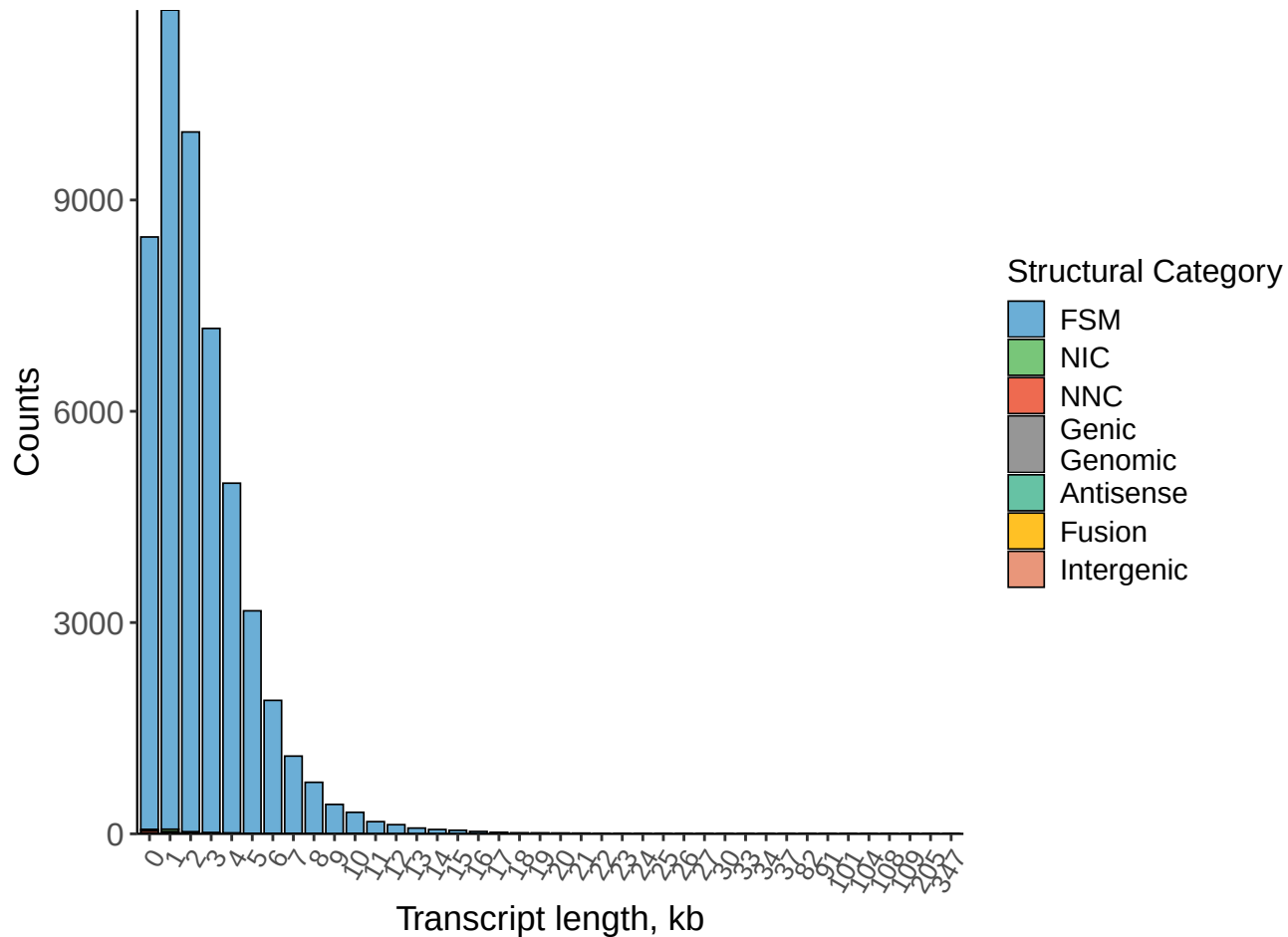
Known vs Novel Genes



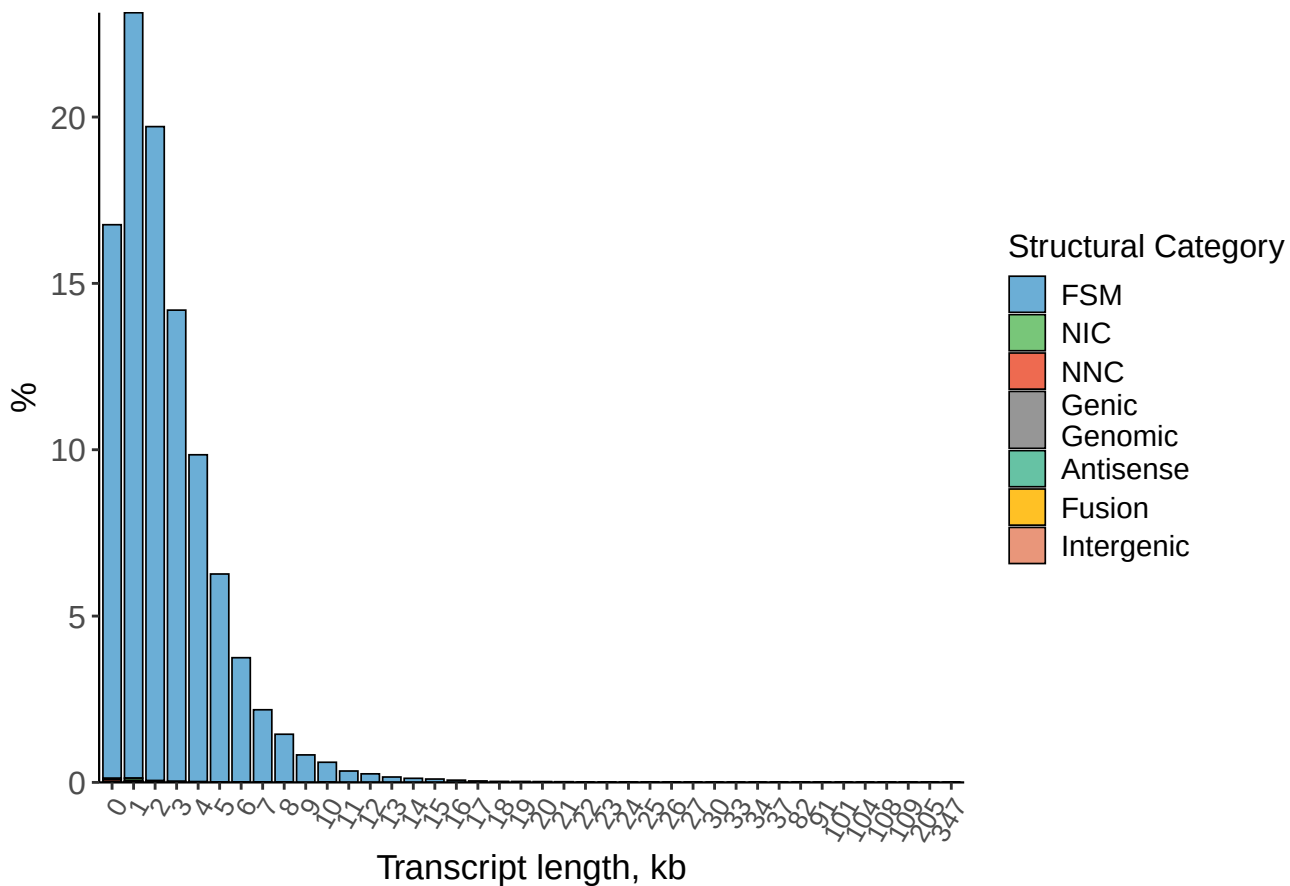
Distribution of Mono- vs Multi-Exon Transcripts



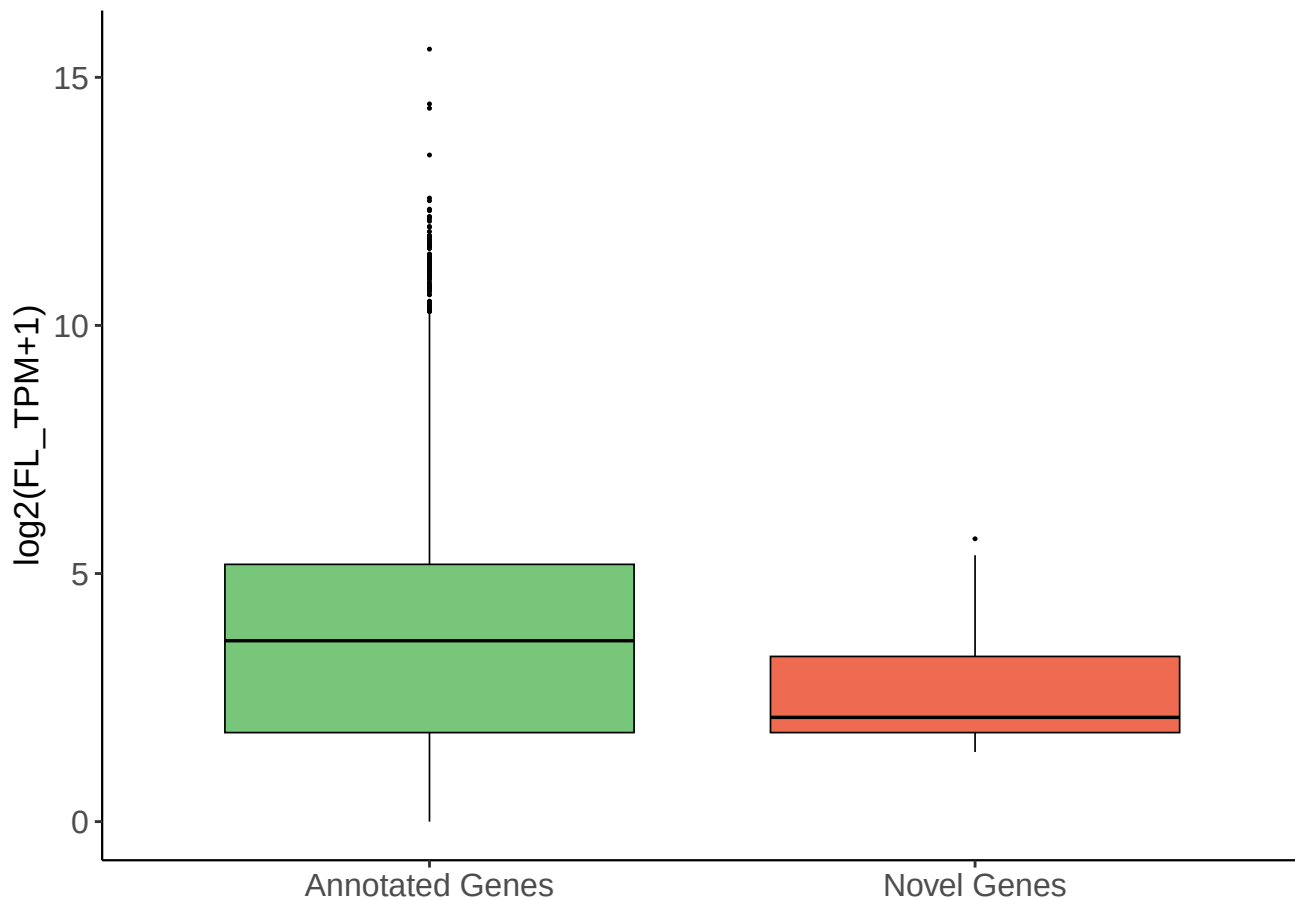
Structural Categories by Transcript Length



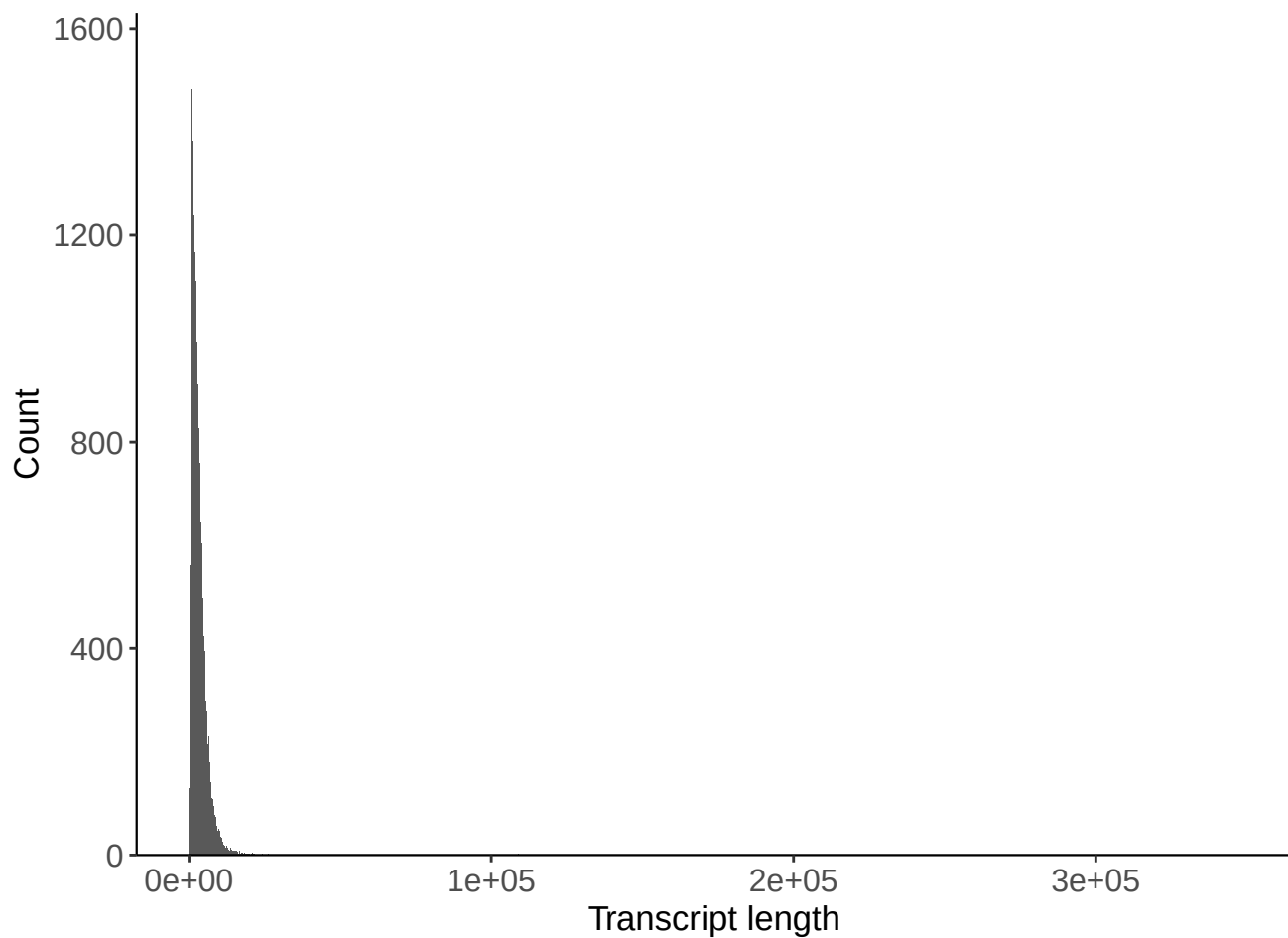
Structural Categories by Transcript Length



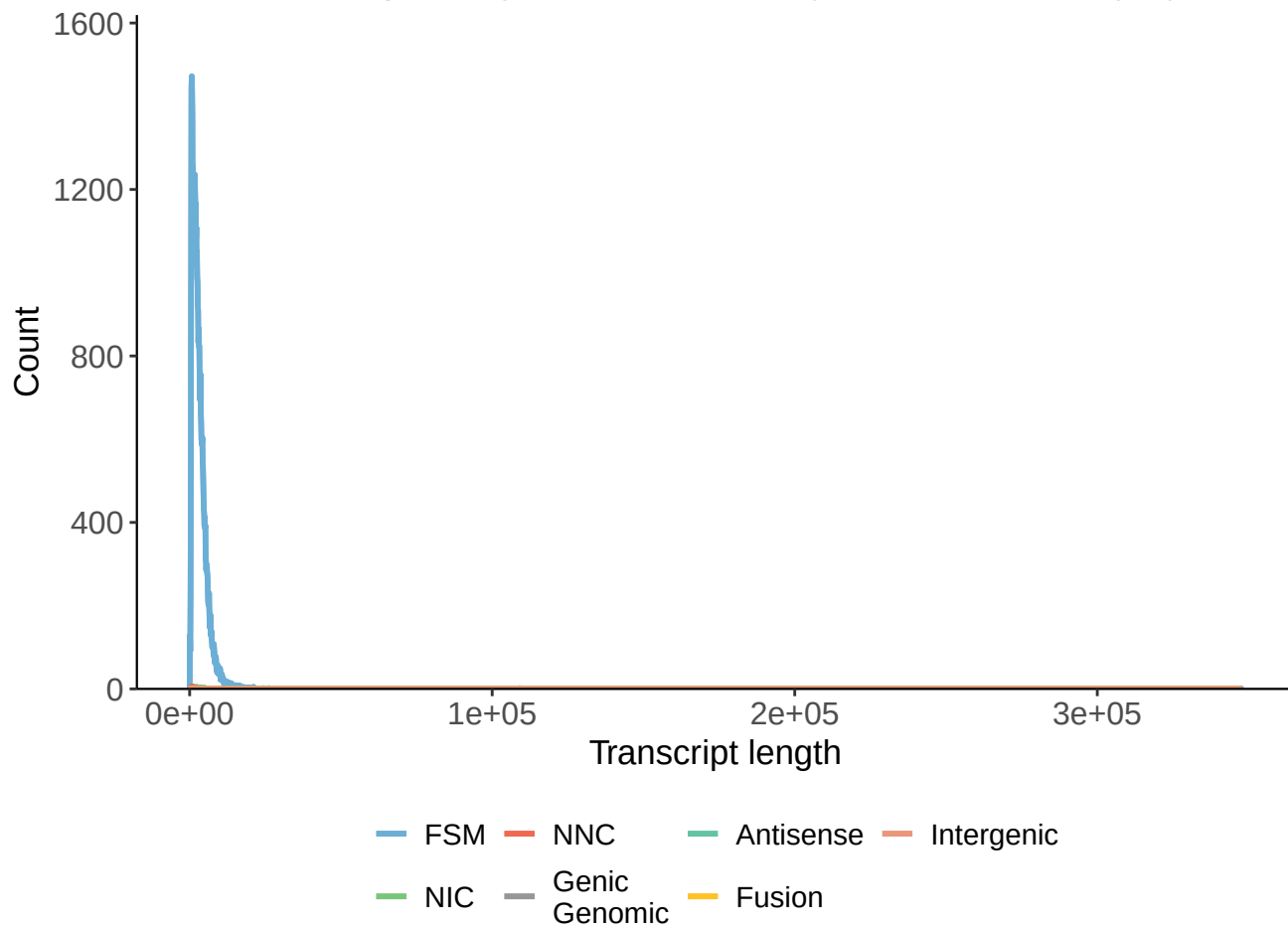
Number of FL reads per Gene by Type of Gene Annotation



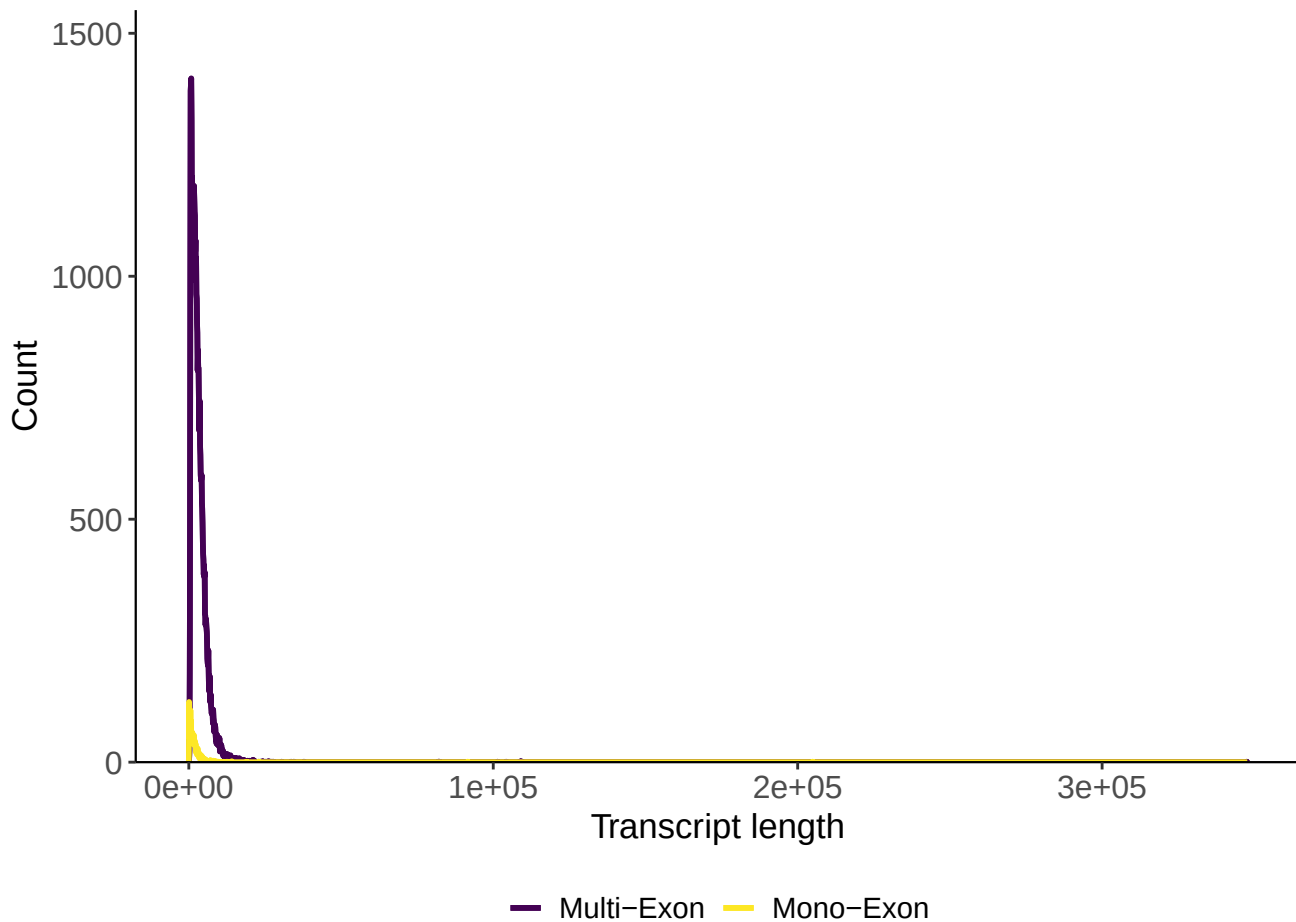
All Transcript Lengths Distribution



Transcript Lengths Distribution by Structural Category

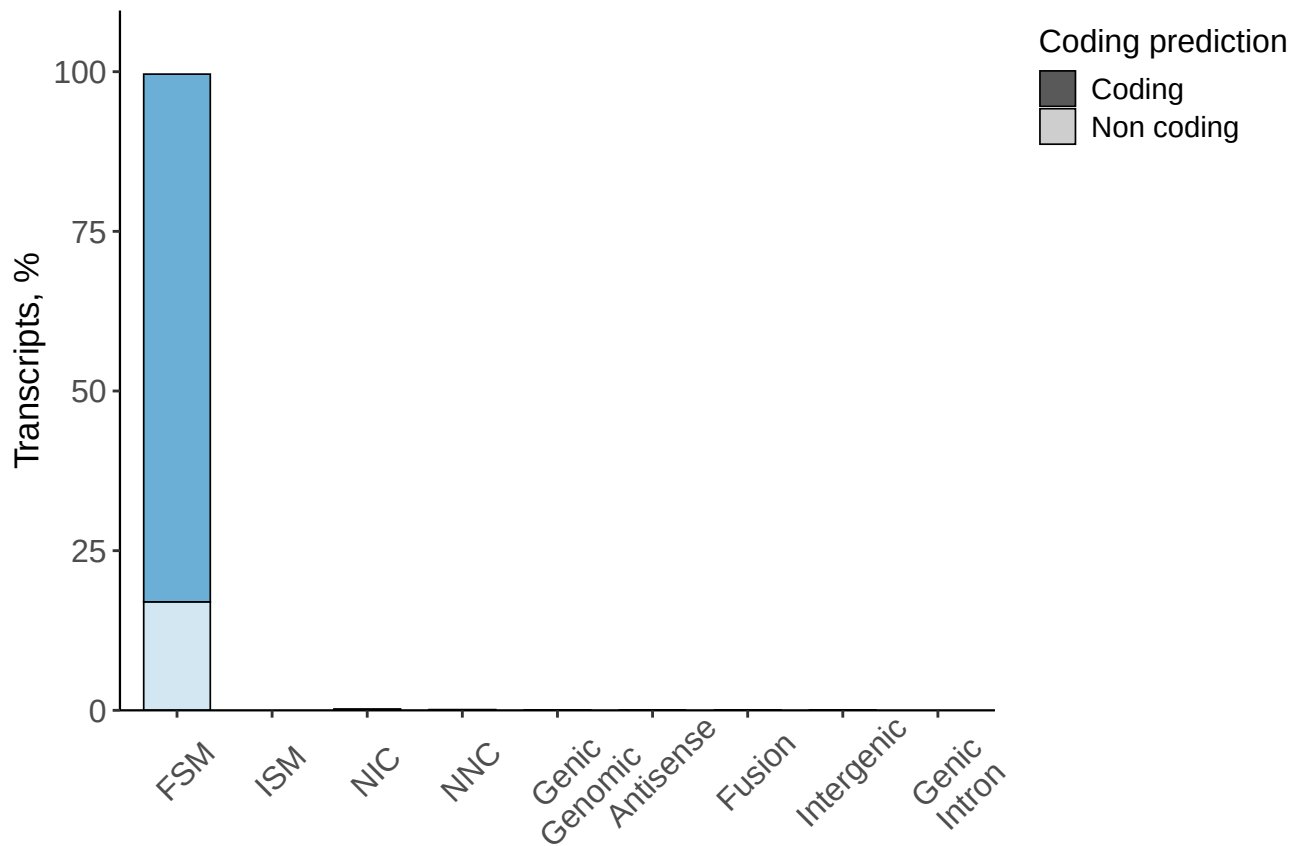


Mono- vs Multi- Exon Transcript Lengths Distribution

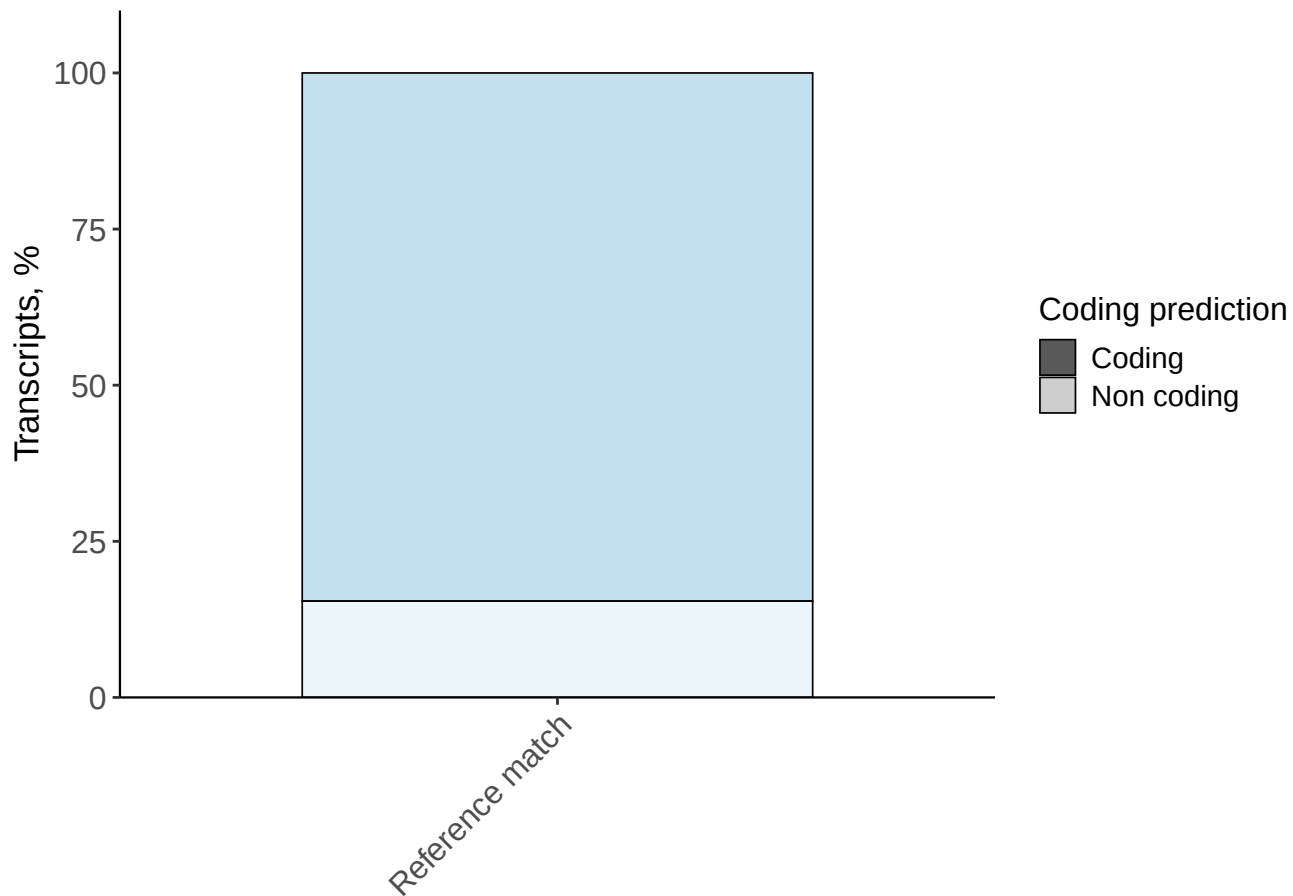


Structural Isoform Characterization

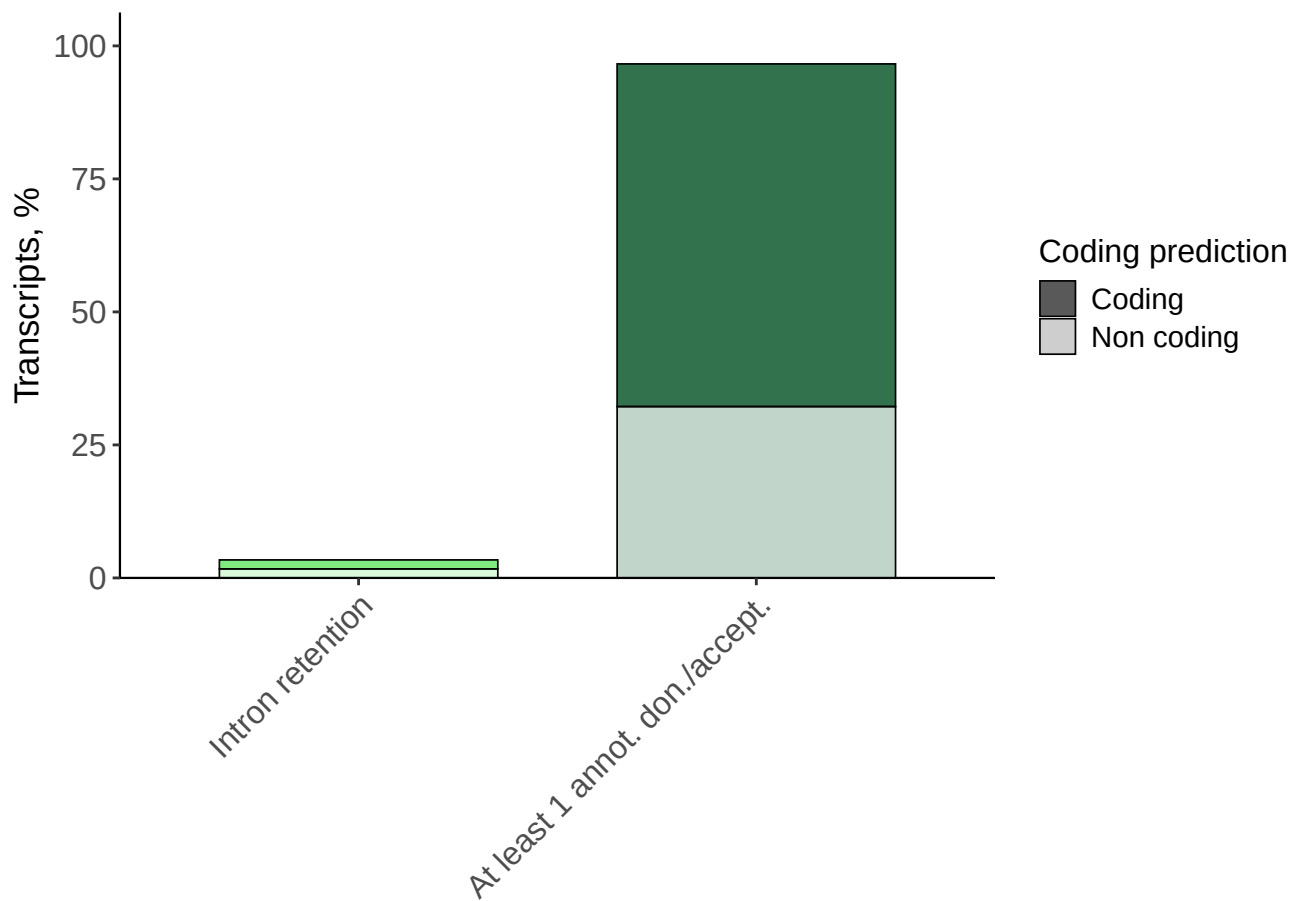
Isoform Distribution Across Structural Categories



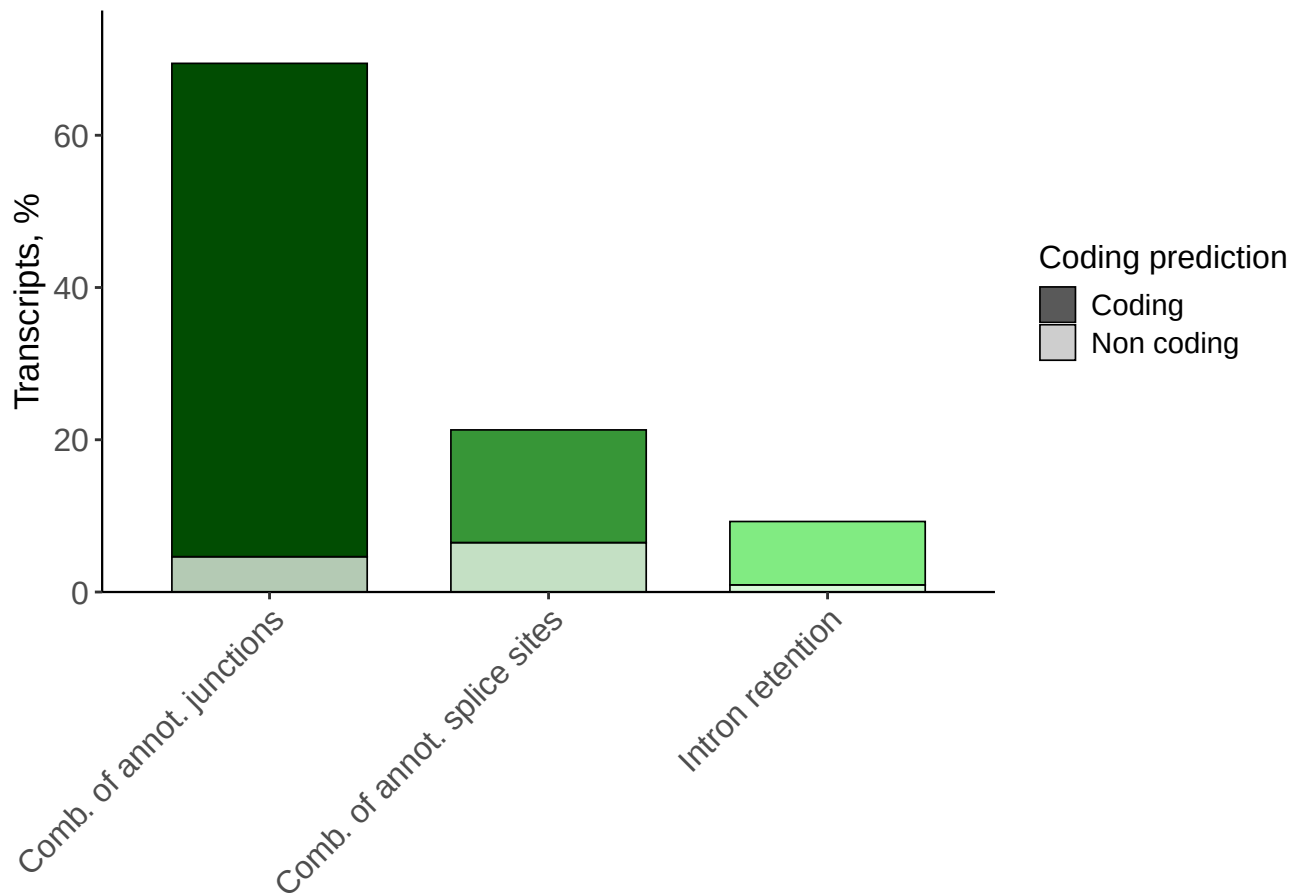
Isoform Distribution Across FSM



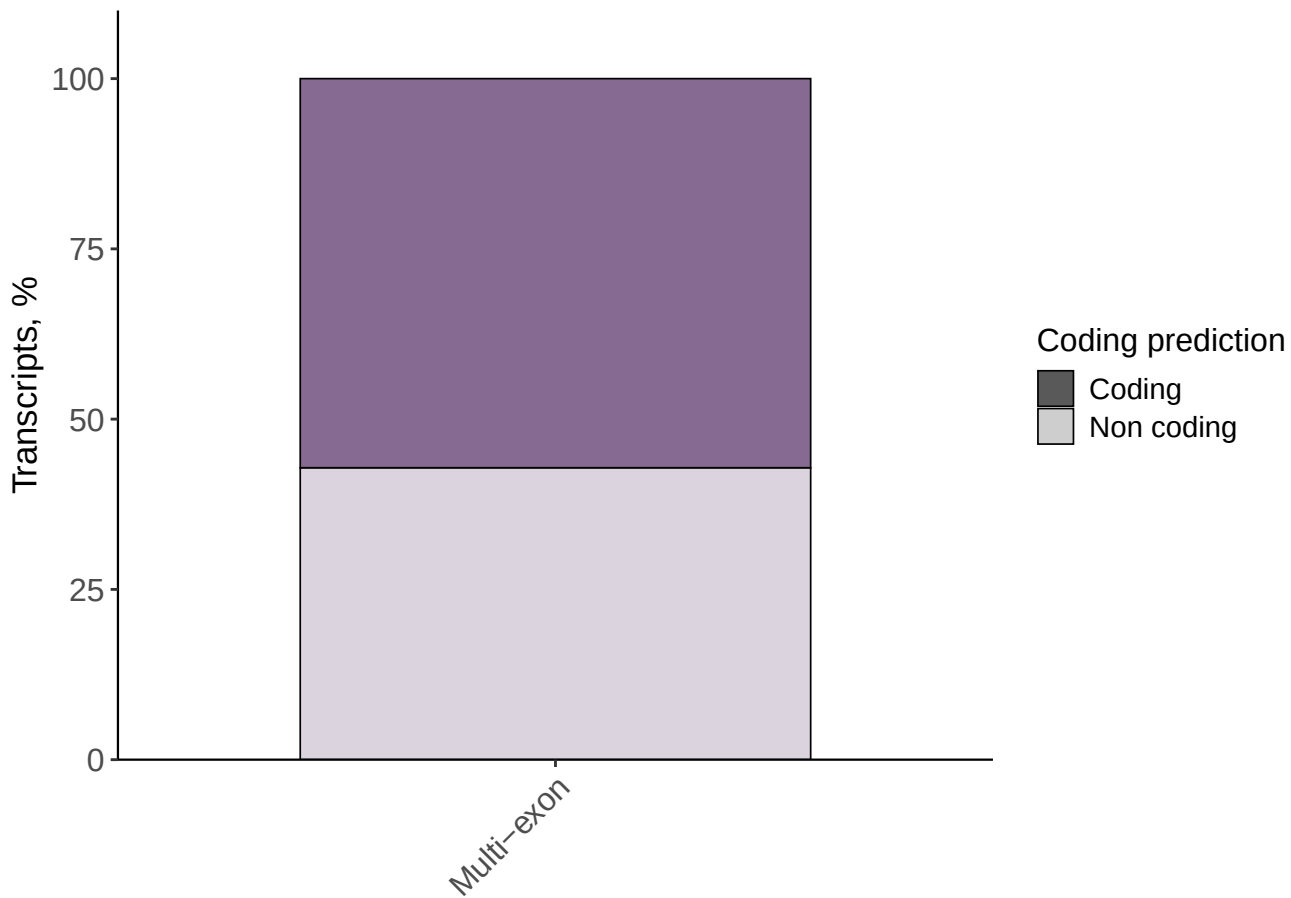
Isoform Distribution Across NNC



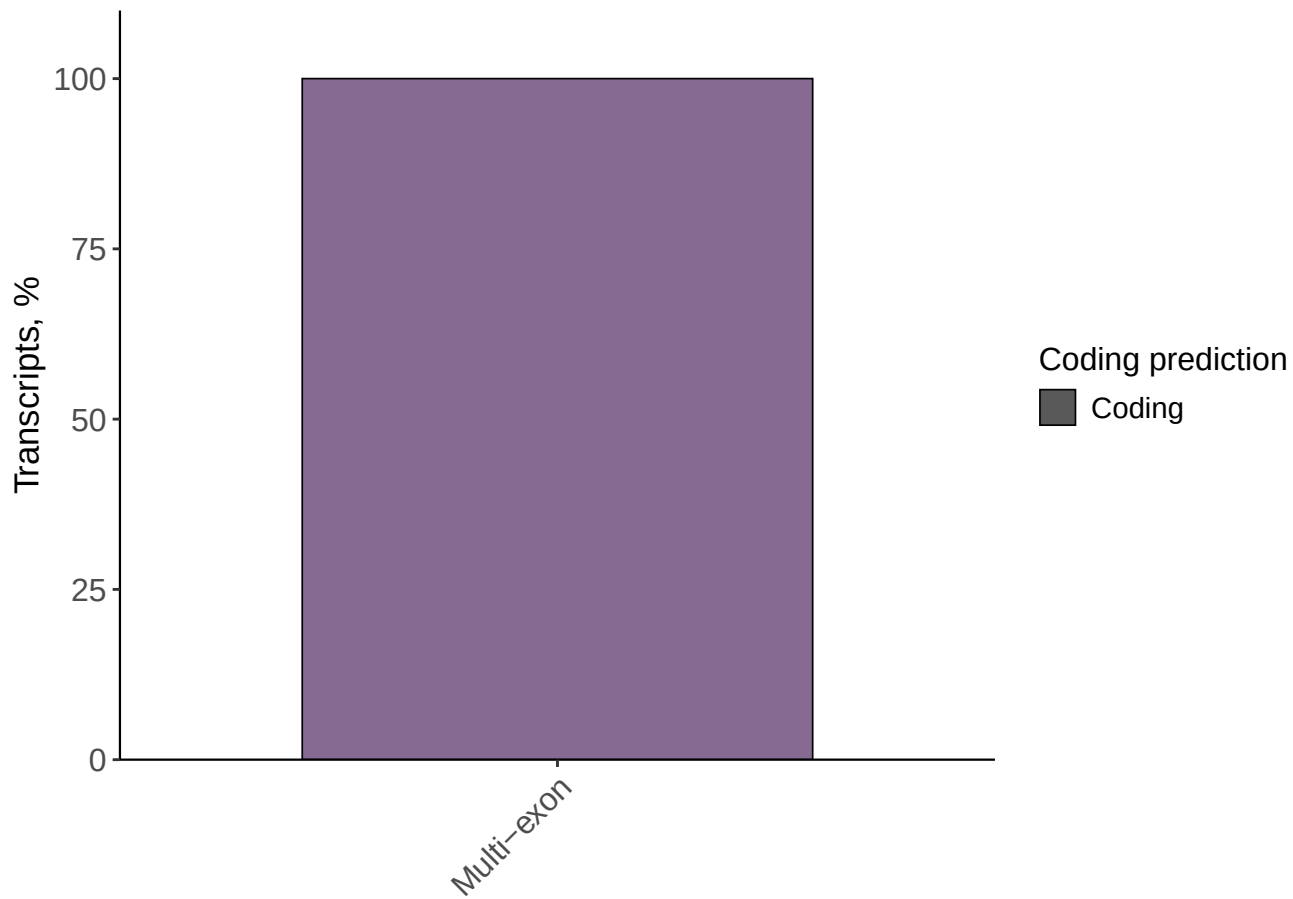
Isoform Distribution Across NIC



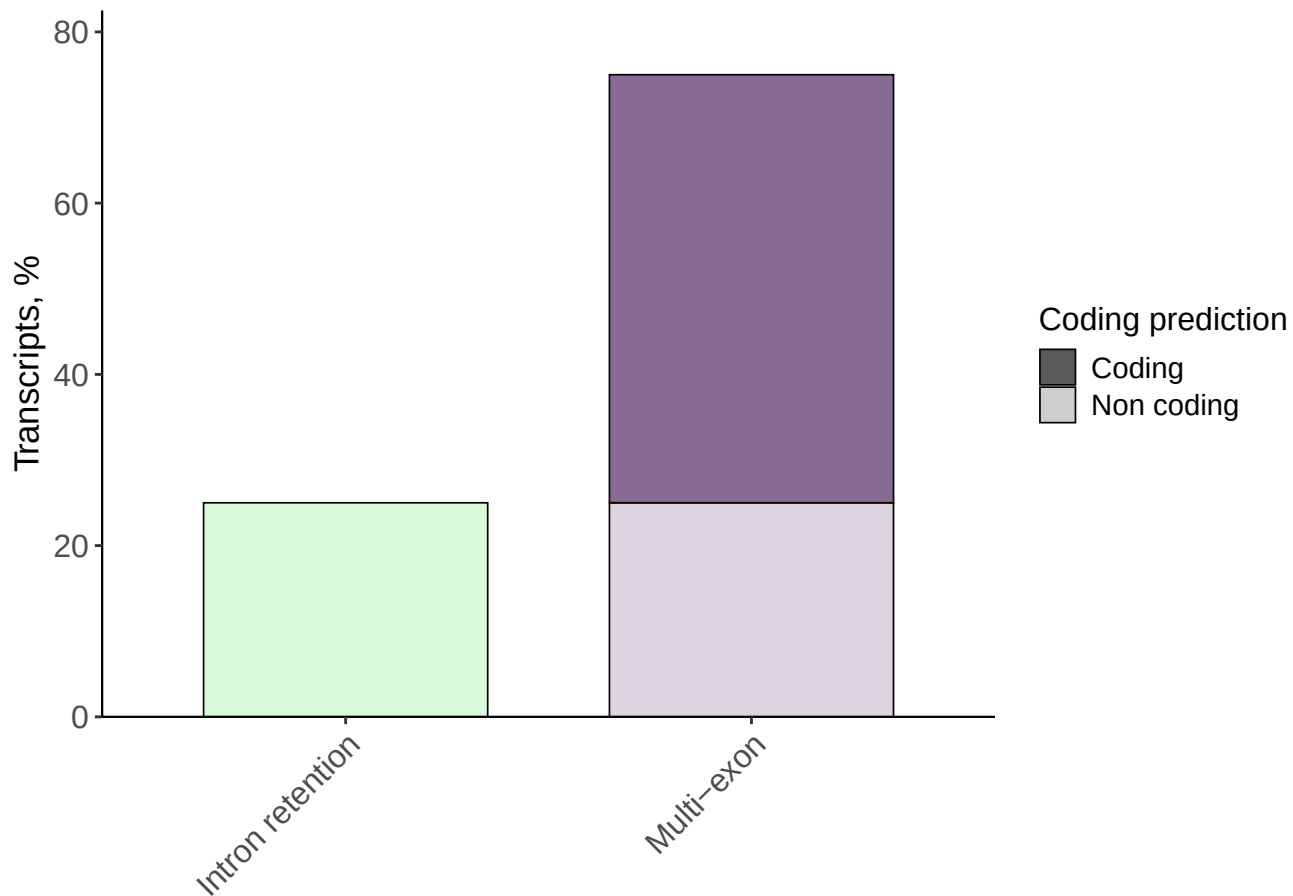
Isoform Distribution Across Genic Genomic



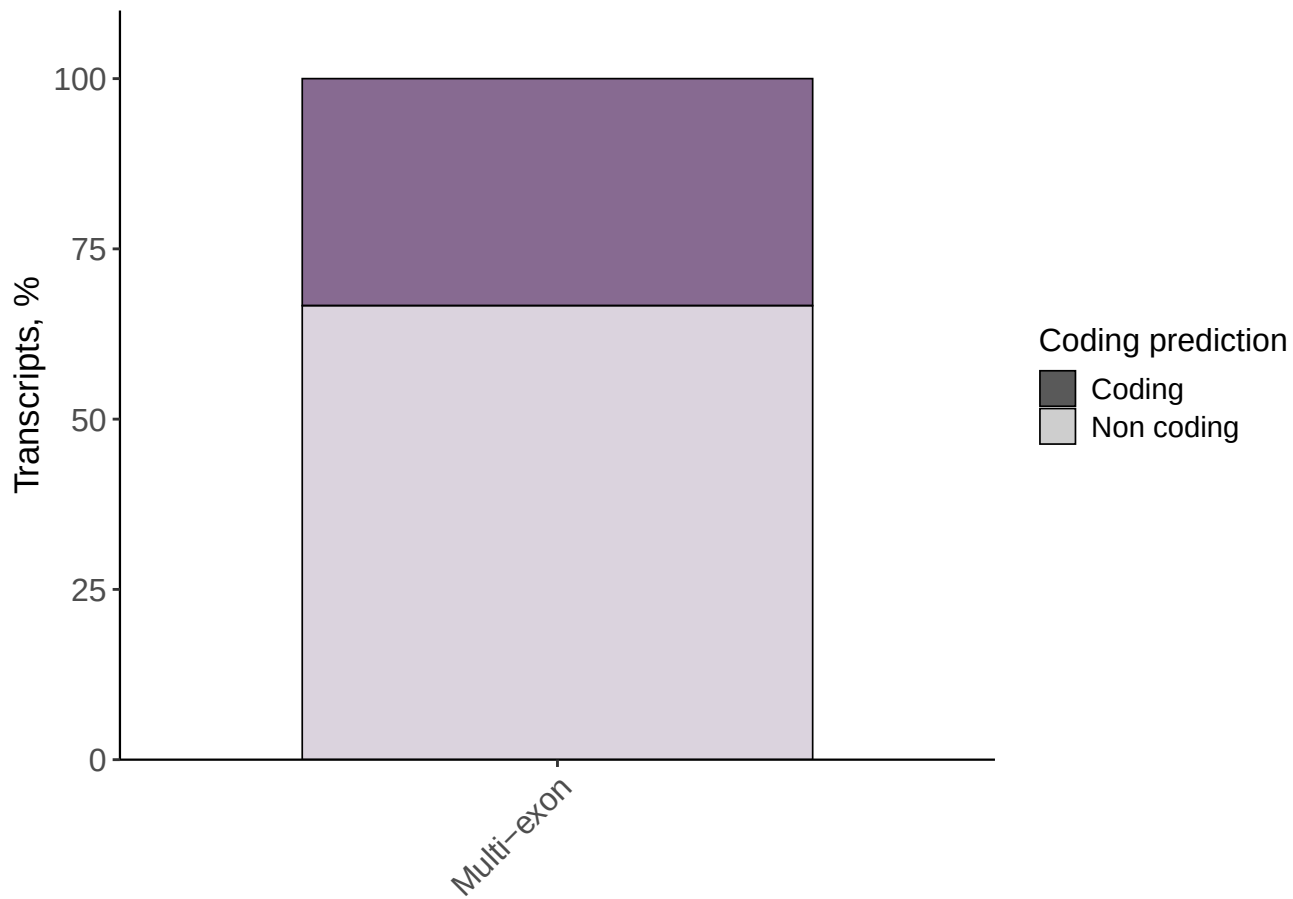
Isoform Distribution Across Antisense



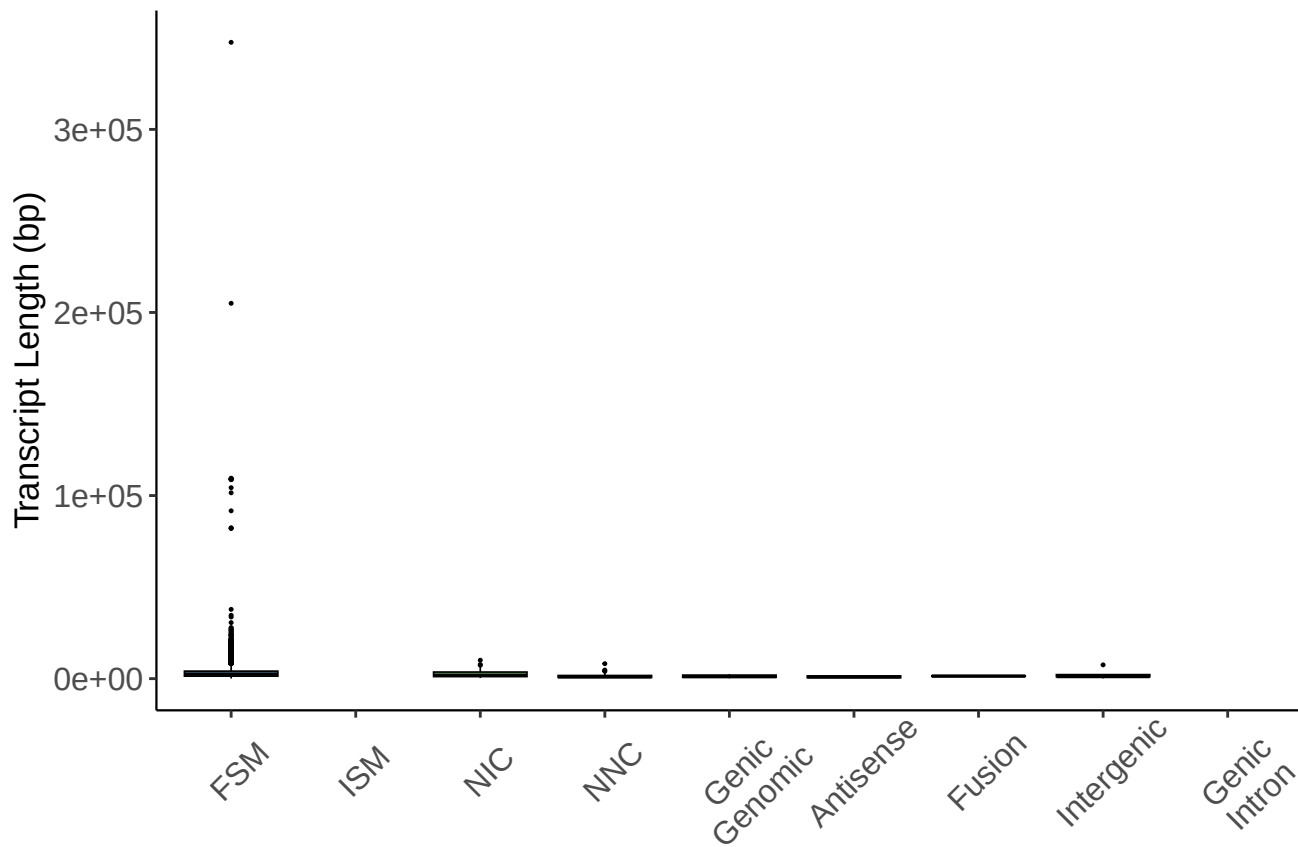
Isoform Distribution Across Fusion



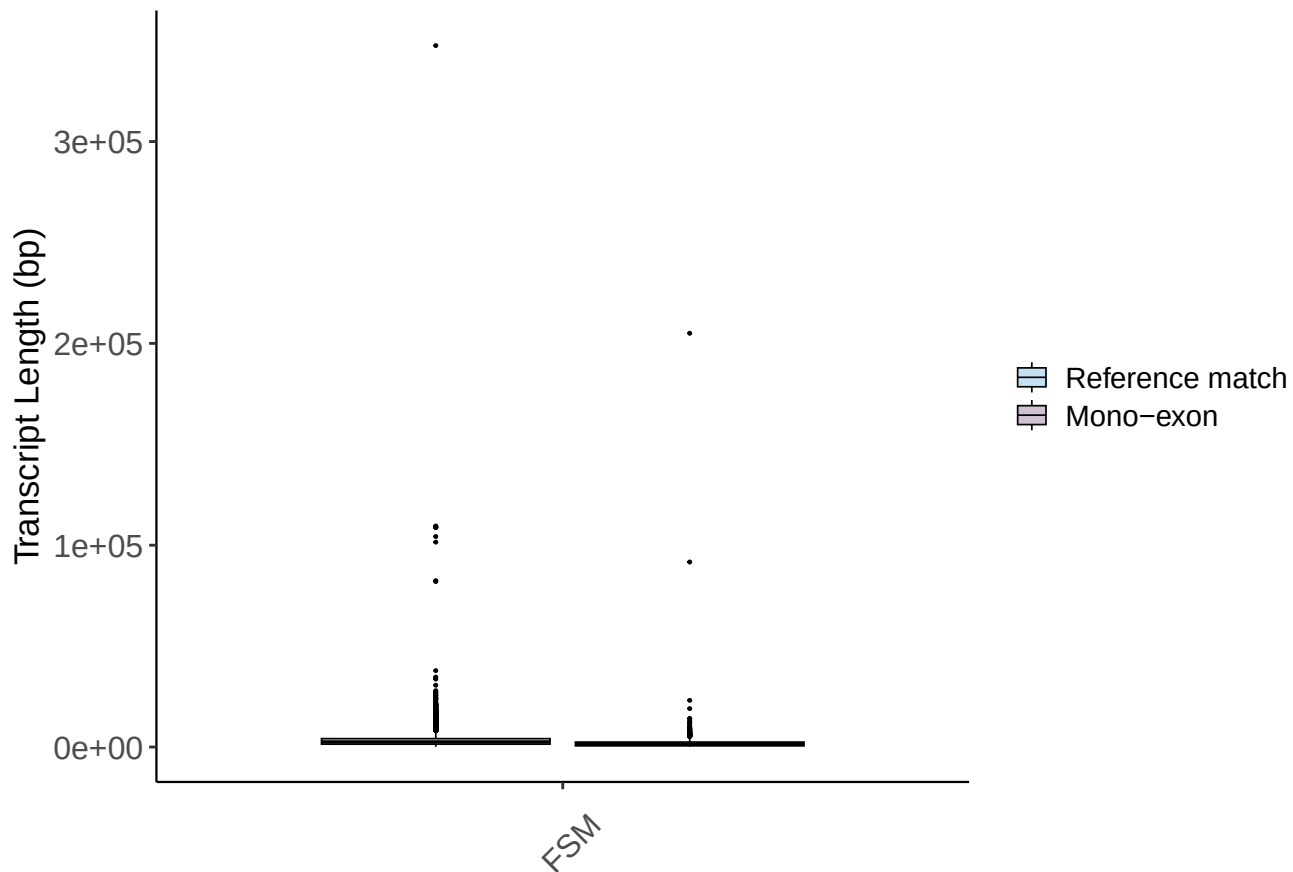
Isoform Distribution Across Intergenic



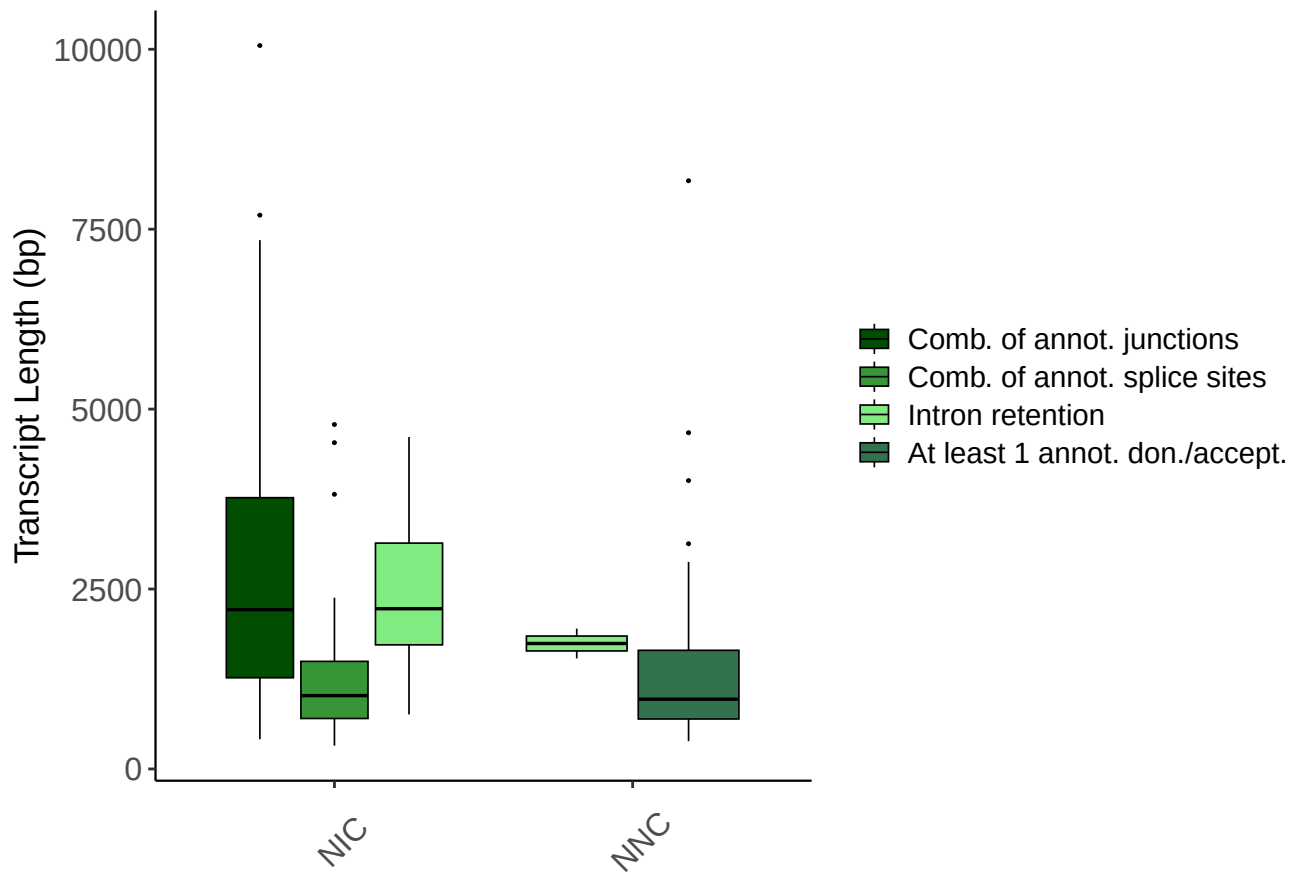
Transcript Lengths by Structural Classification



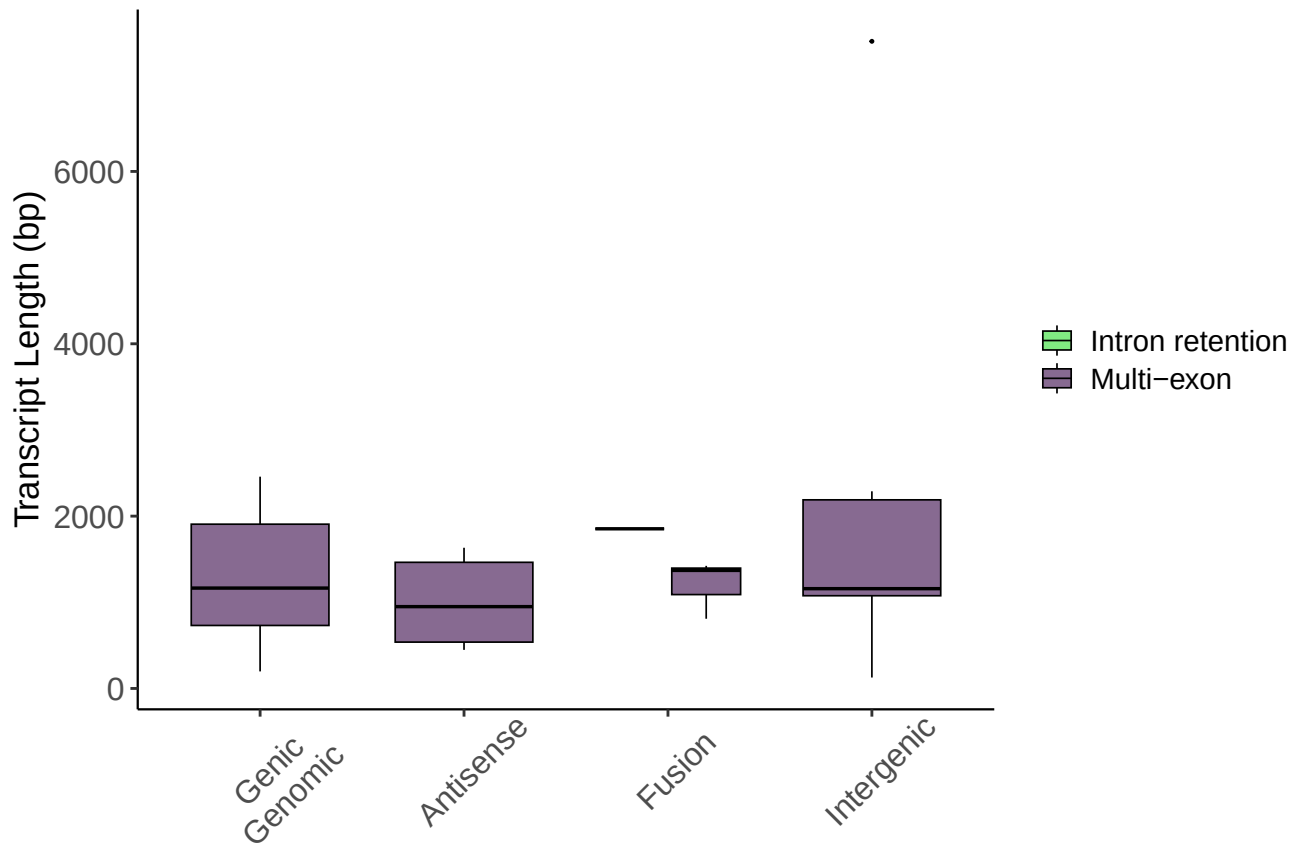
Transcript Lengths by Subcategory



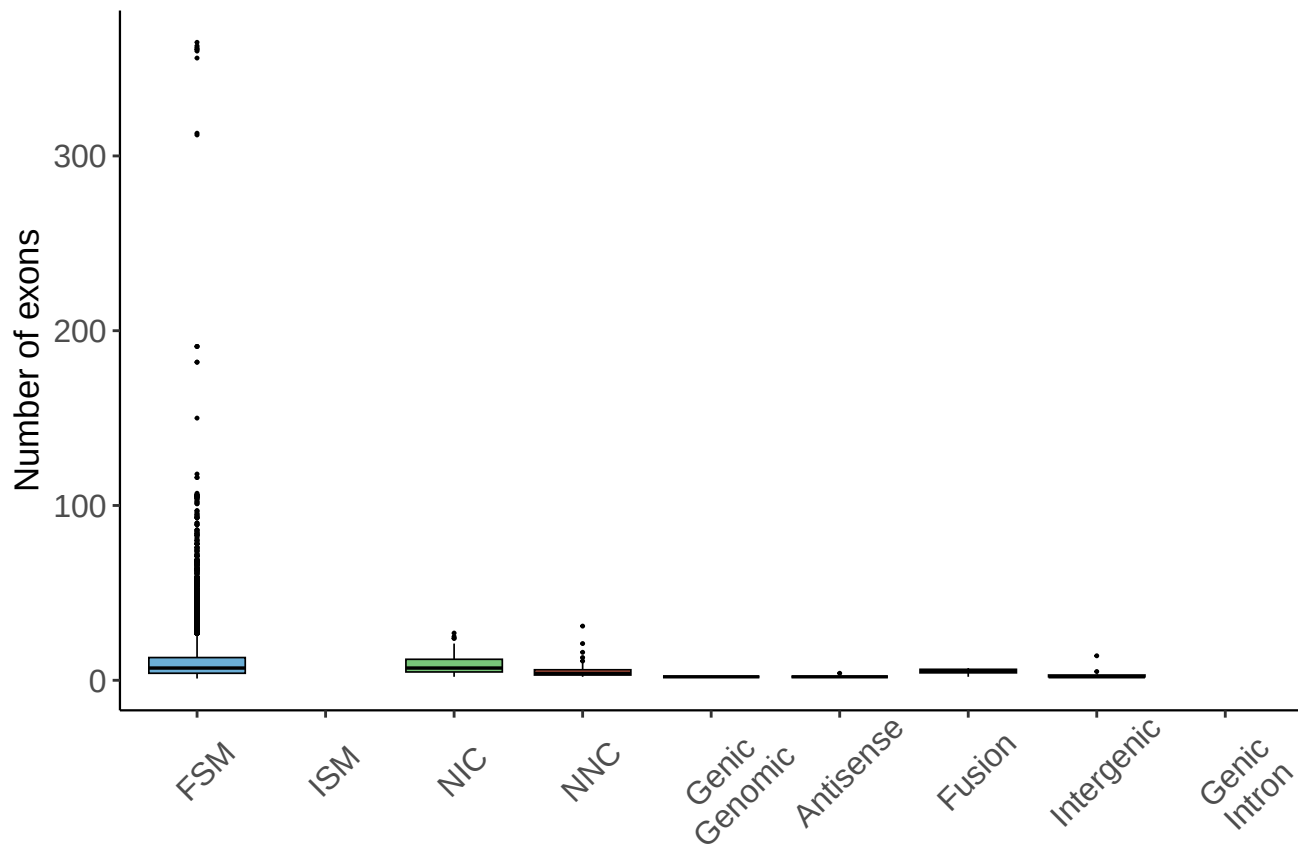
Transcript Lengths by Subcategory



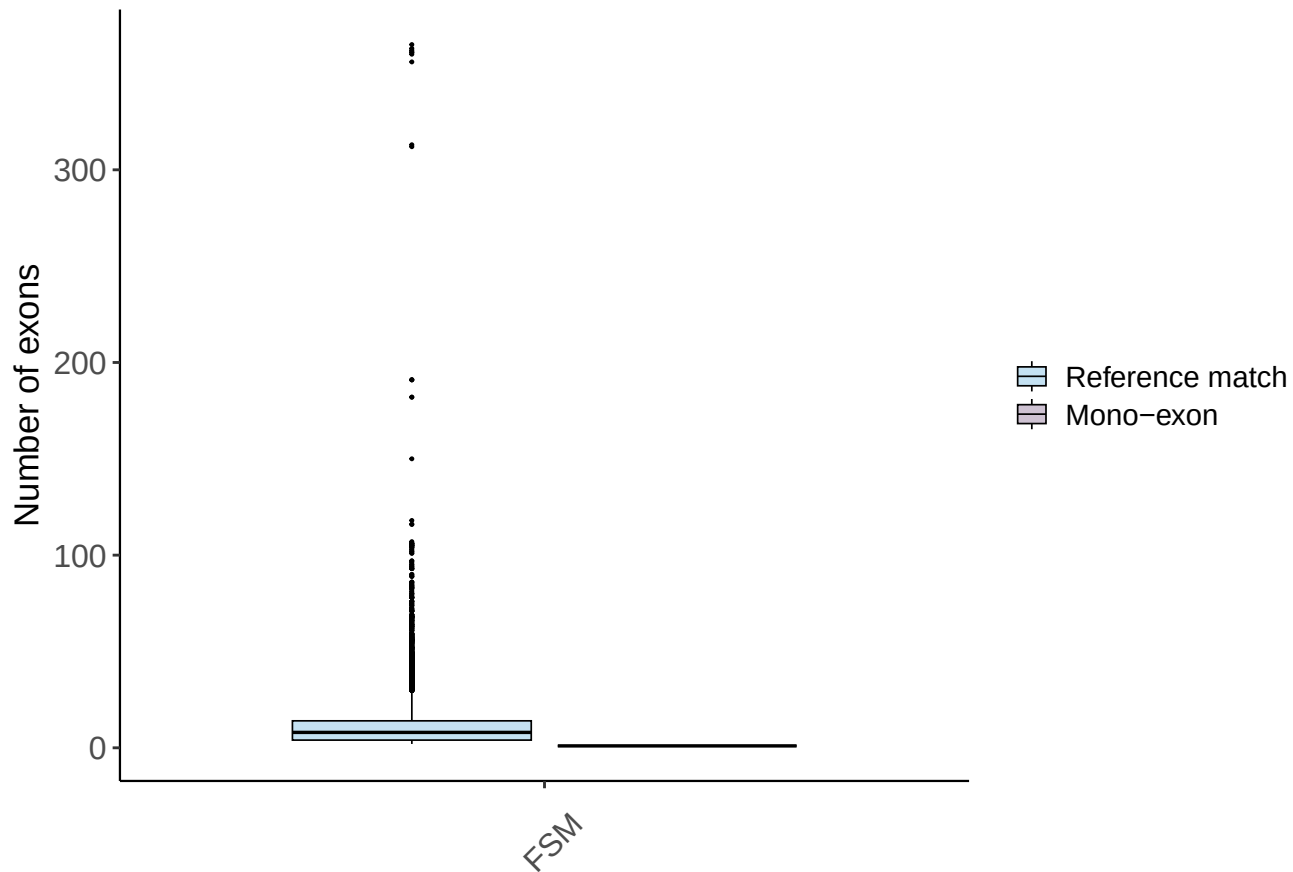
Transcript Lengths by Subcategory



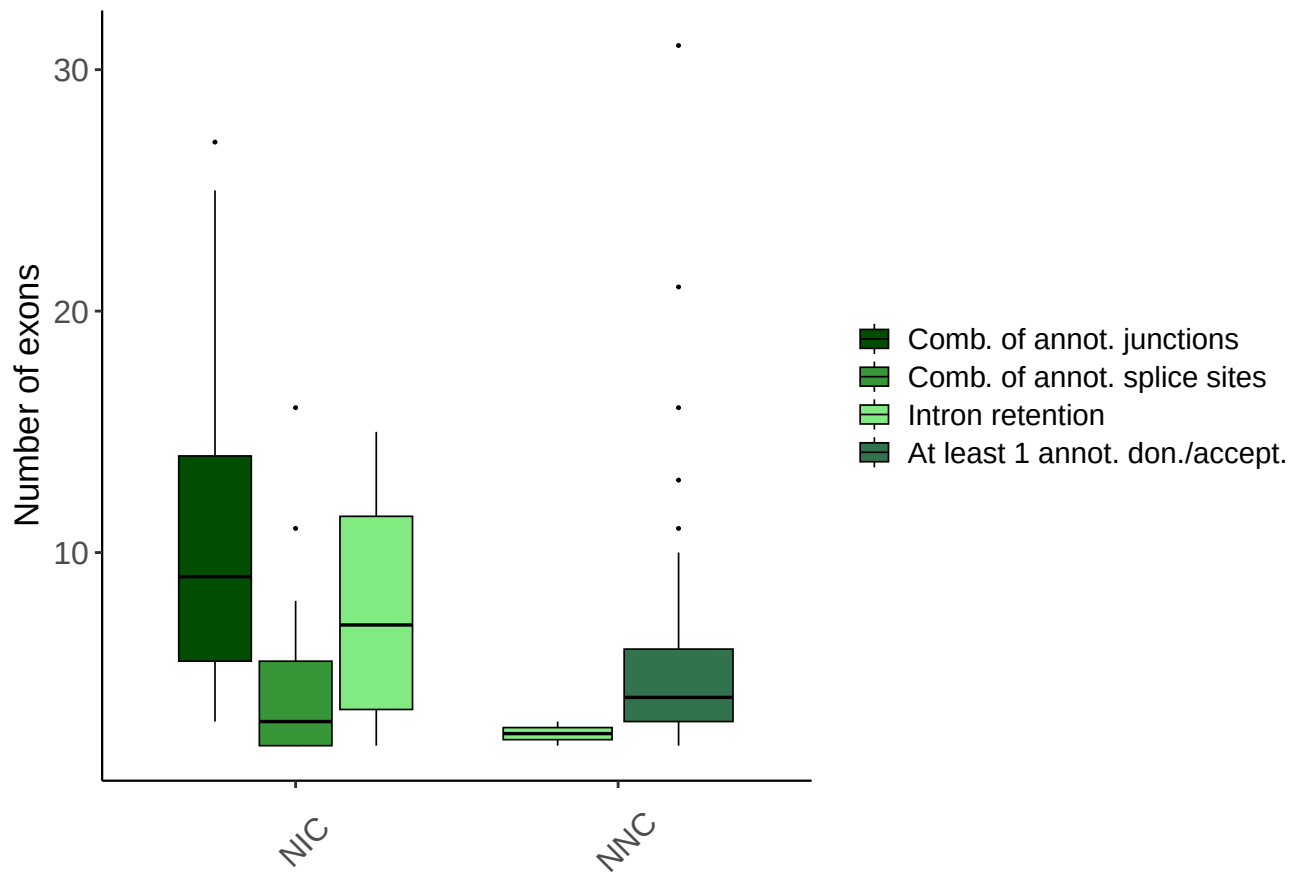
Exon Counts by Structural Classification



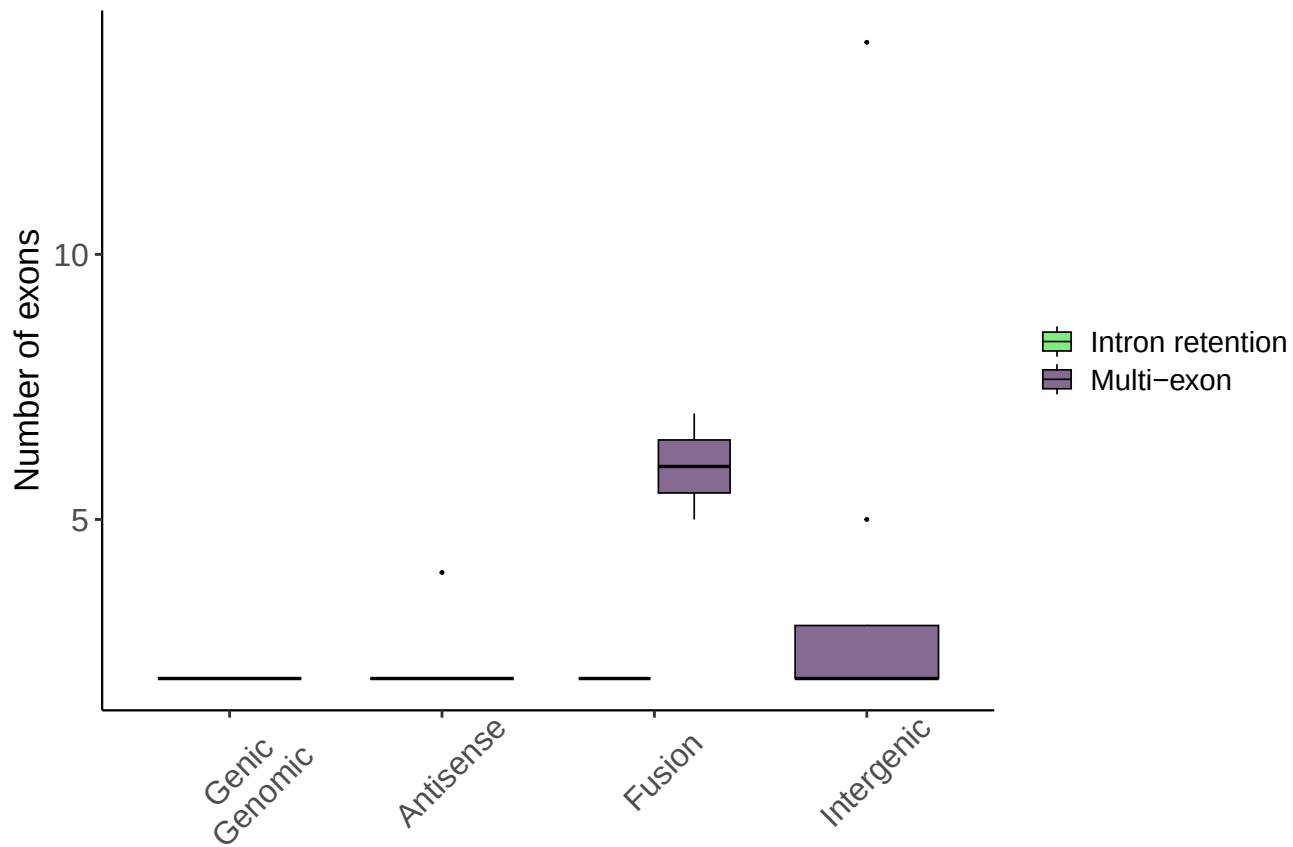
Exon Counts by Subcategory



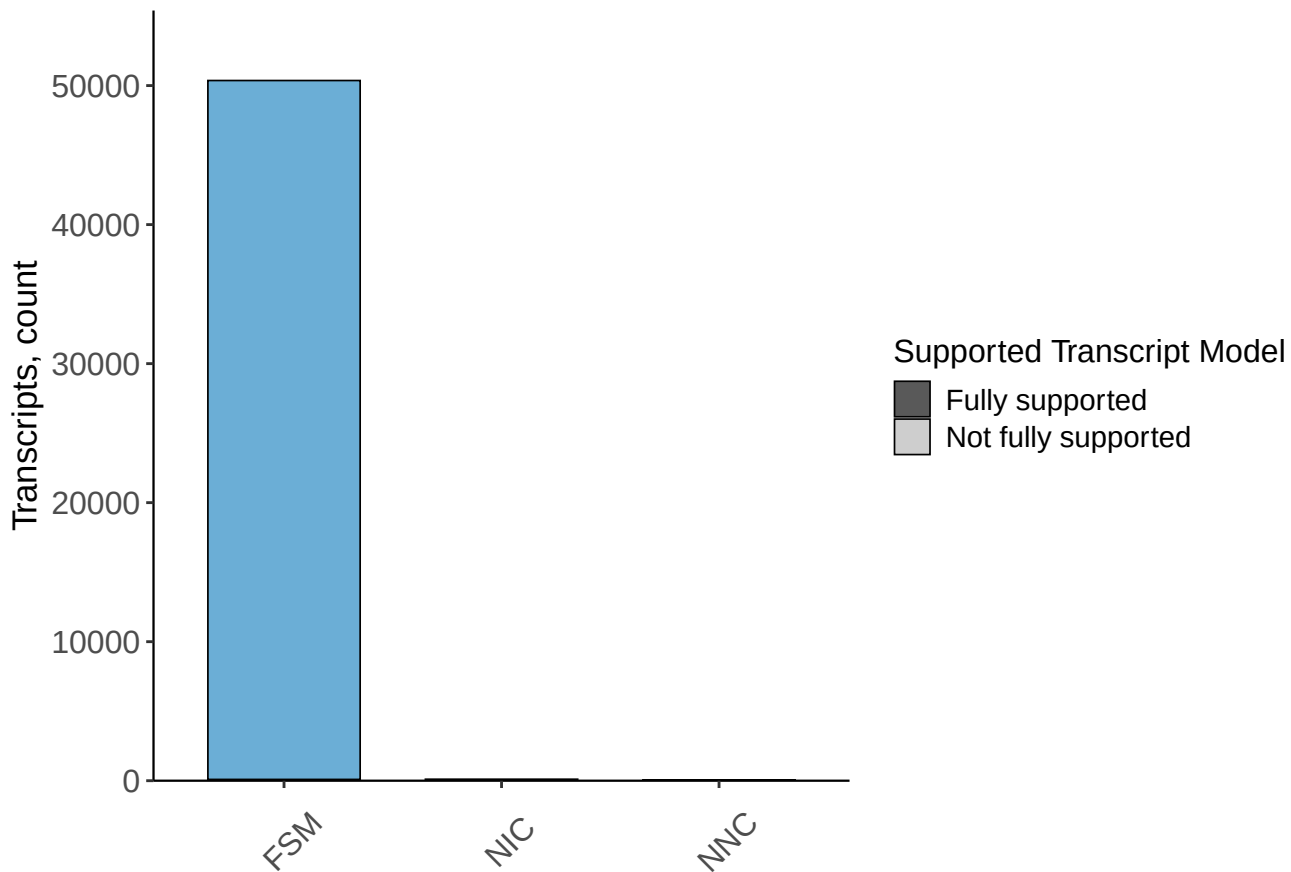
Exon Counts by Subcategory



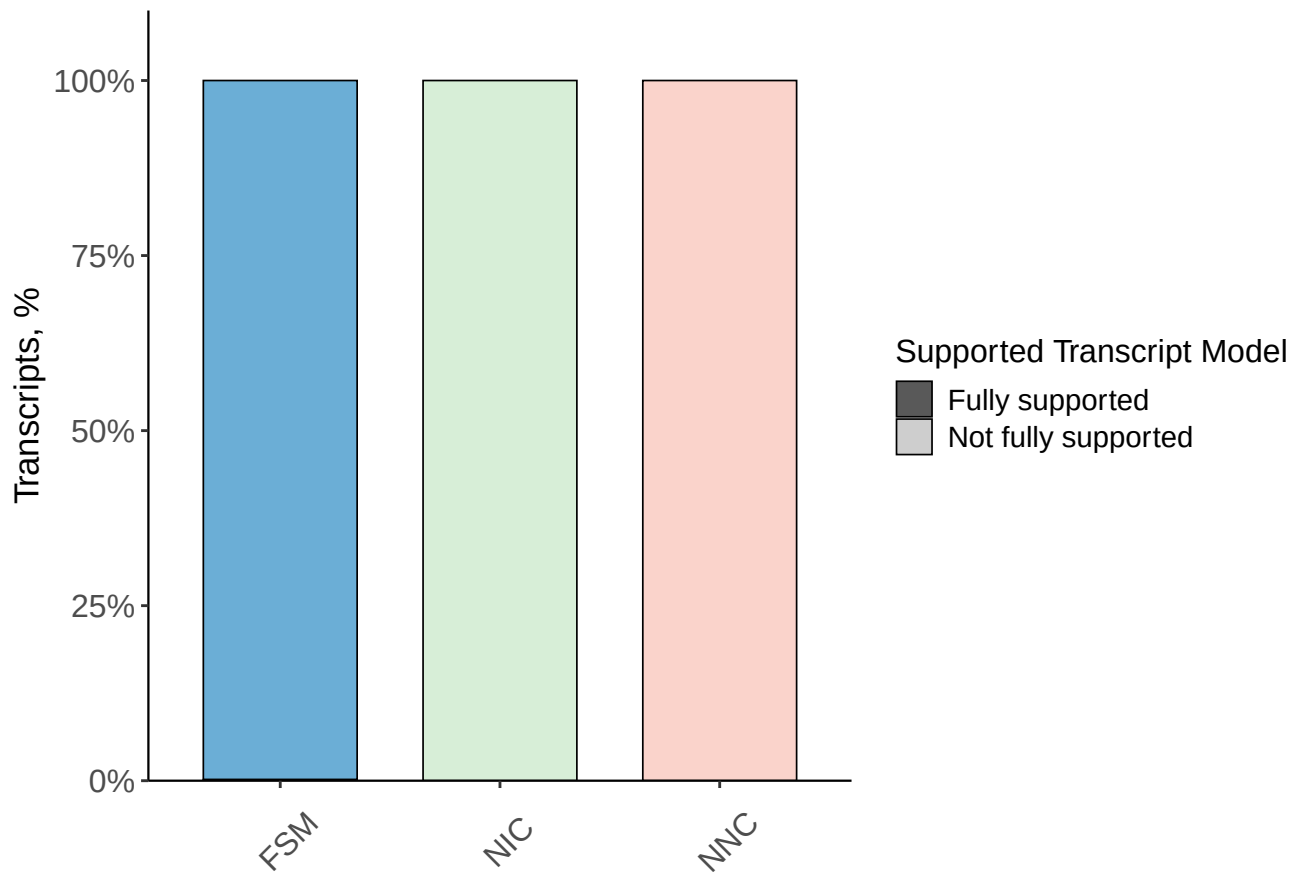
Exon Counts by Subcategory



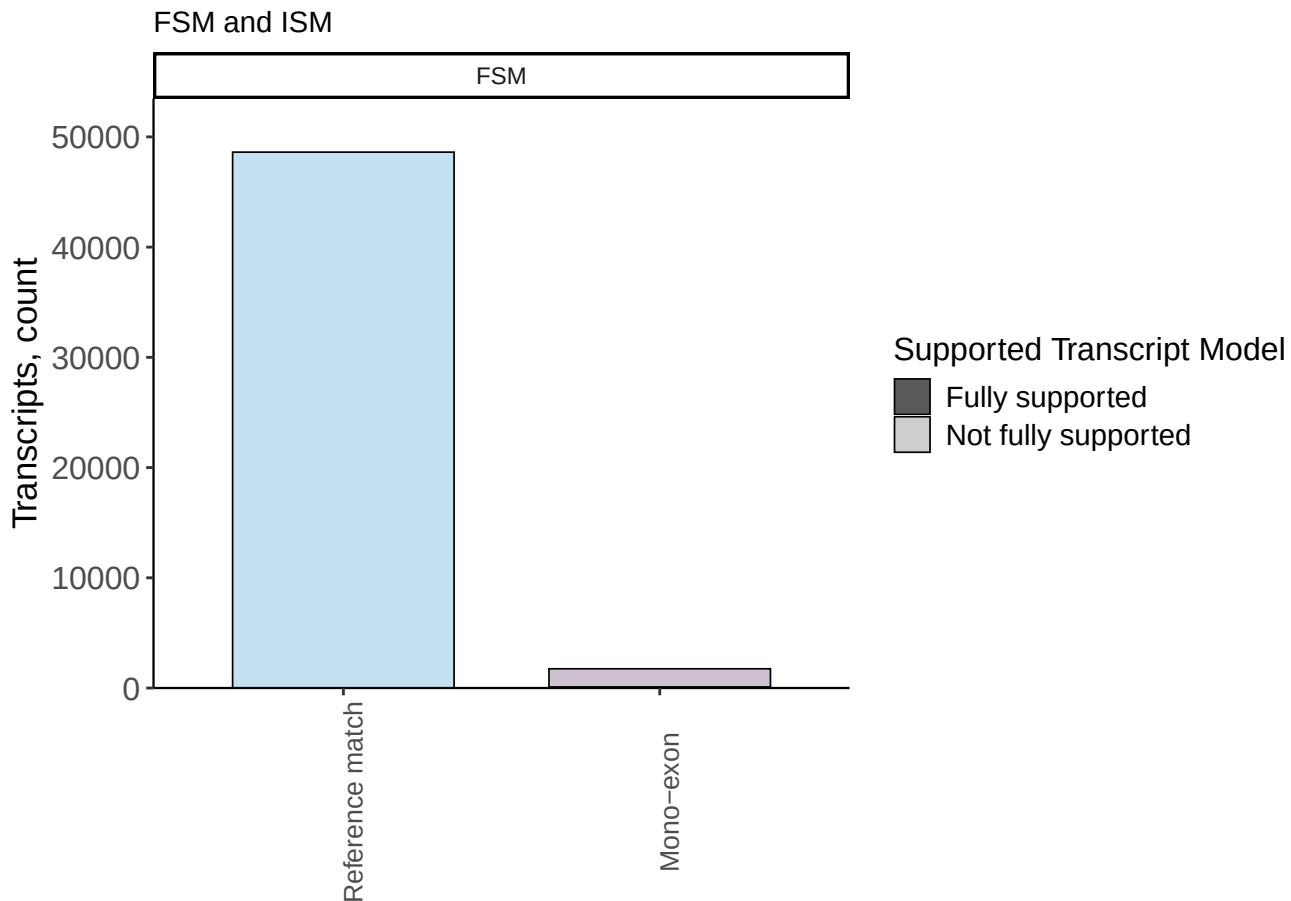
Isoform Distribution Across Structural Categories



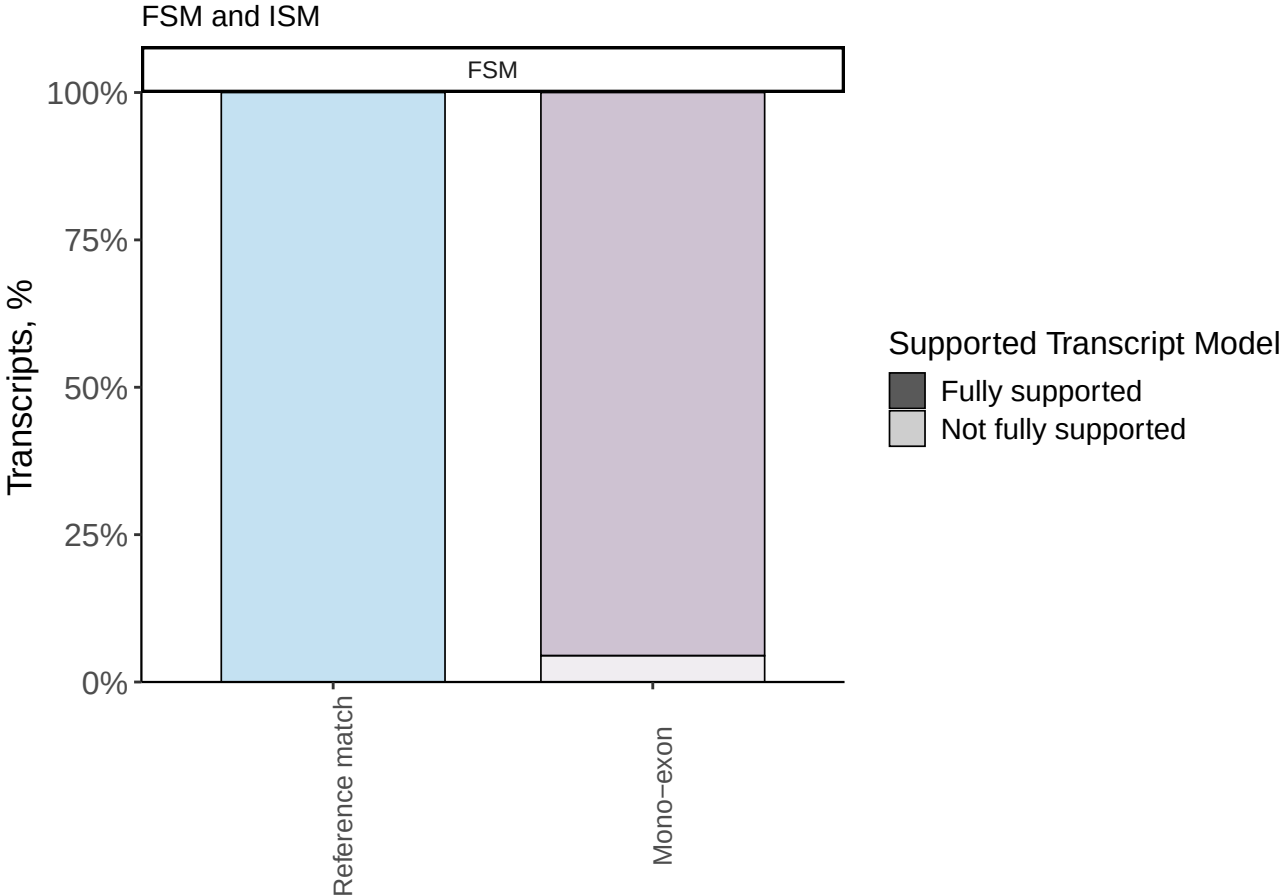
Isoform Distribution Across Structural Categories



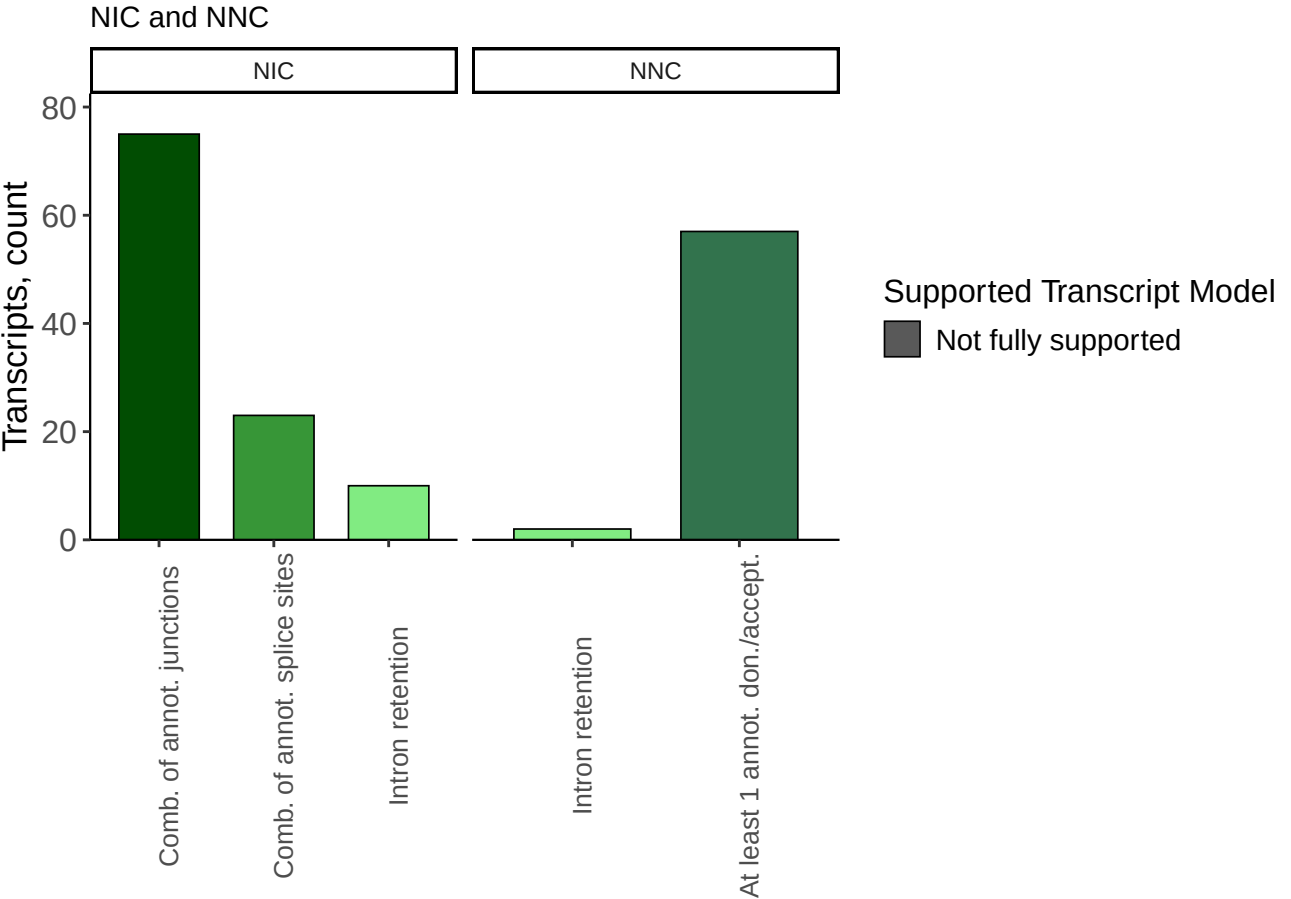
Isoform Distribution Across Structural Subcategories



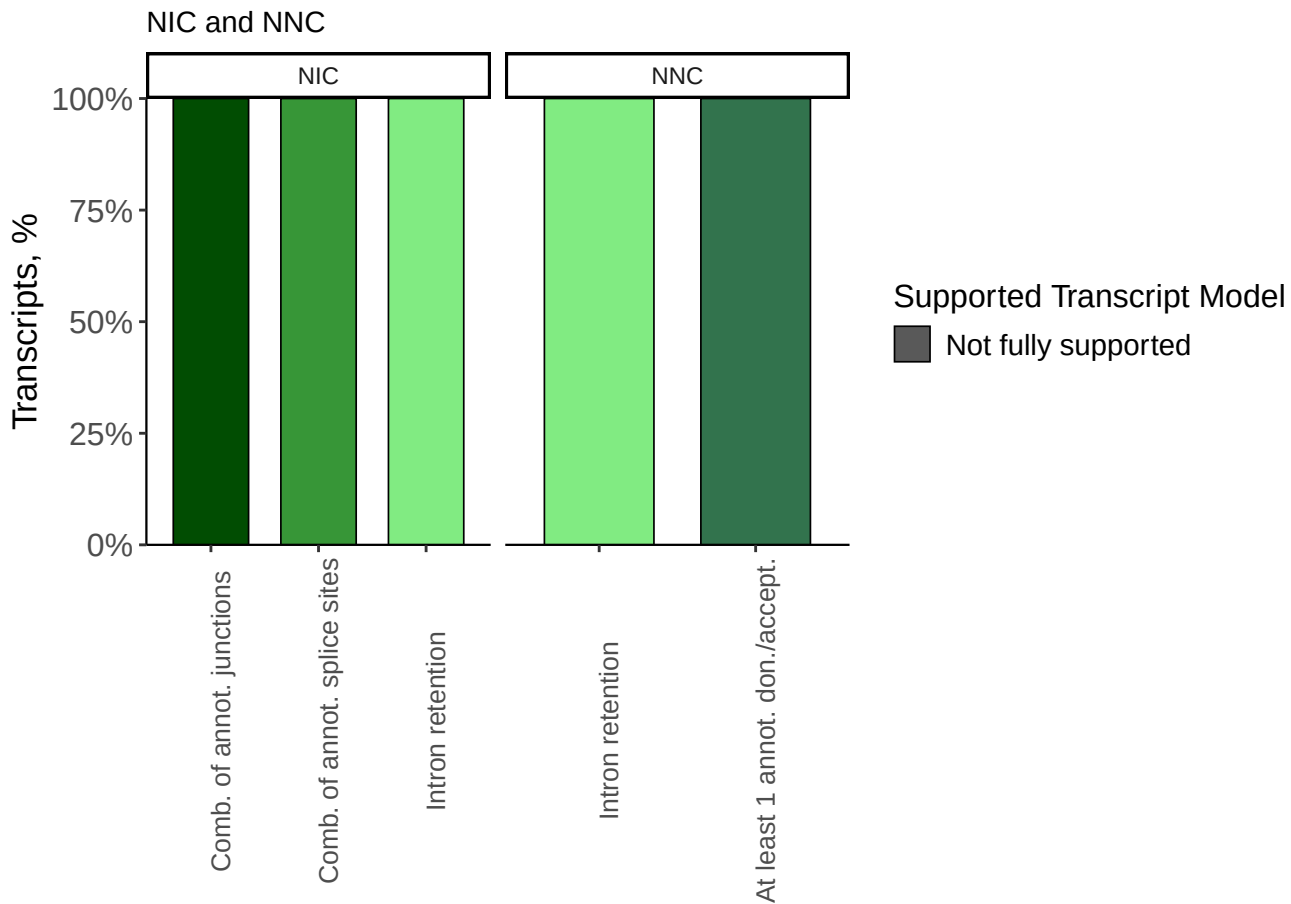
Isoform Distribution Across Structural Subcategories



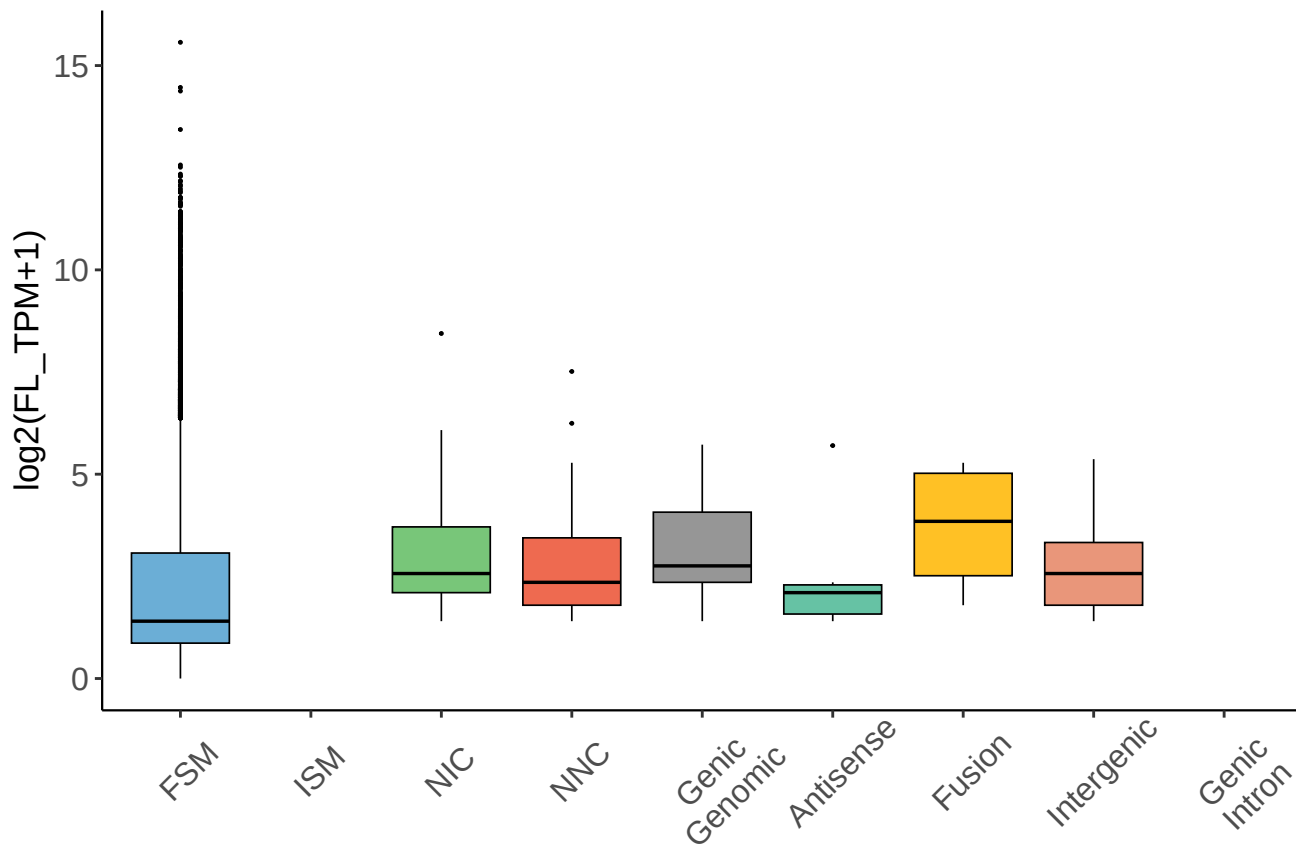
Isoform Distribution Across Structural Subcategories



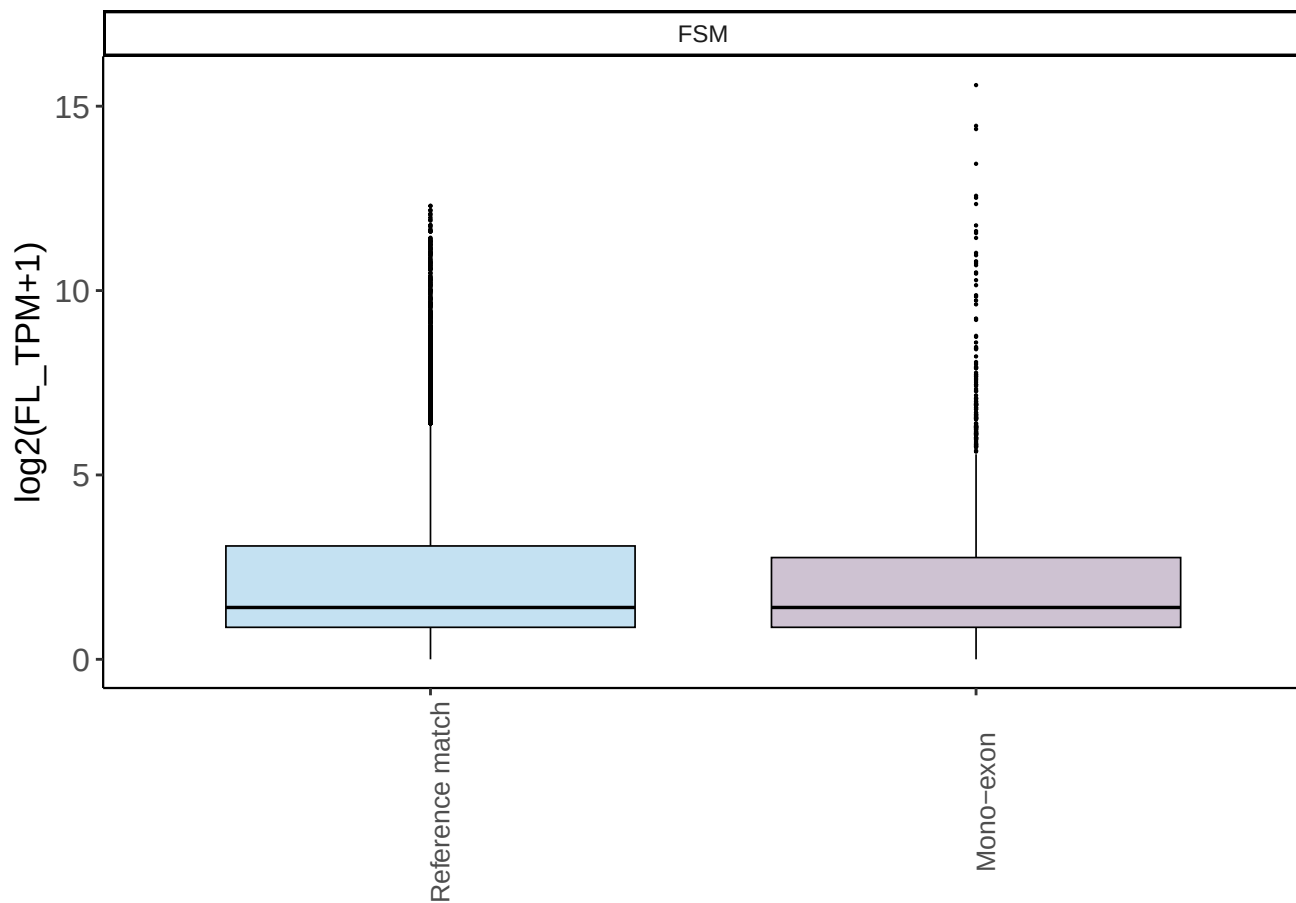
Isoform Distribution Across Structural Subcategories



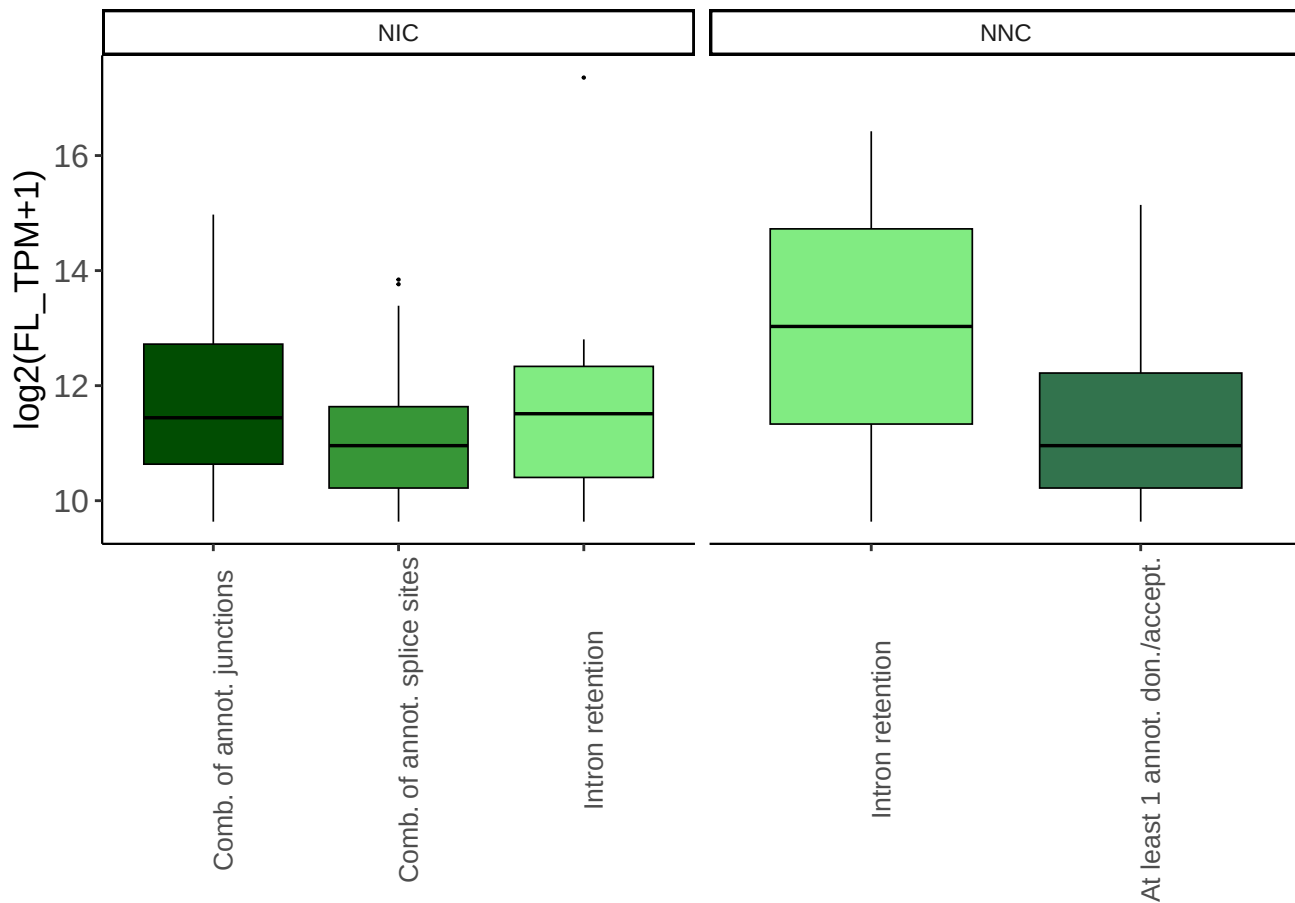
Long Reads Count by Structural Category



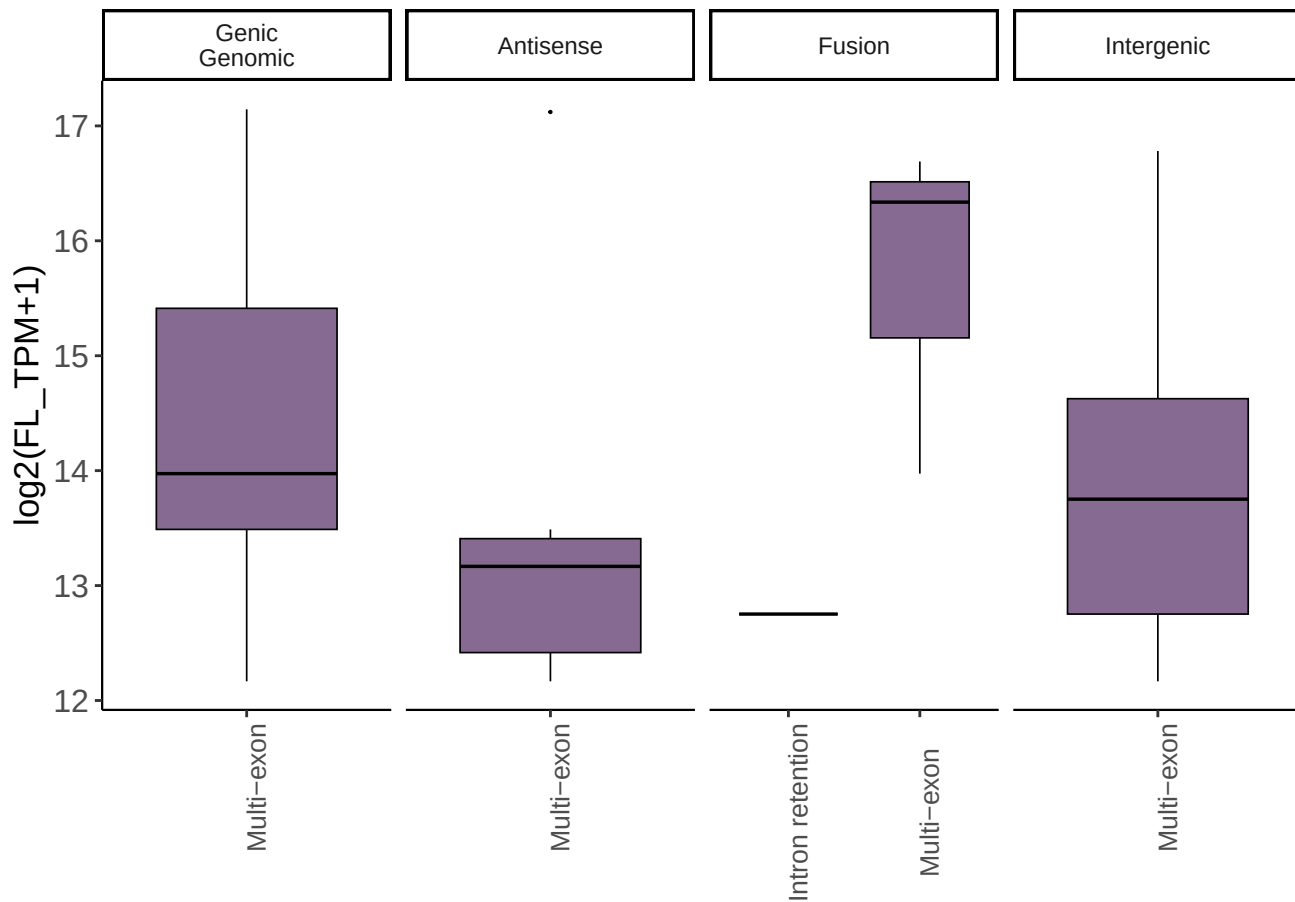
Long Reads Count by Subcategory



Long Reads Count by Subcategory

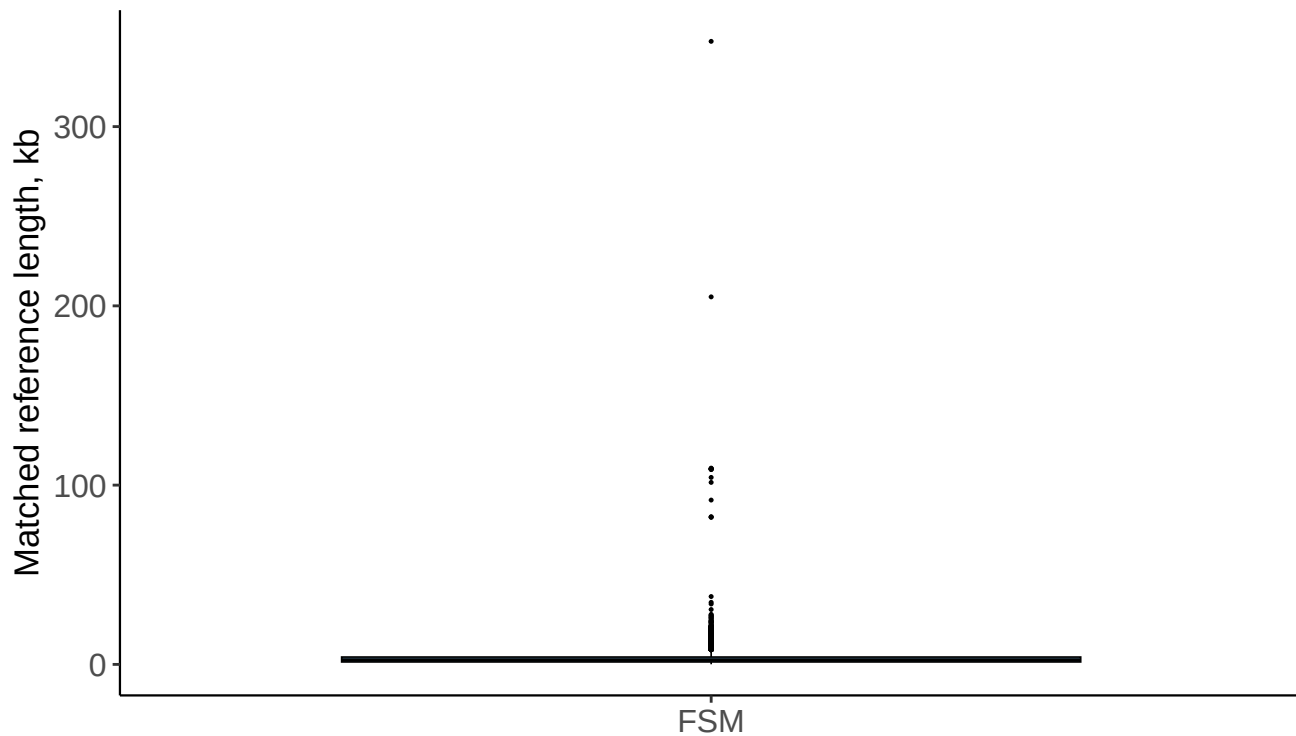


Long Reads Count by Subcategory



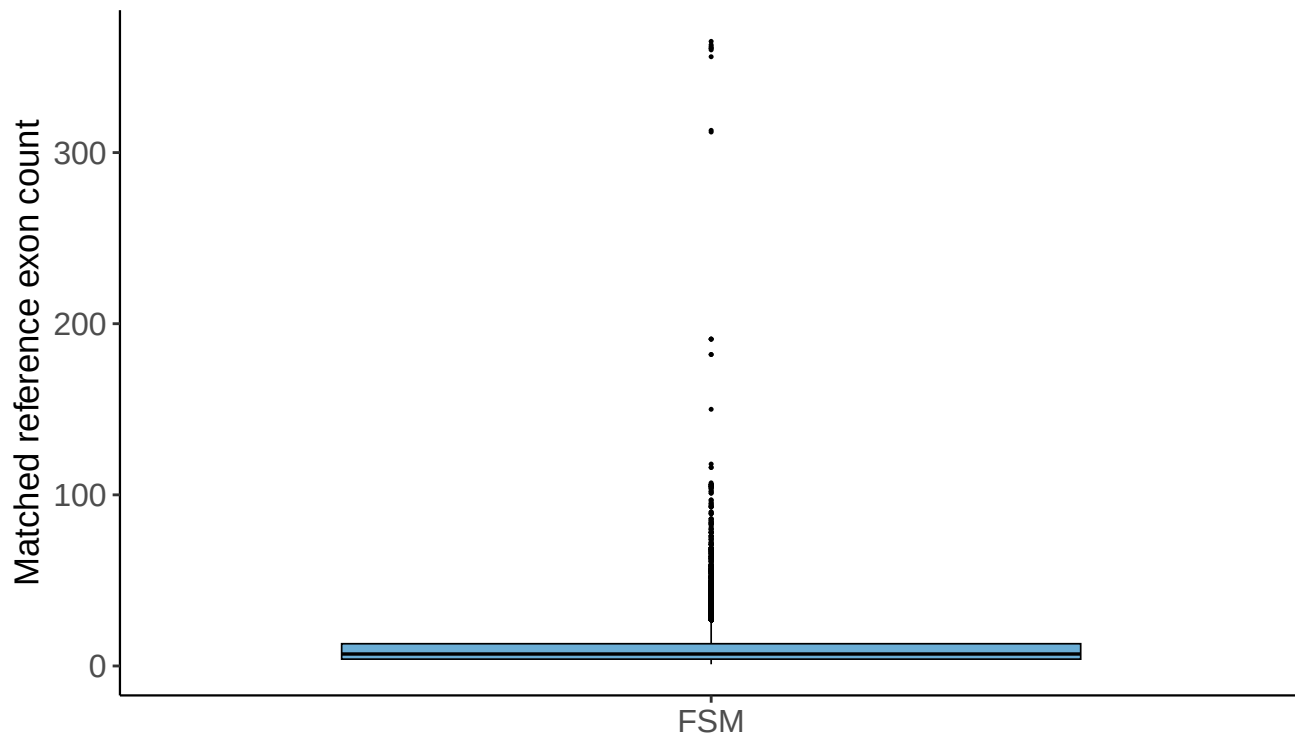
Length Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories



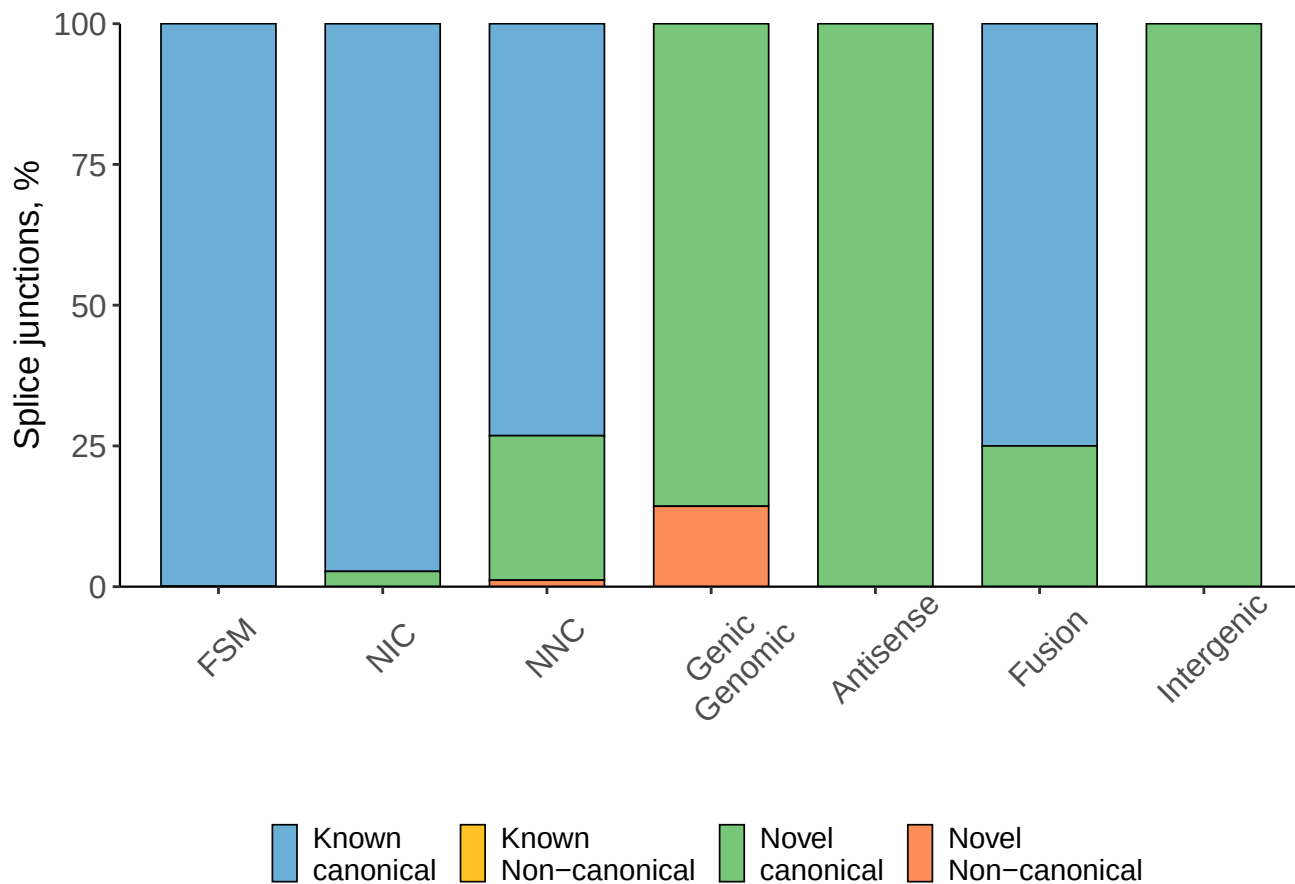
Exon Count Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories

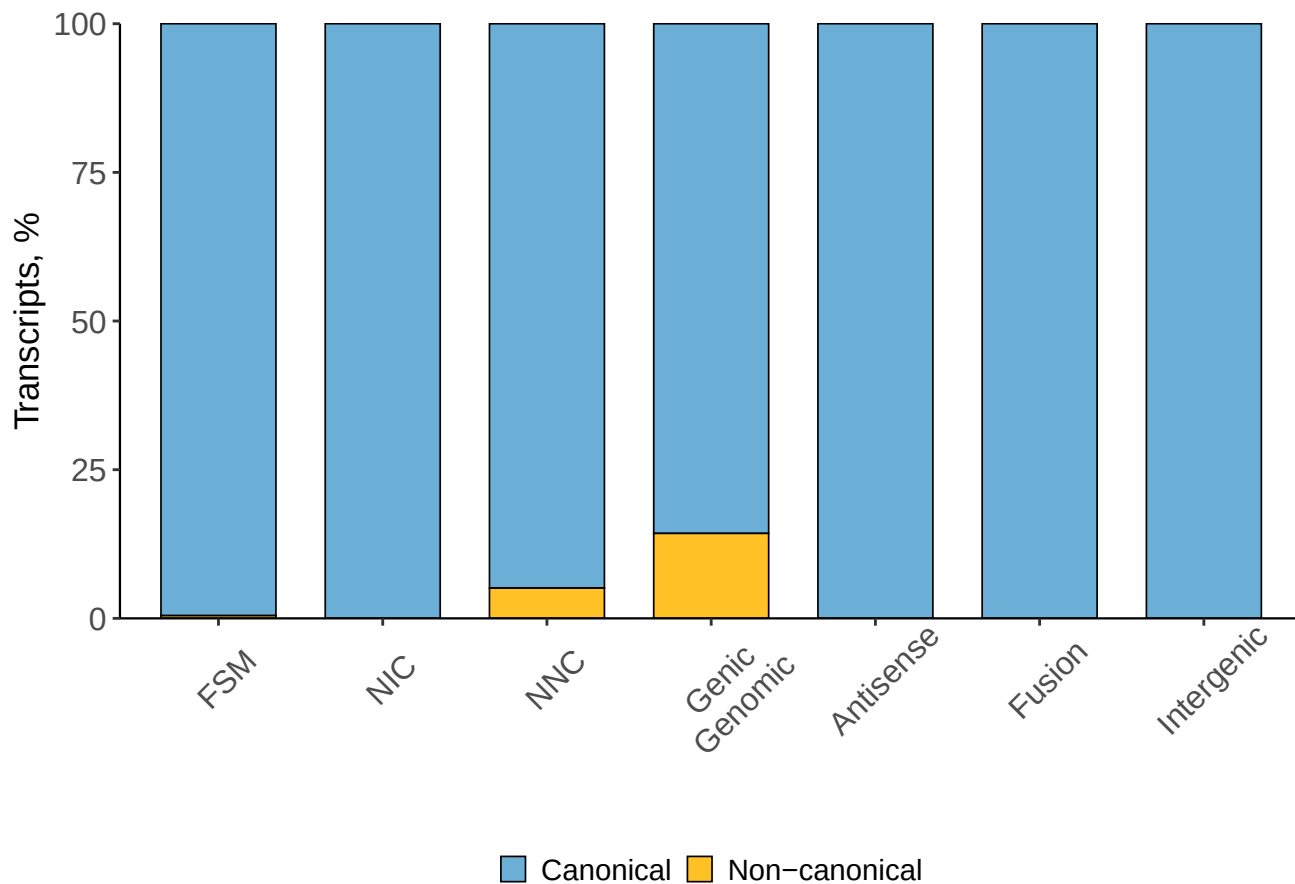


Splice Junction Characterization

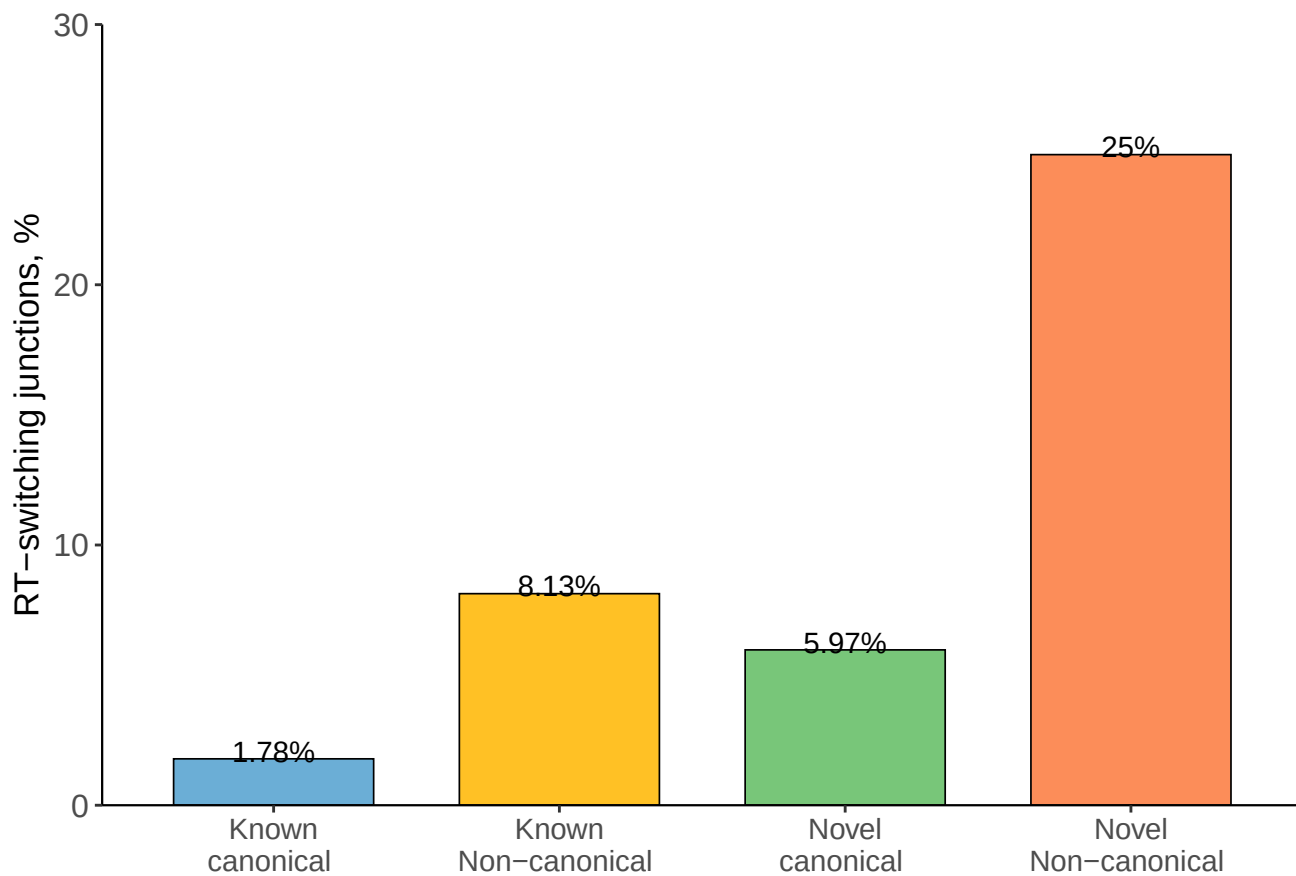
Distribution of Splice Junctions by Structural Classification



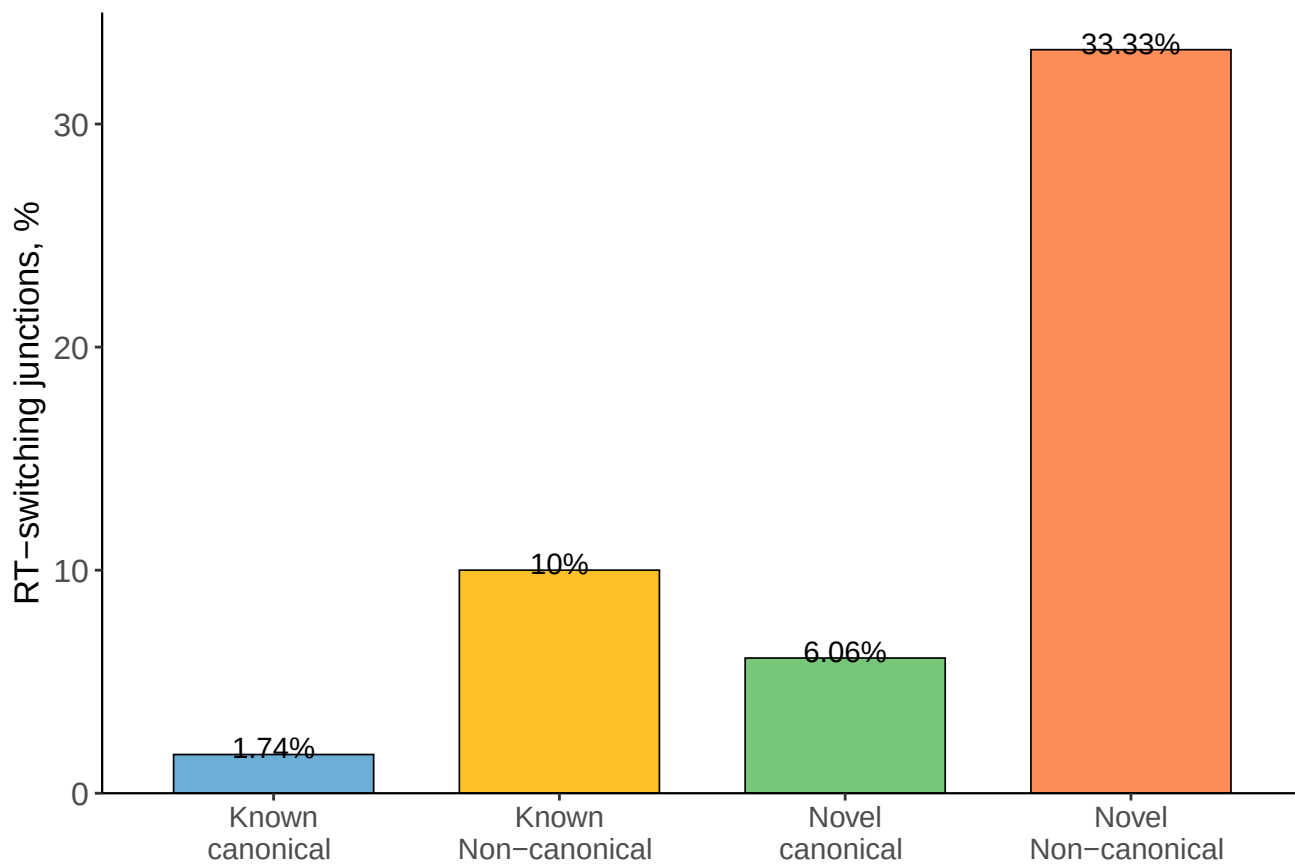
Distribution of Transcripts by Splice Junctions



RT-Switching All Junctions



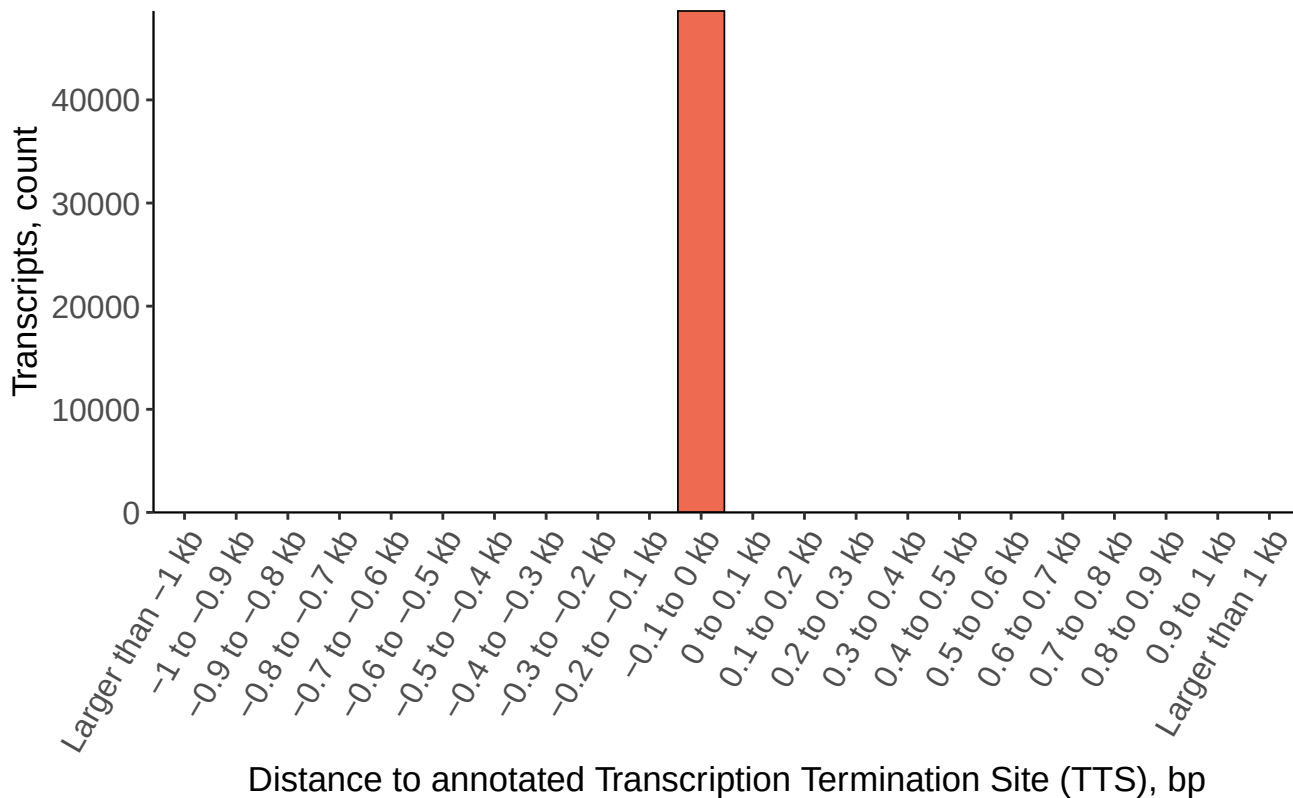
Unique Junctions RT-switching



Comparison With Annotated TSS and TTS

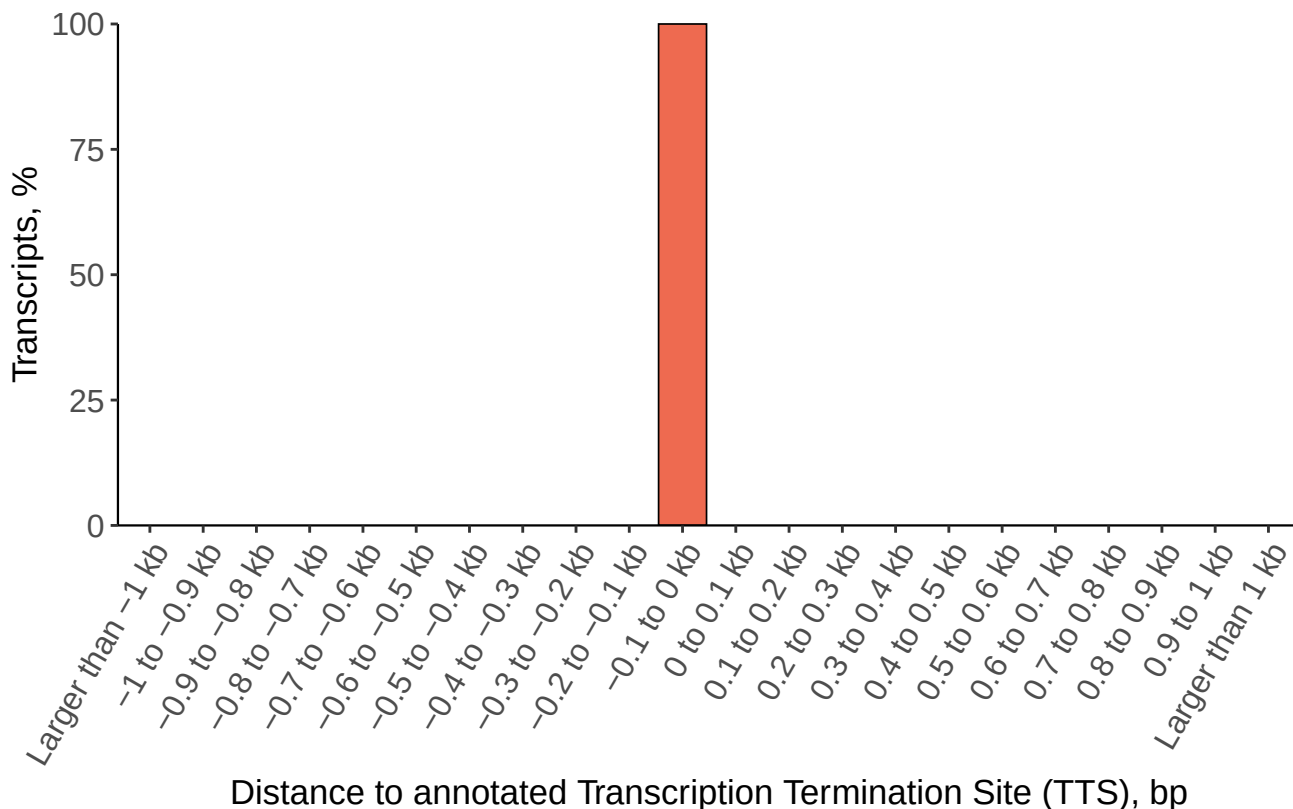
Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



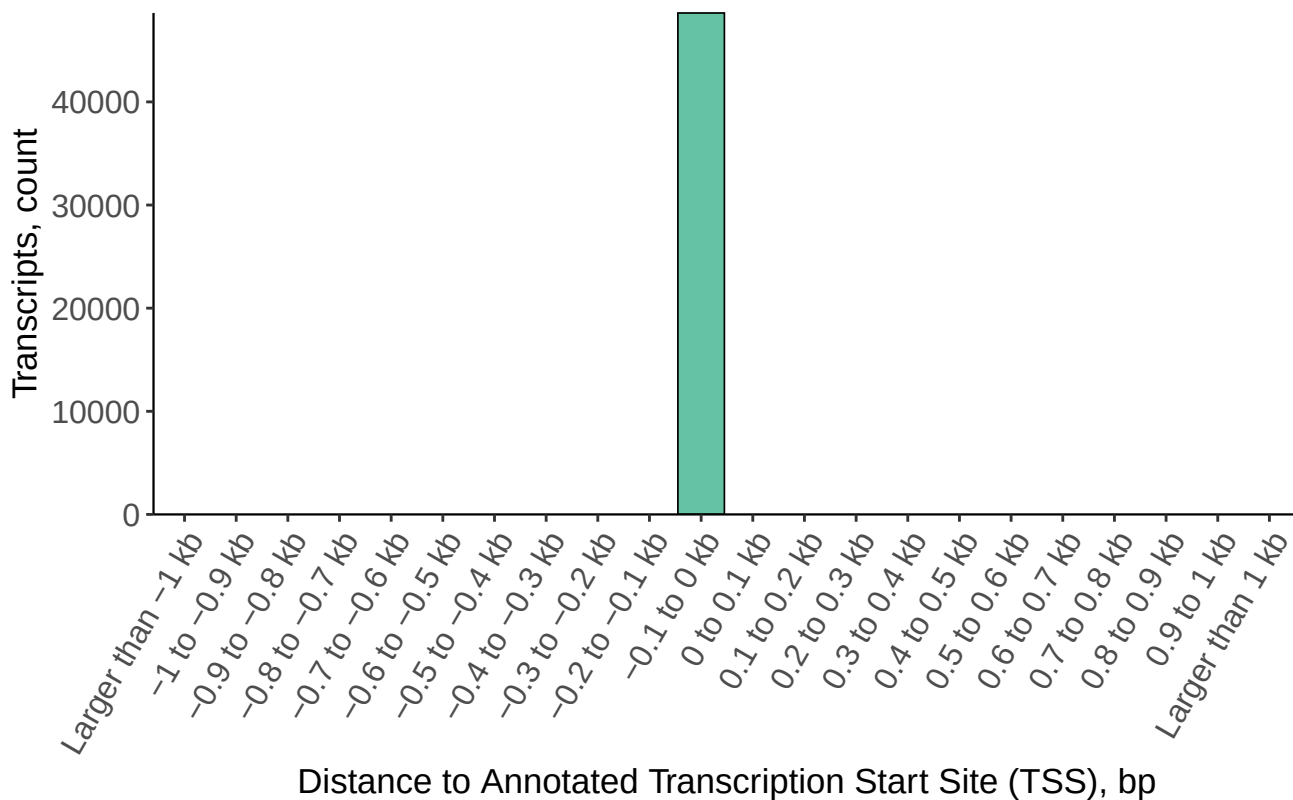
Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



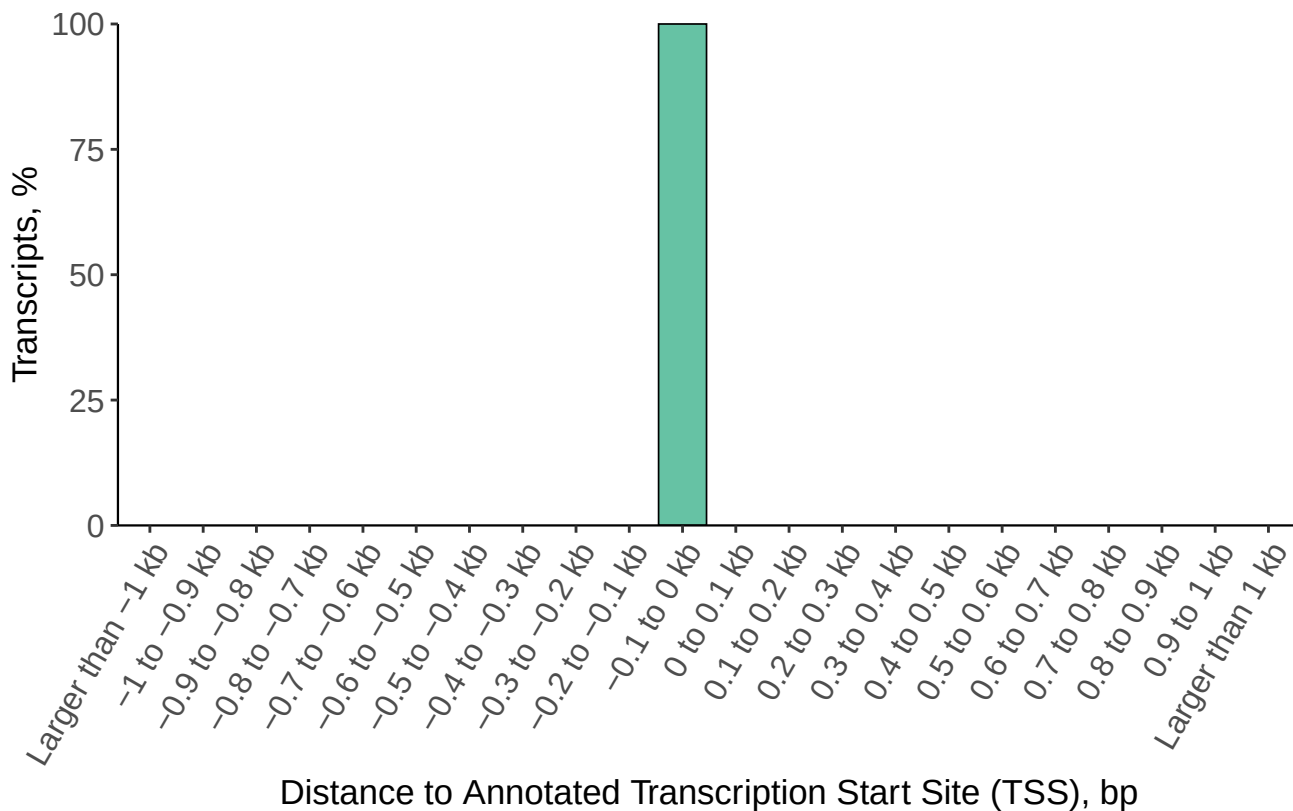
Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS



Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS

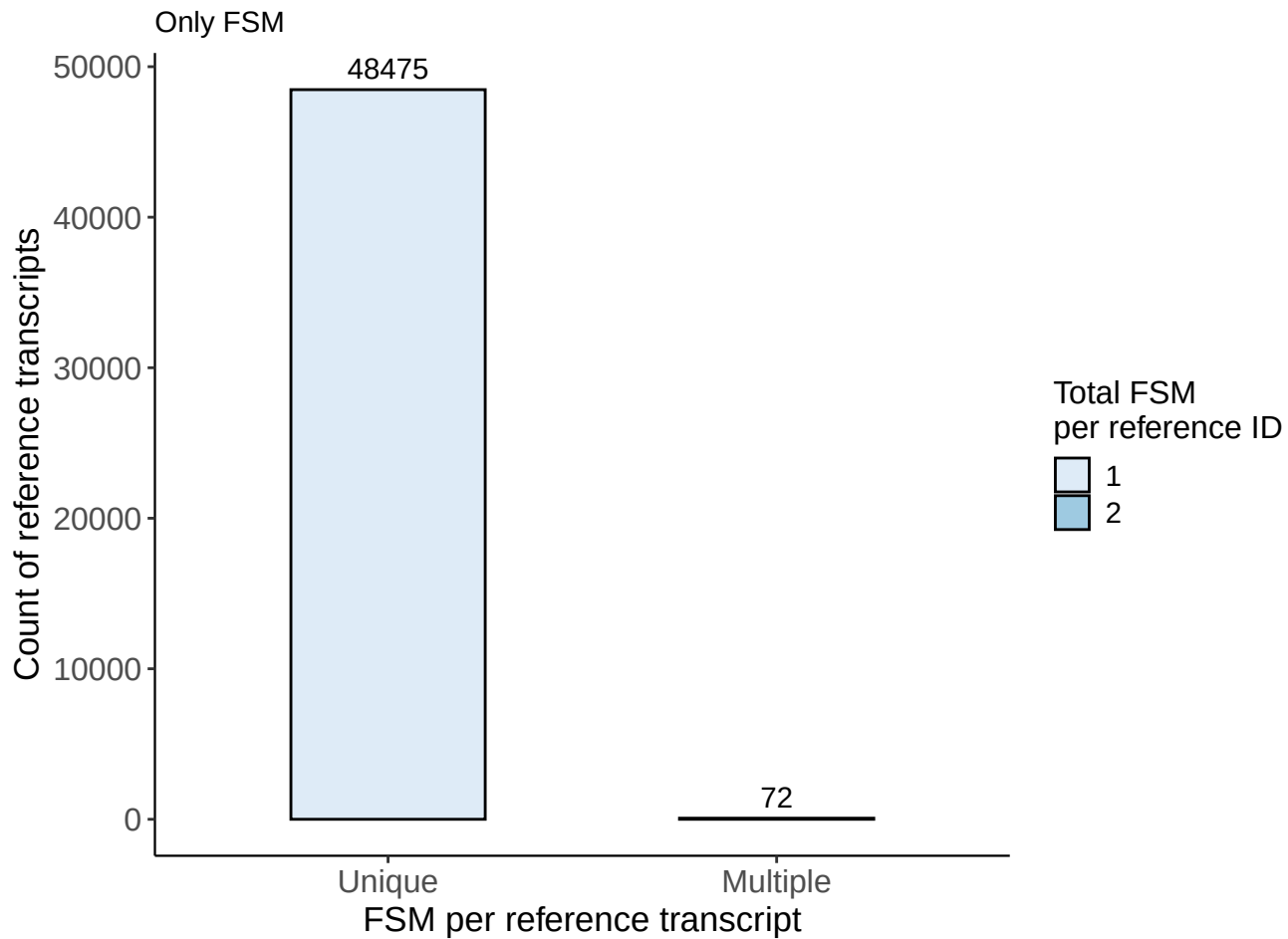


*Comparison With Annotated TSS and TTS
by Subcategories*

PolyA Distance Analysis

Redundancy Analysis

Reference Transcript Redundancy



Reference Transcript Redundancy

Only ISM

Count of Reference Transcripts

10.

8

6

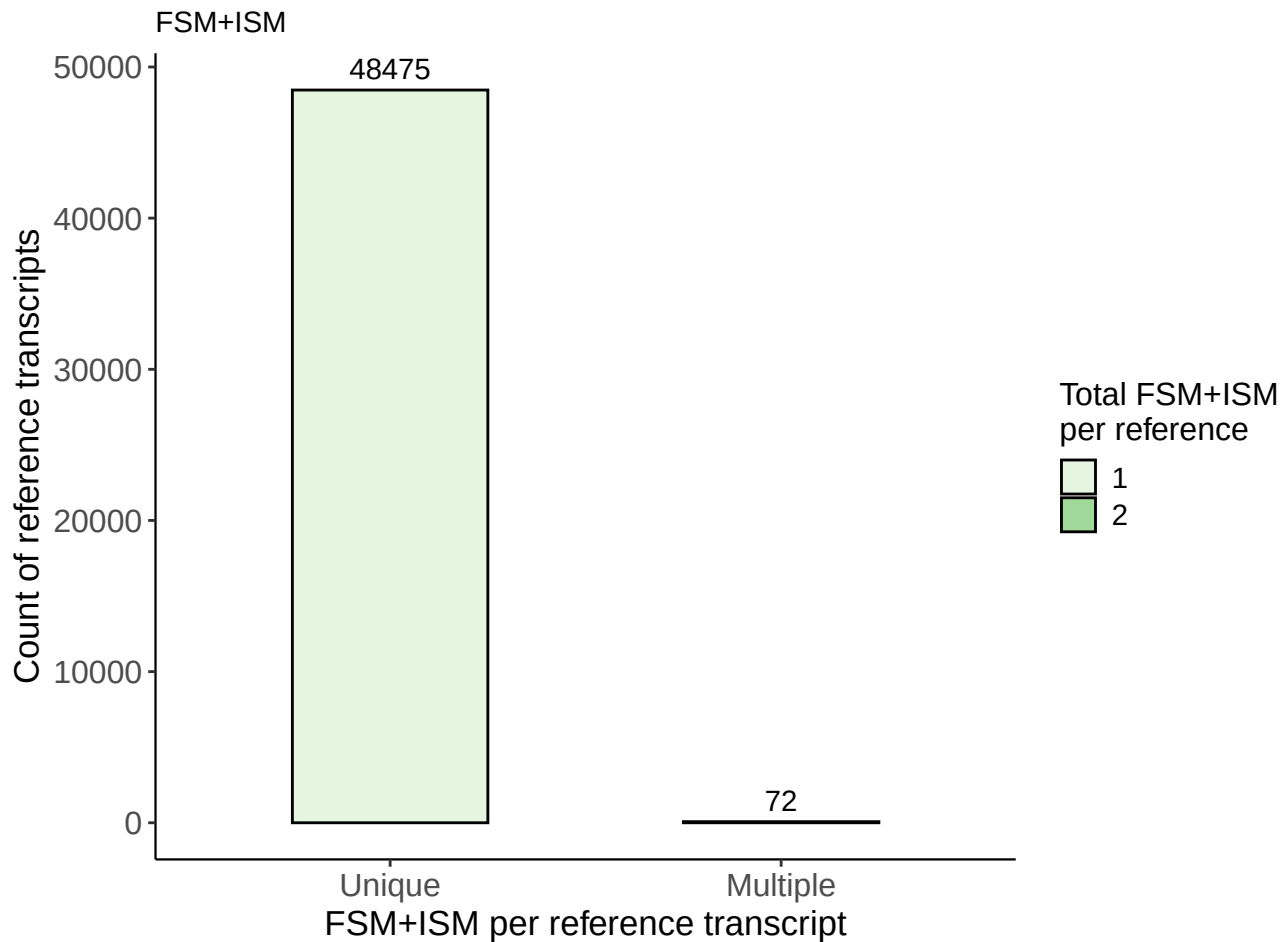
4

2

0

ISM per reference transcript

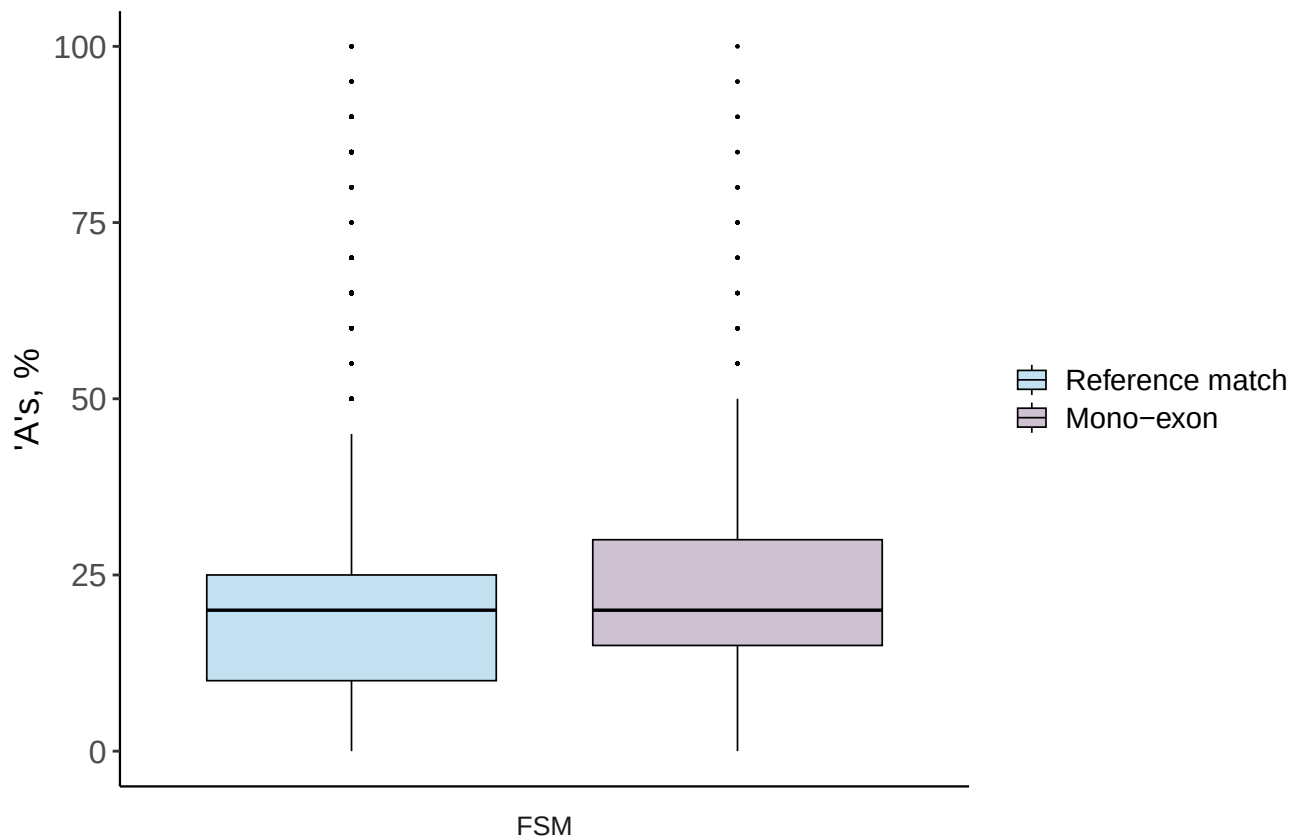
Reference Transcript Redundancy



Intra-Priming Quality Check

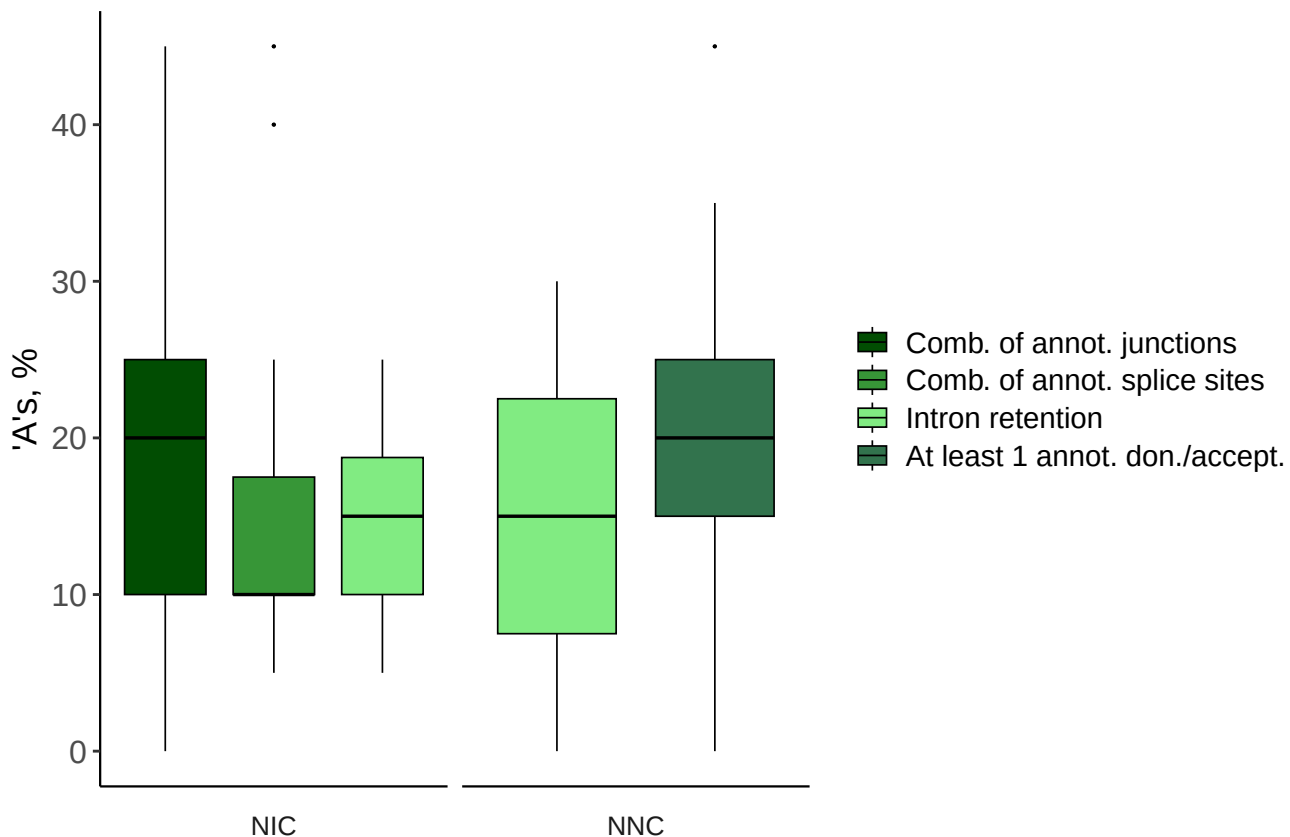
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



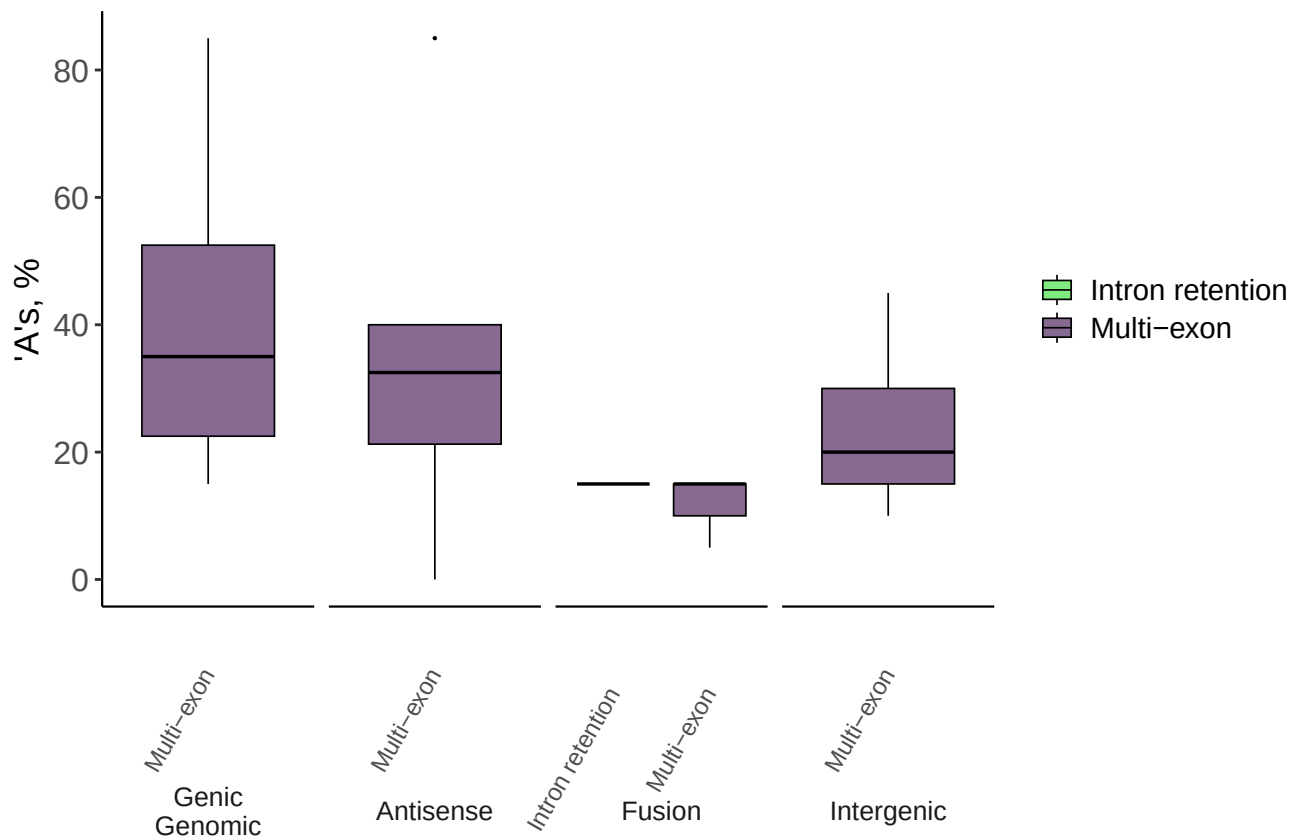
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



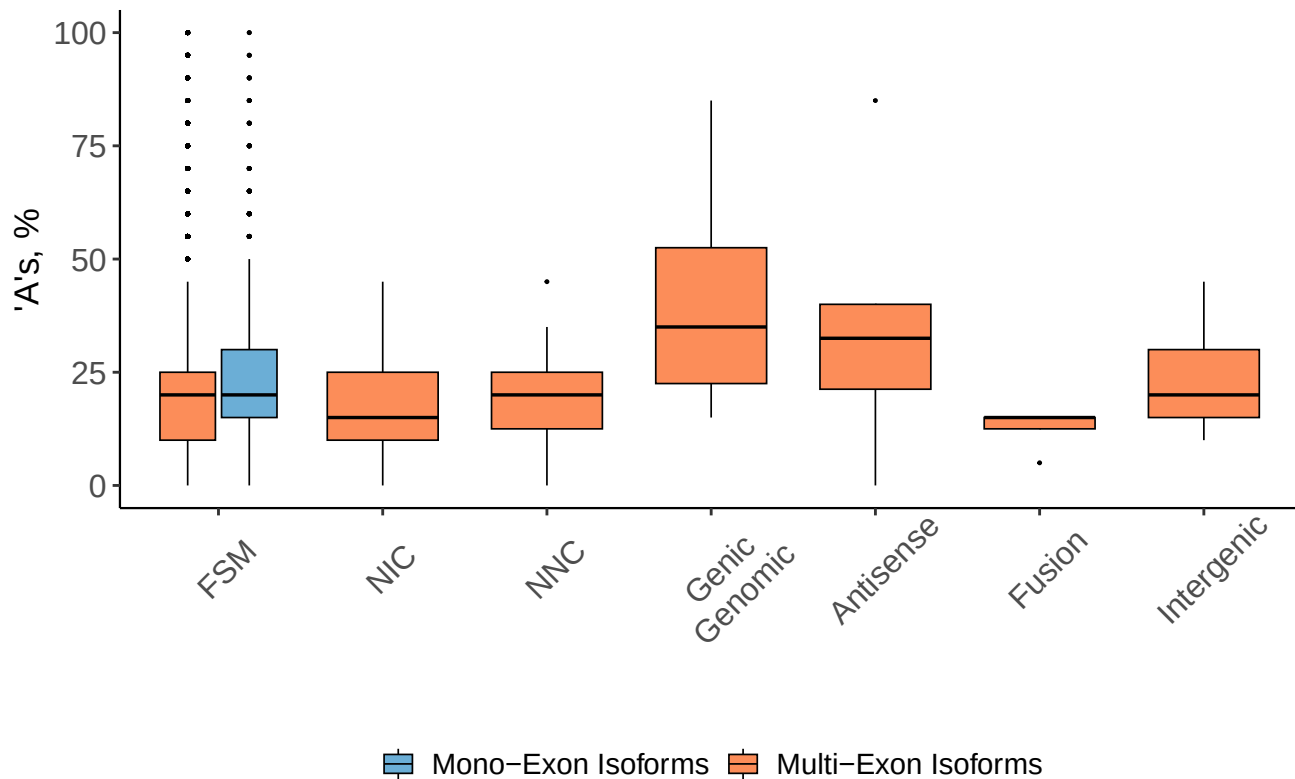
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



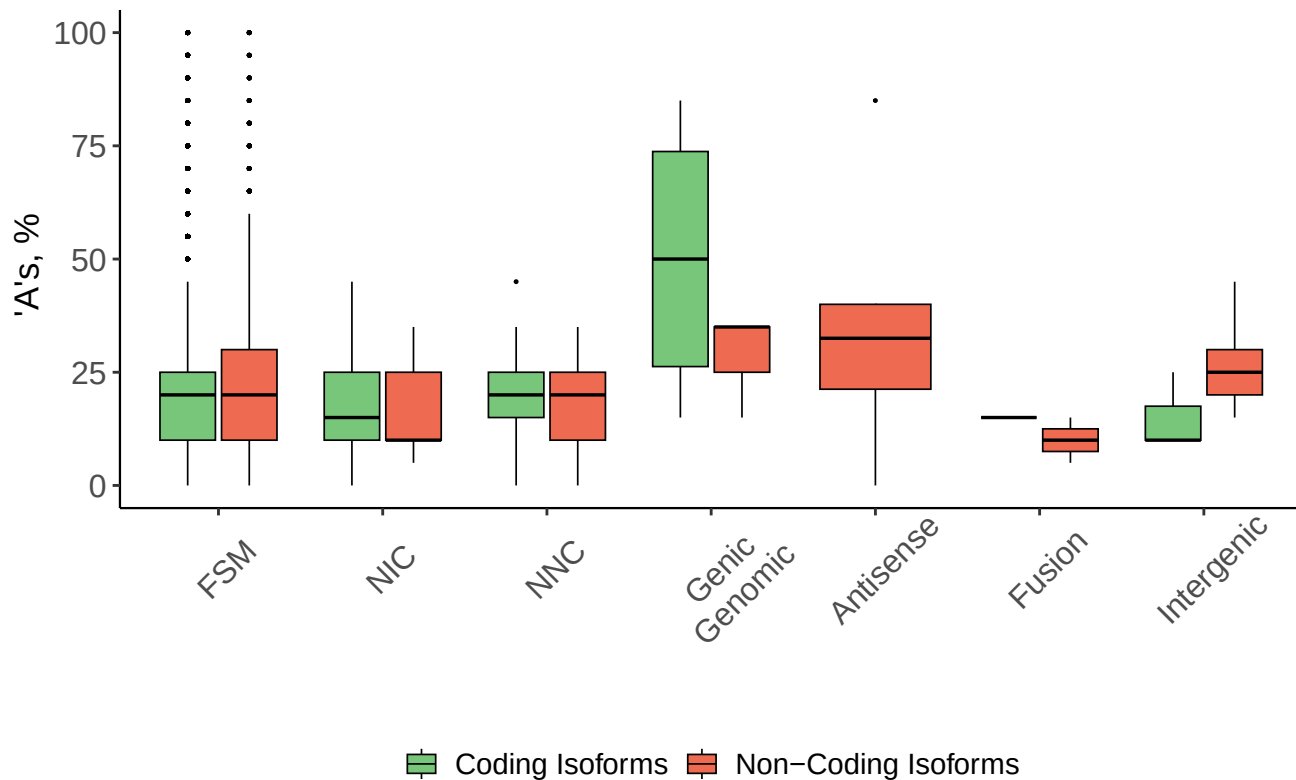
Mono- vs Multi-Exon Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp



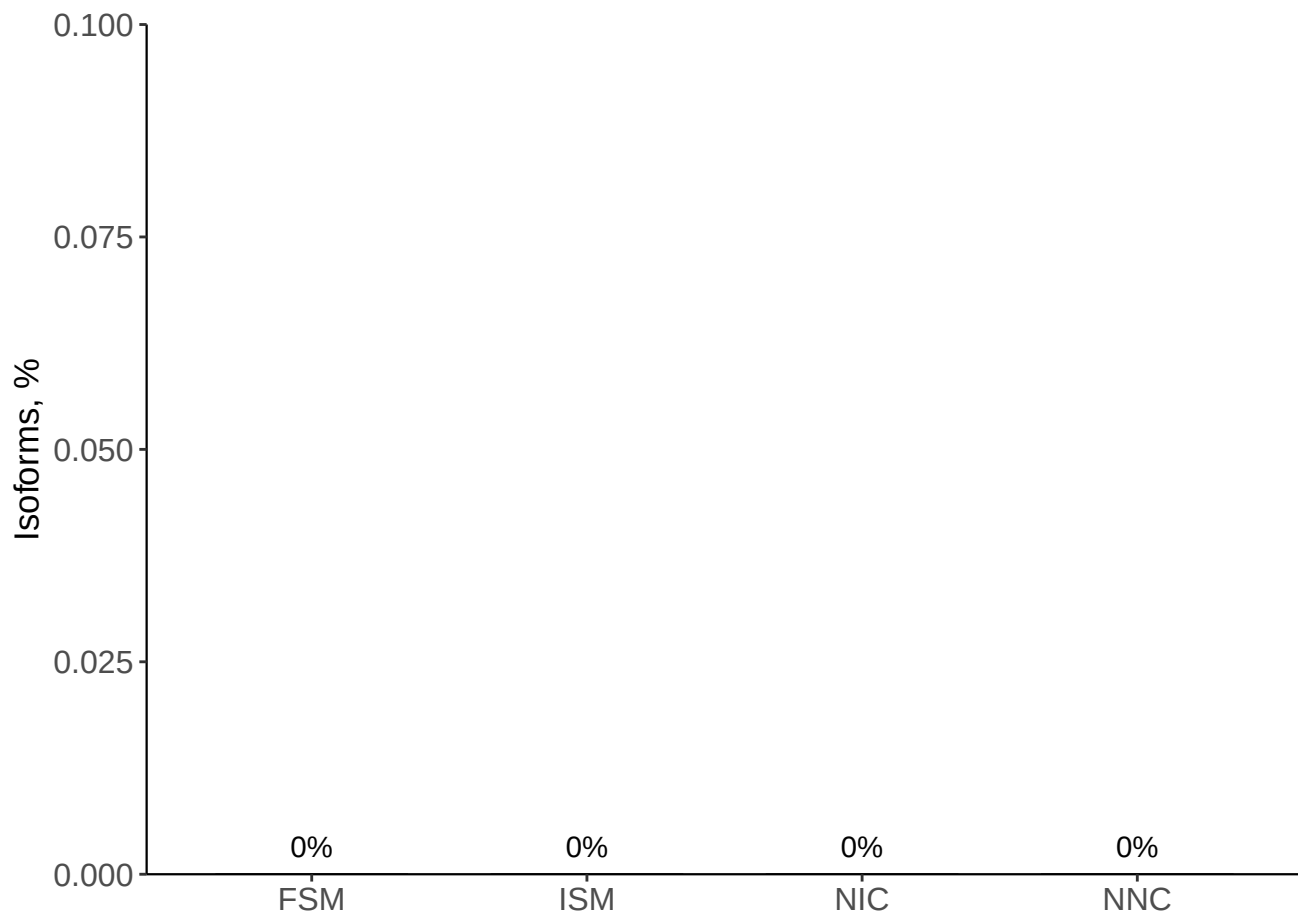
Coding vs Non-Coding Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp

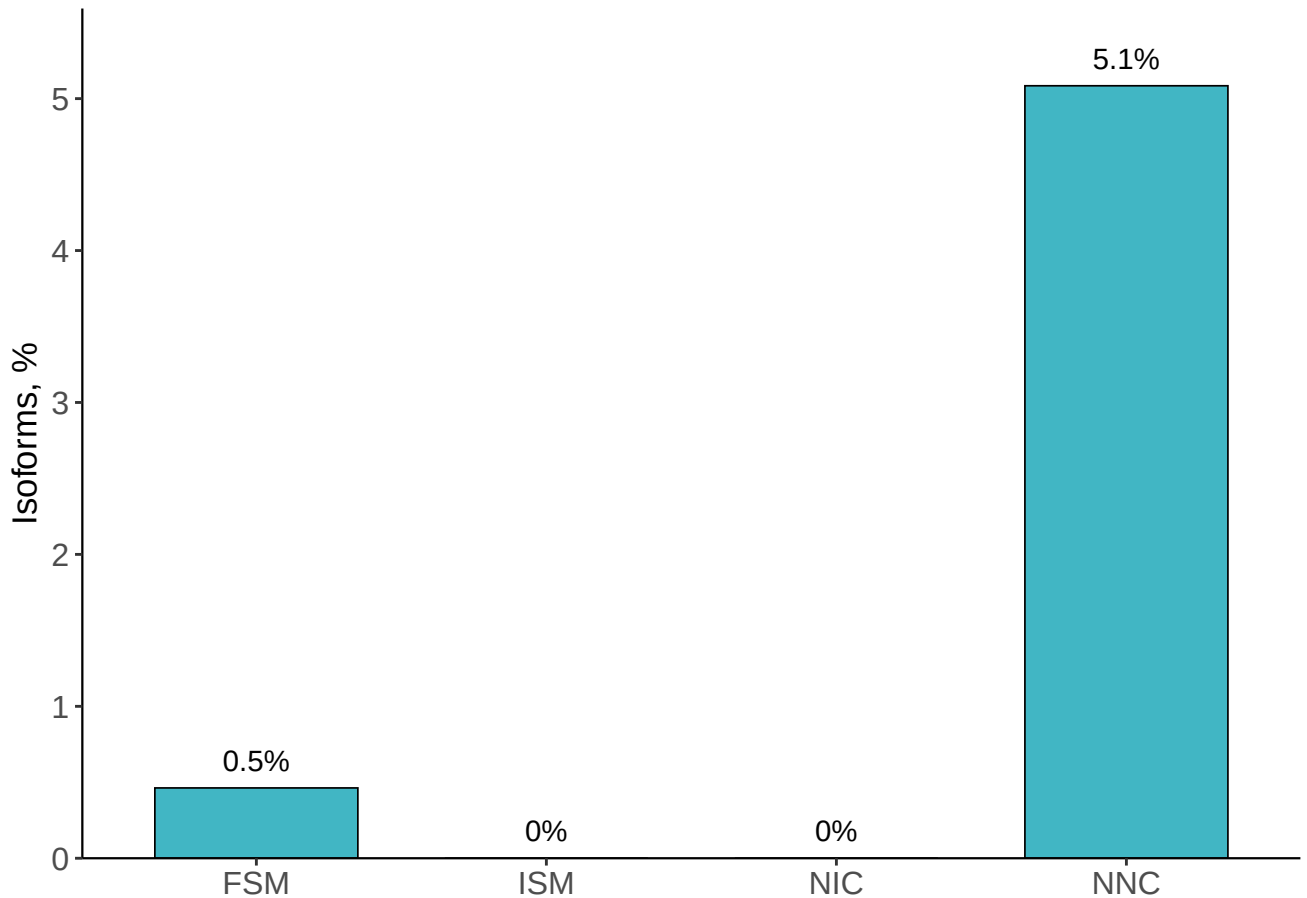


Features of Bad Quality

RT-switching

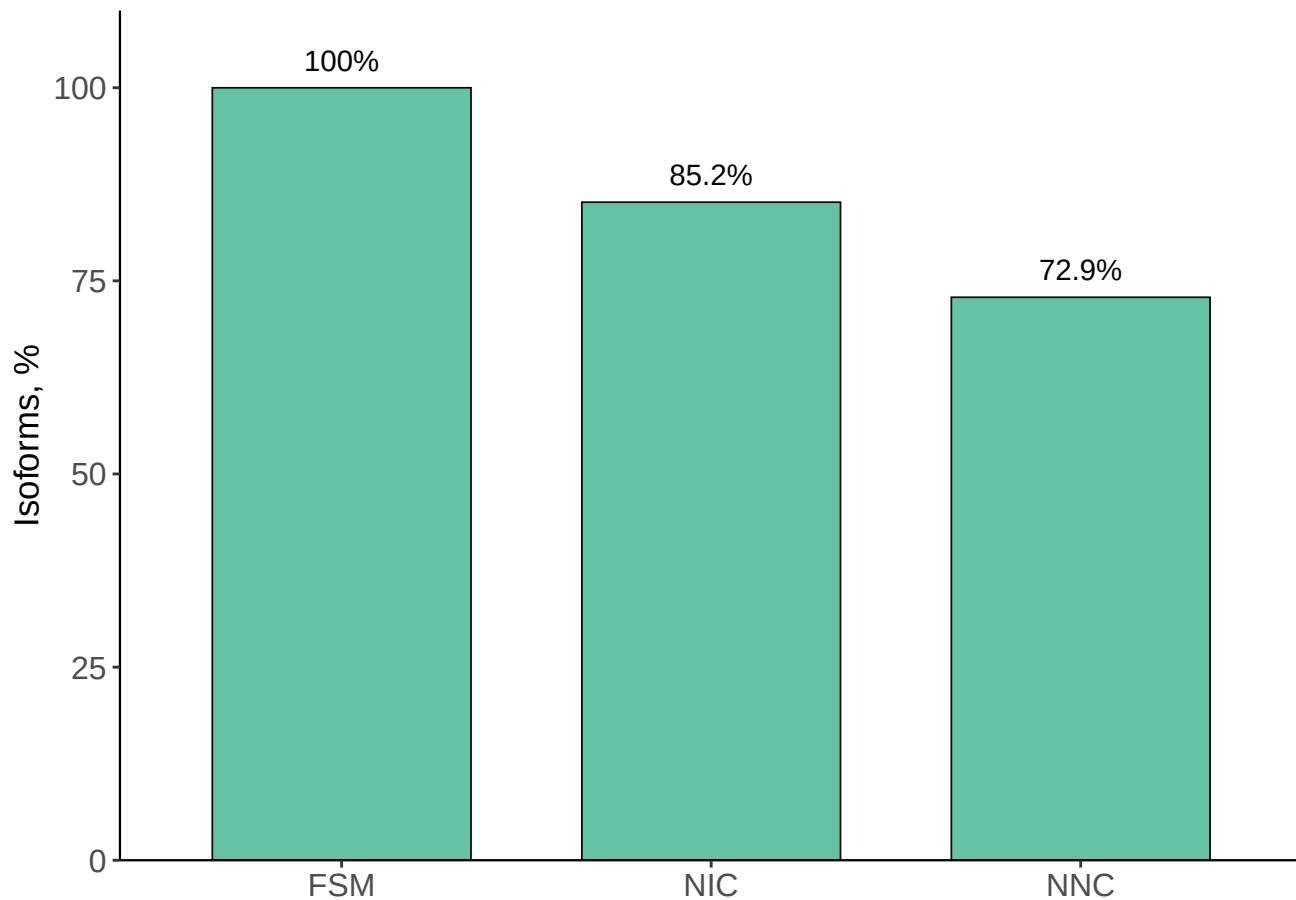


Non-Canonical Junctions

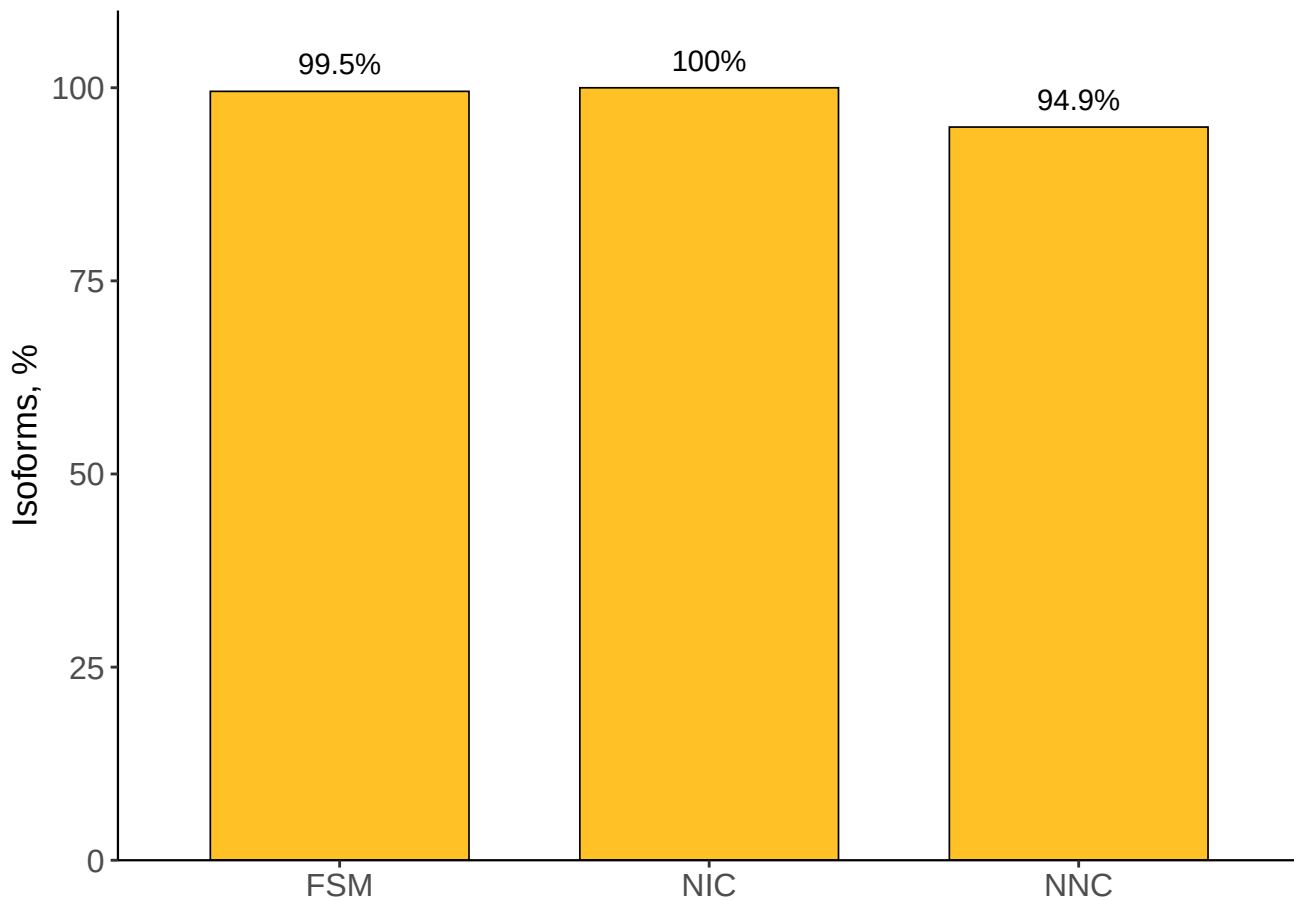


Features of Good Quality

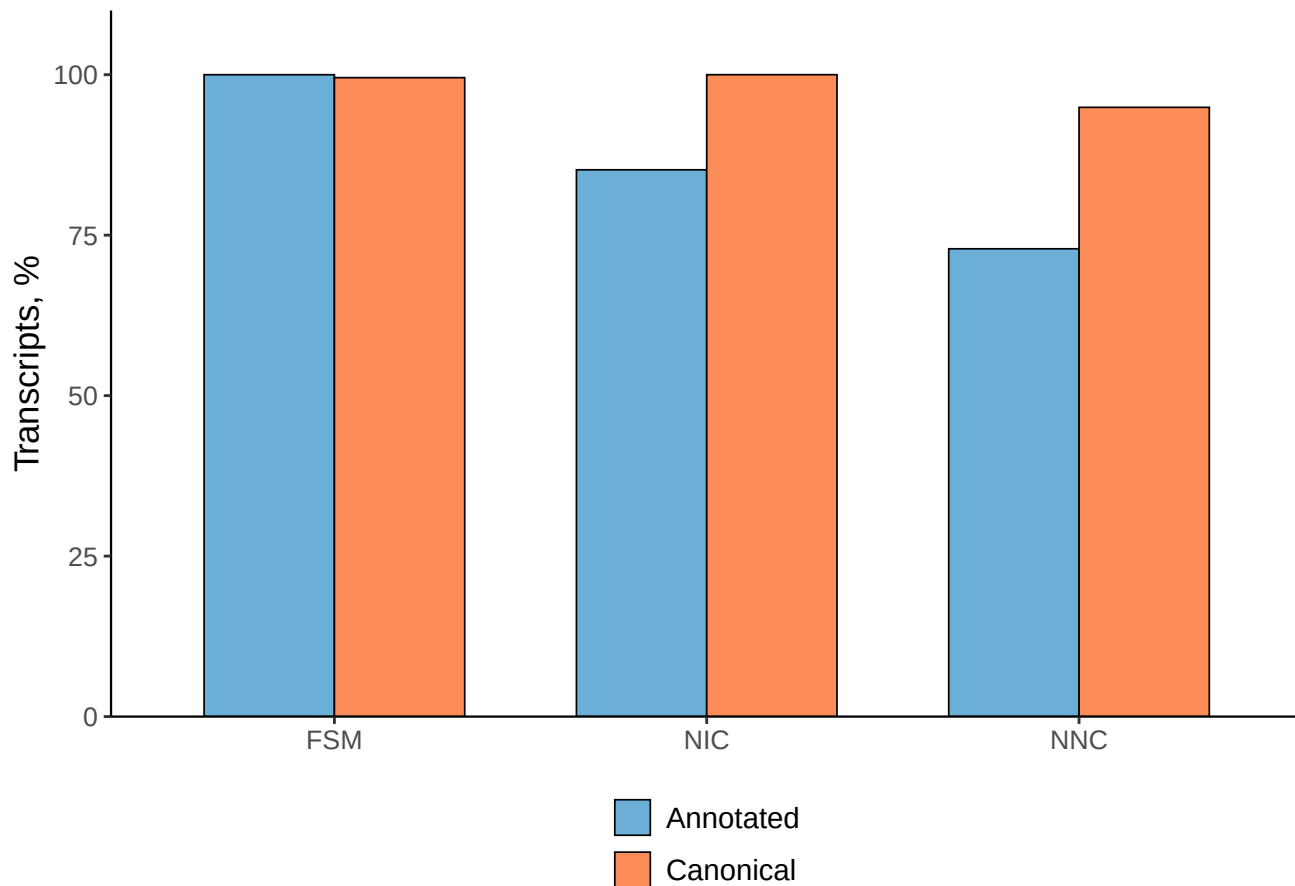
Annotation Support



All Canonical Junctions

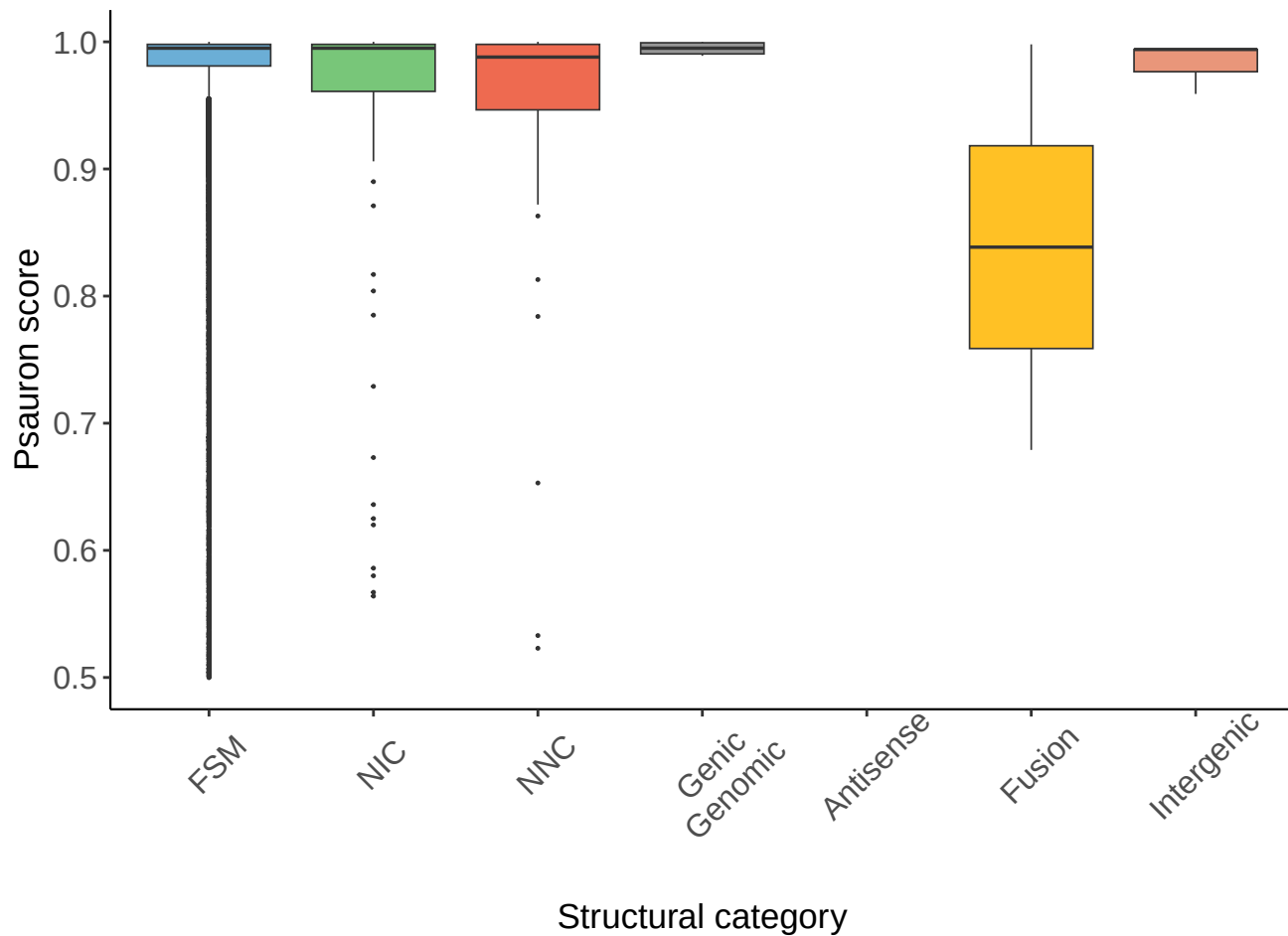


Good Quality Control Attributes Across Structural Categories

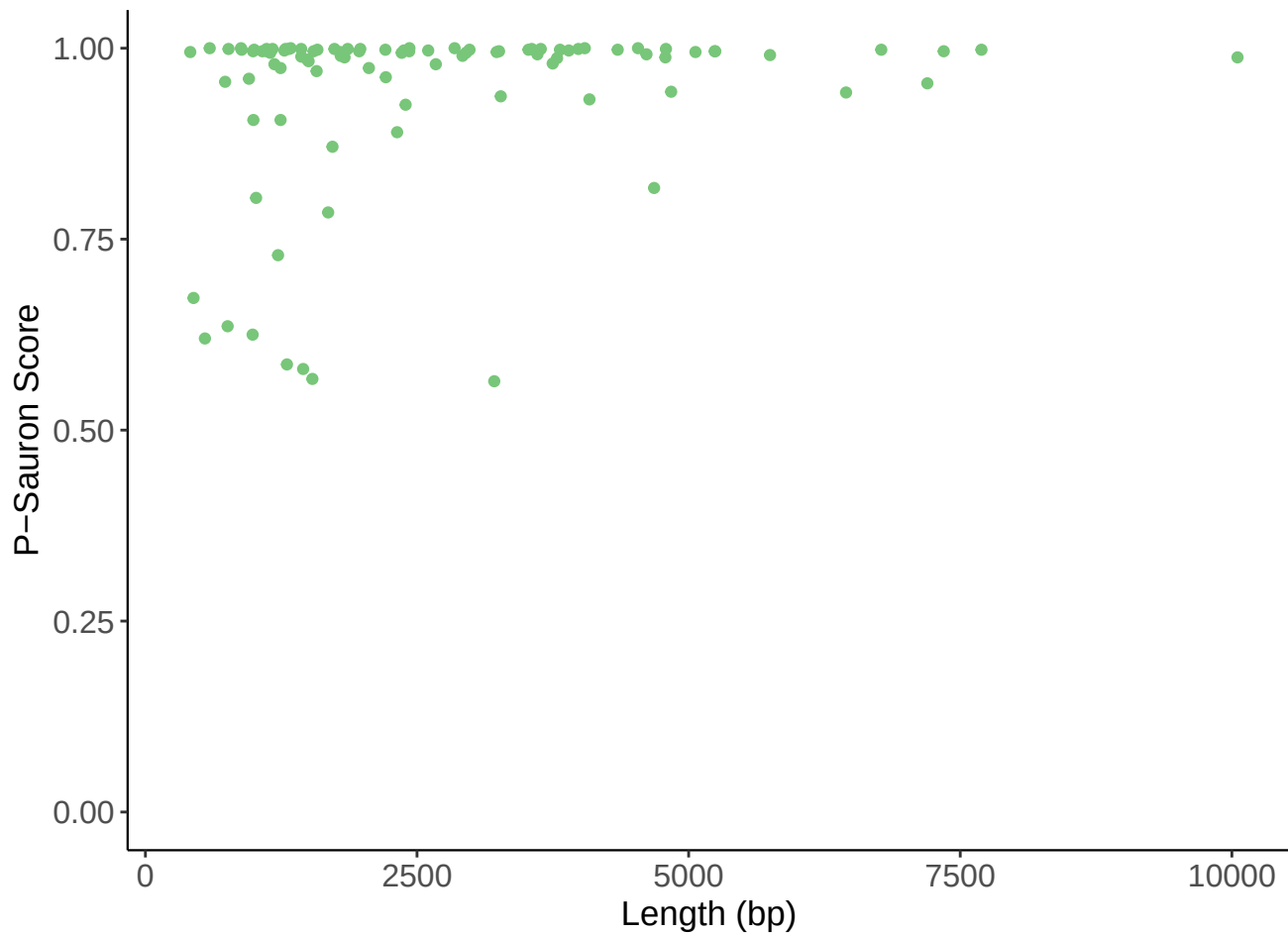


TransDecoder2 CDS prediction statistics

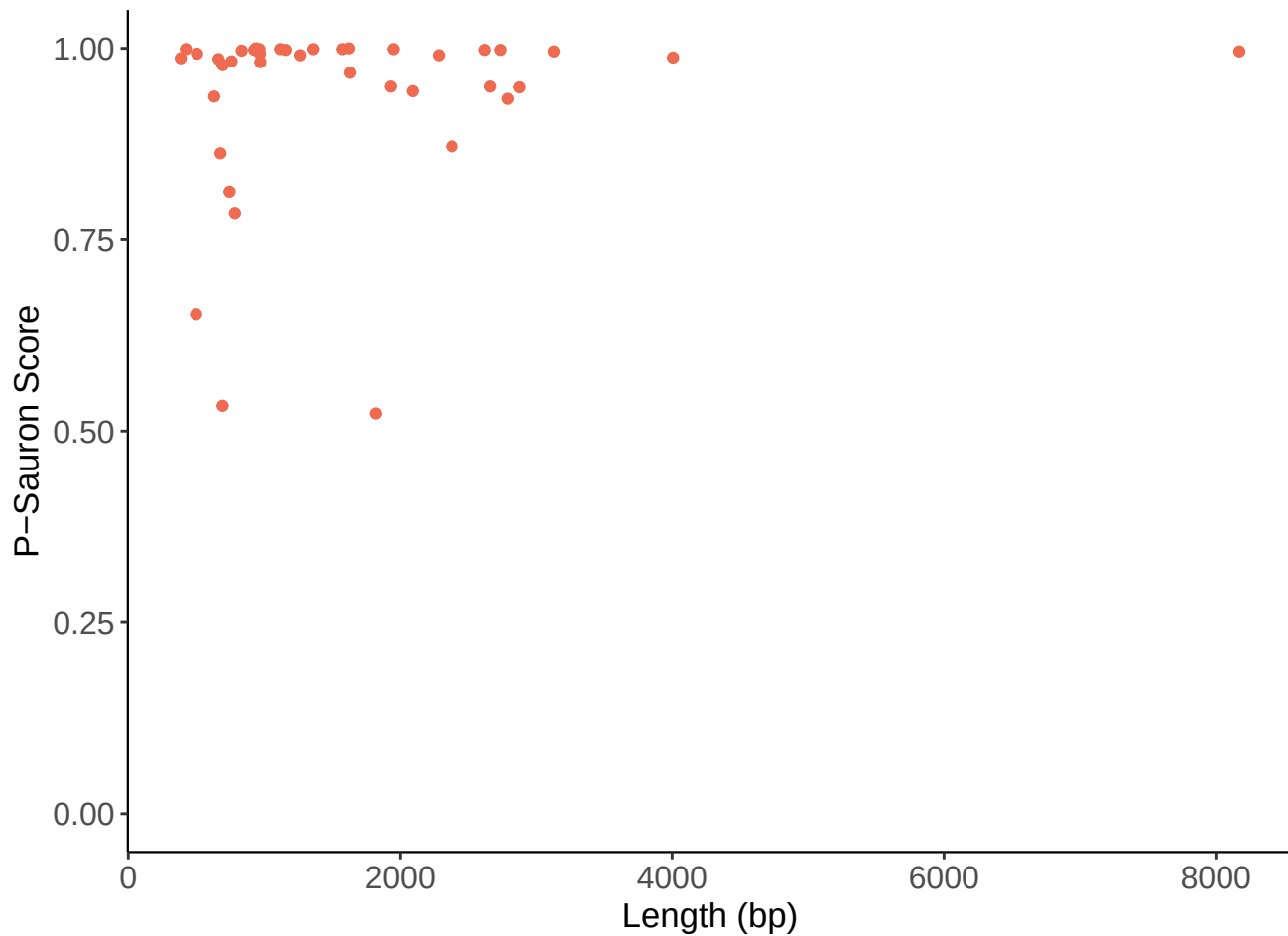
Psauron score per structural category



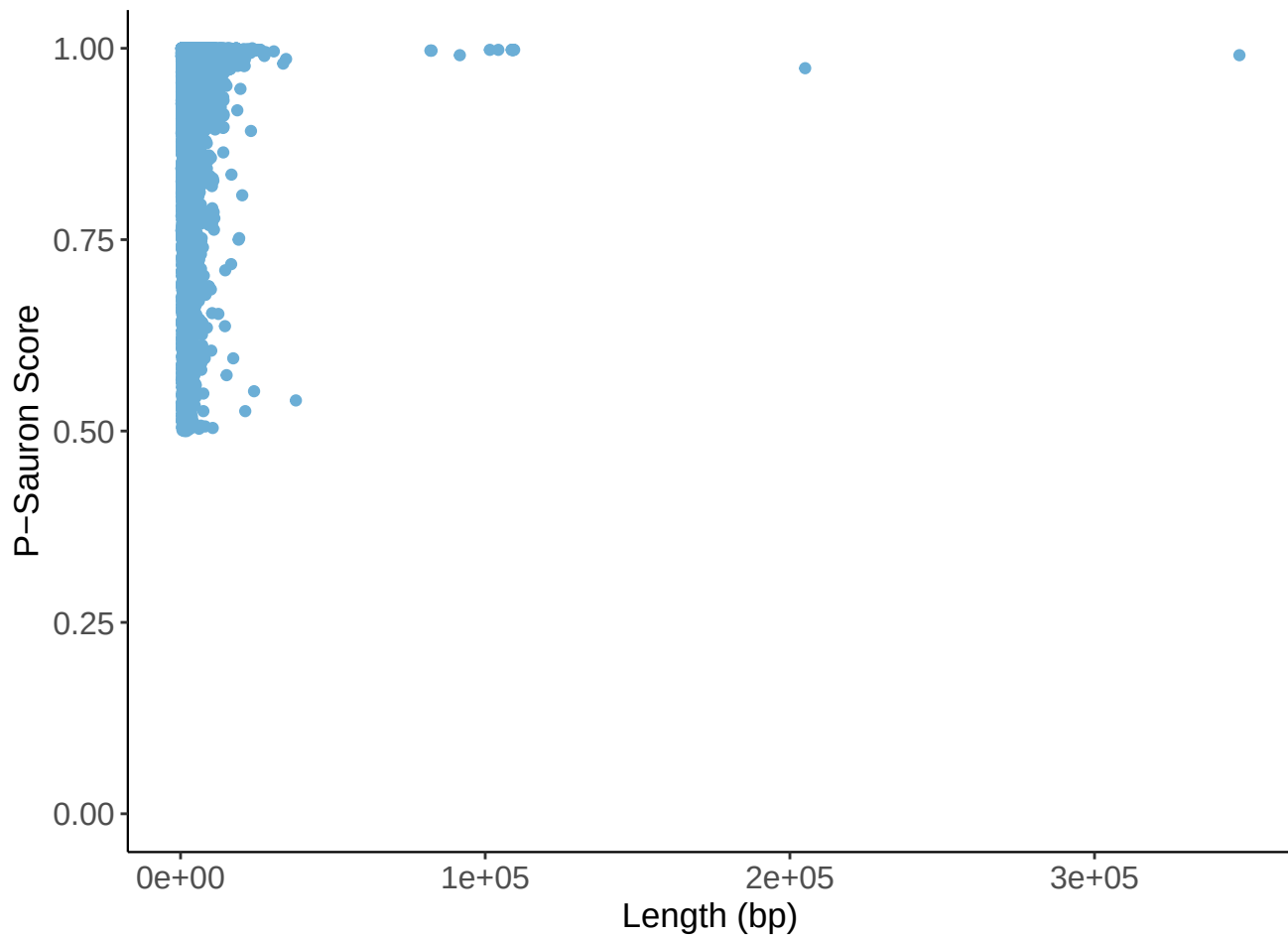
P-Sauron Score vs Length (NIC)



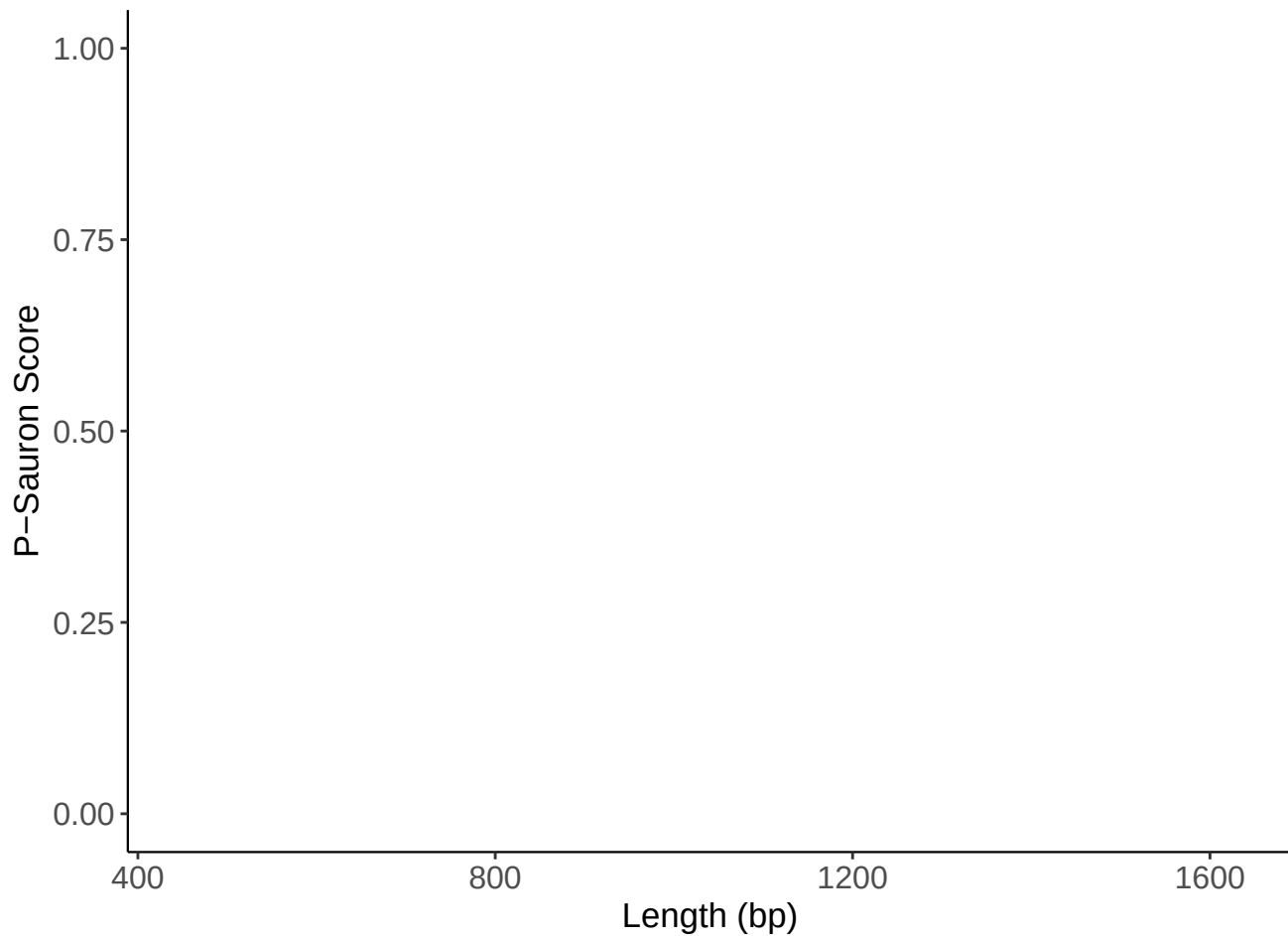
P-Sauron Score vs Length (NNC)



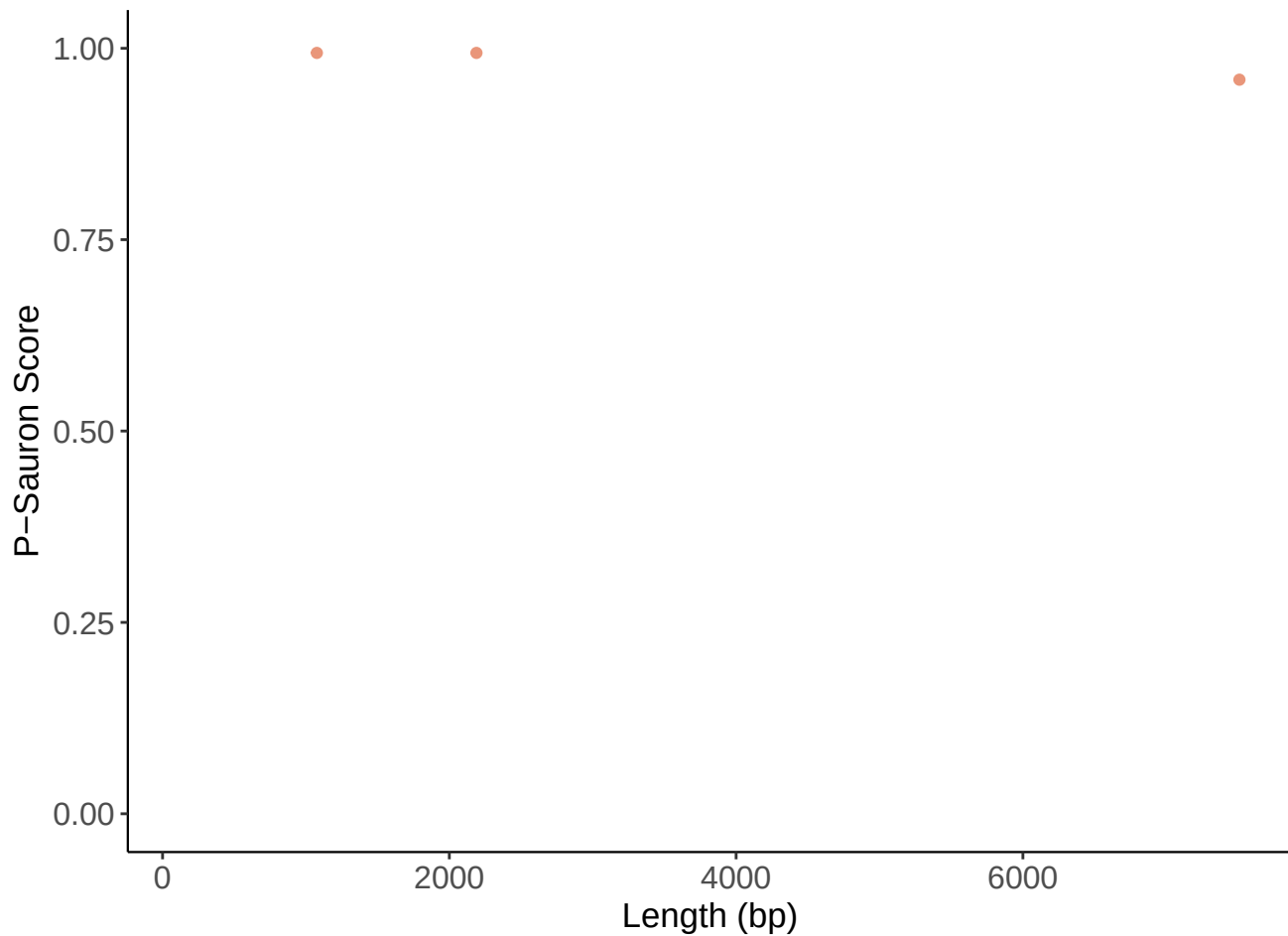
P-Sauron Score vs Length (FSM)



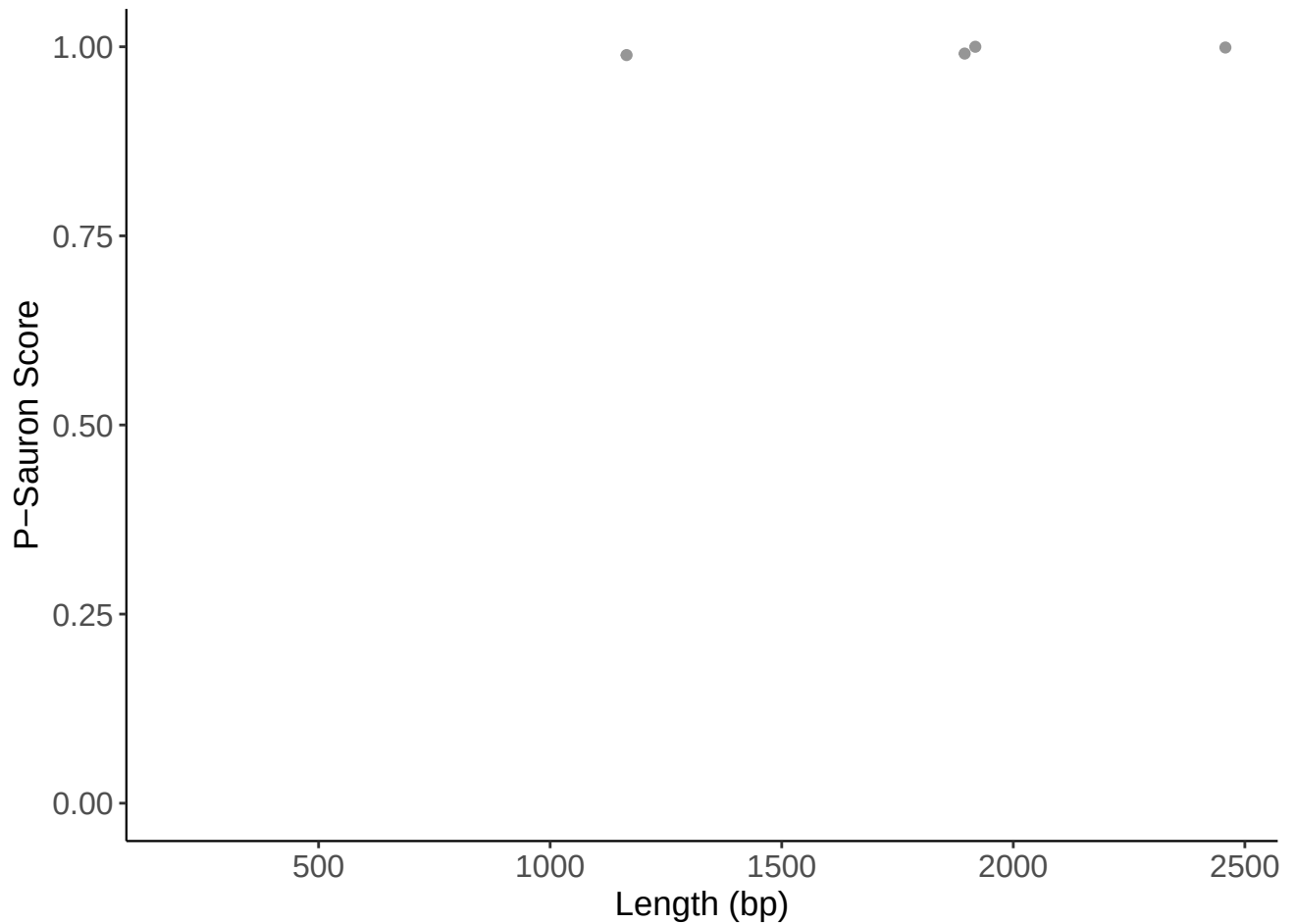
P-Sauron Score vs Length (Antisense)



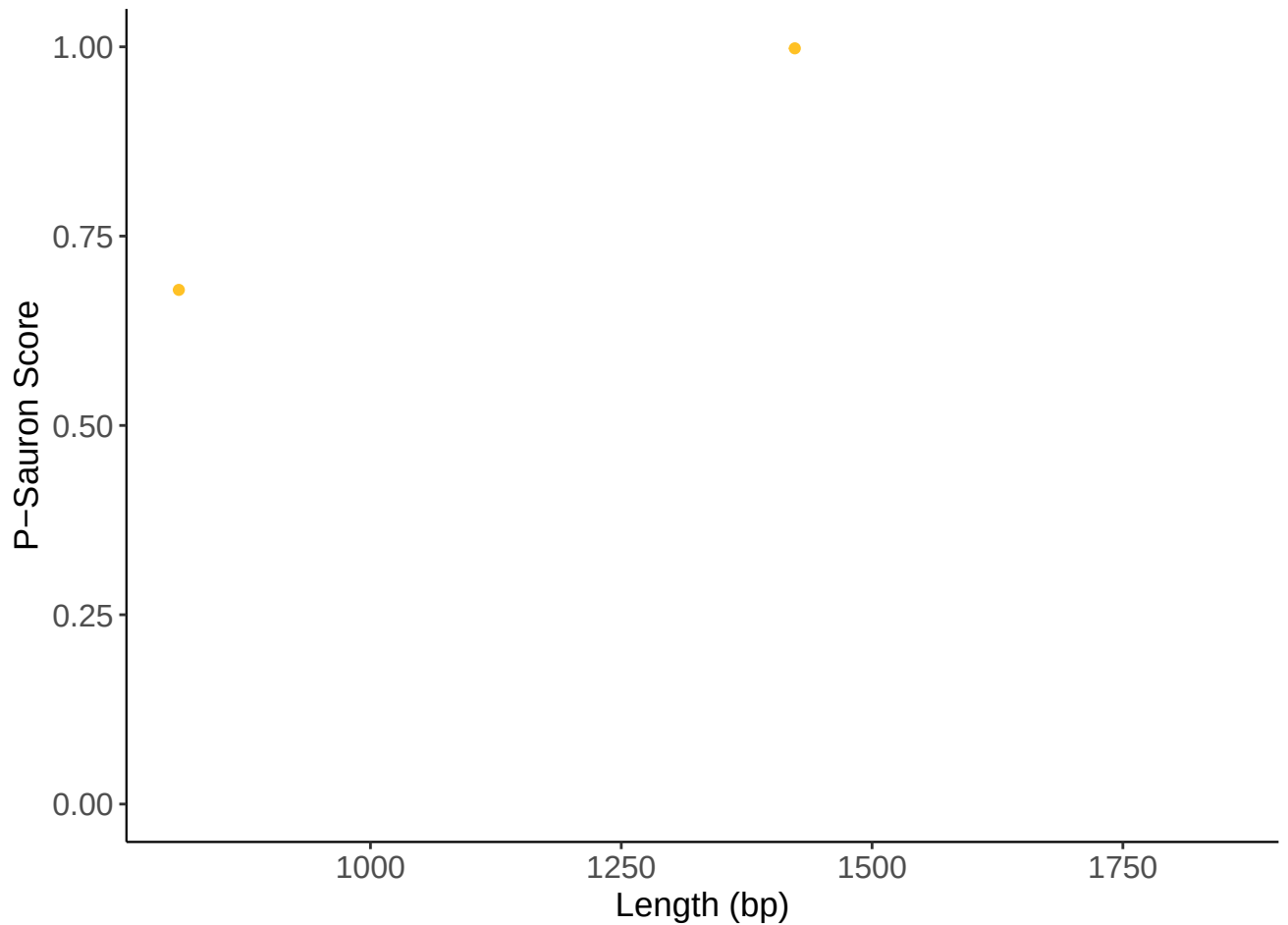
P-Sauron Score vs Length (Intergenic)



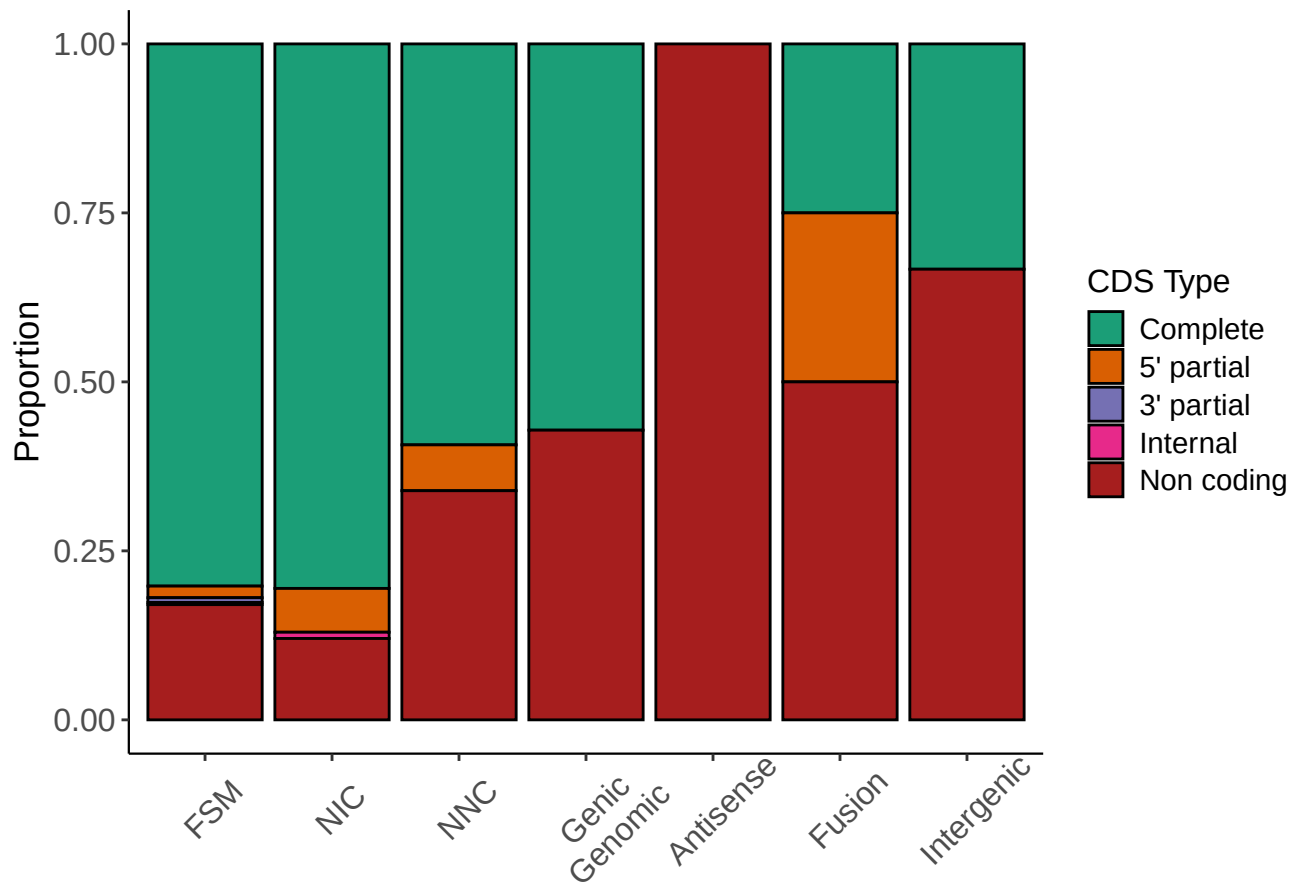
P-Sauron Score vs Length (Genic Genomic)



P-Sauron Score vs Length (Fusion)



CDS type by subcategory



Psaaron score per ORF type

